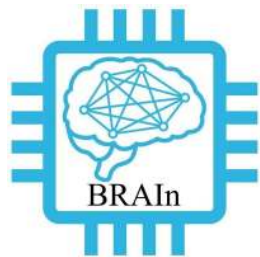




IMT Atlantique
Bretagne-Pays de la Loire
École Mines-Télécom



Examiners:

- Céline Hudelot
- Ismail Ben Ayed
- Frédéric Jurie

PhD Defense

A Data Modeling Perspective on Few-Shot Image Classification

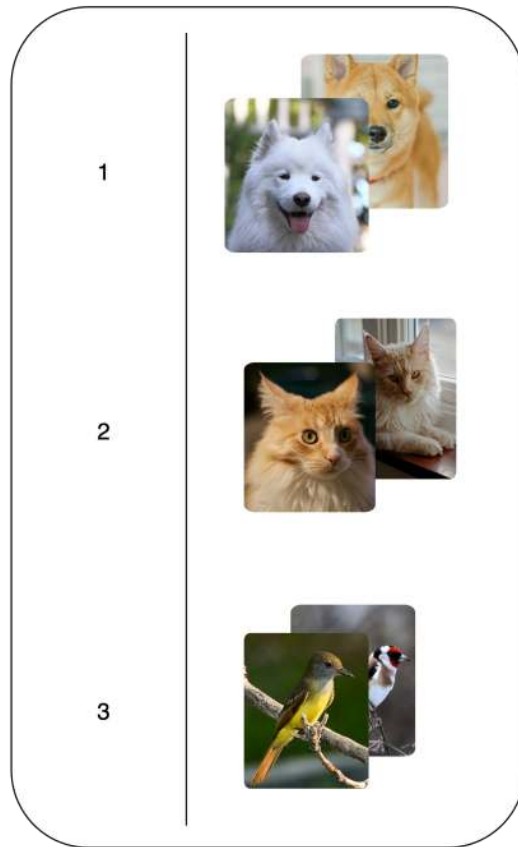
Yassir Bendou

Brest, 2nd December 2024

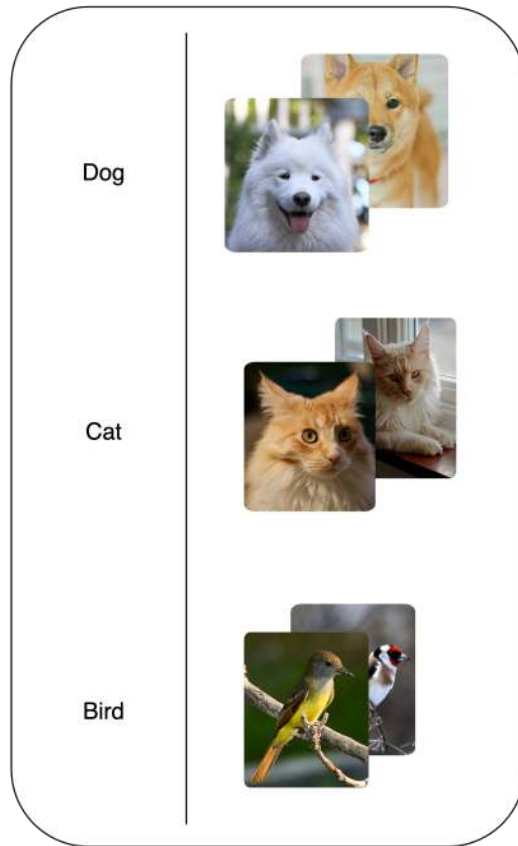
Supervisors:

- Vincent Gripon
- Bastien Pasdeloup
- Giulia Lioi

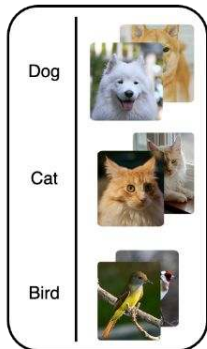
An overview of few-shot learning methods in image classification



An overview of few-shot learning methods in image classification

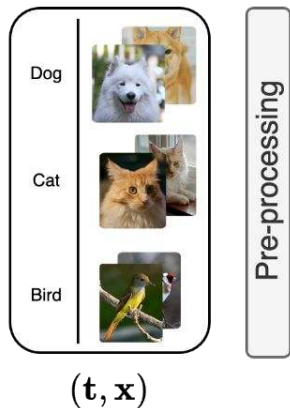


An overview of few-shot learning methods in image classification

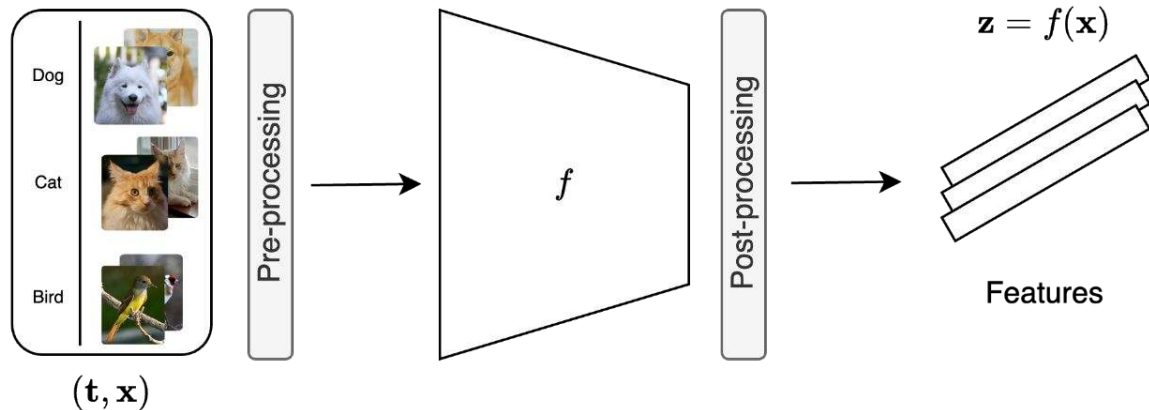


$(\mathbf{t}, \mathbf{x})$

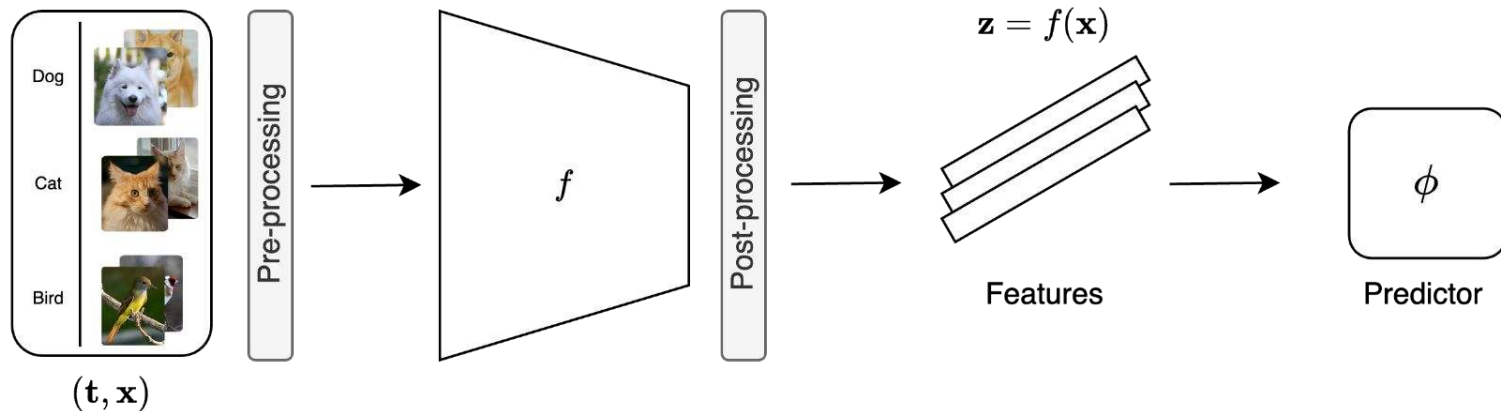
An overview of few-shot learning methods in image classification



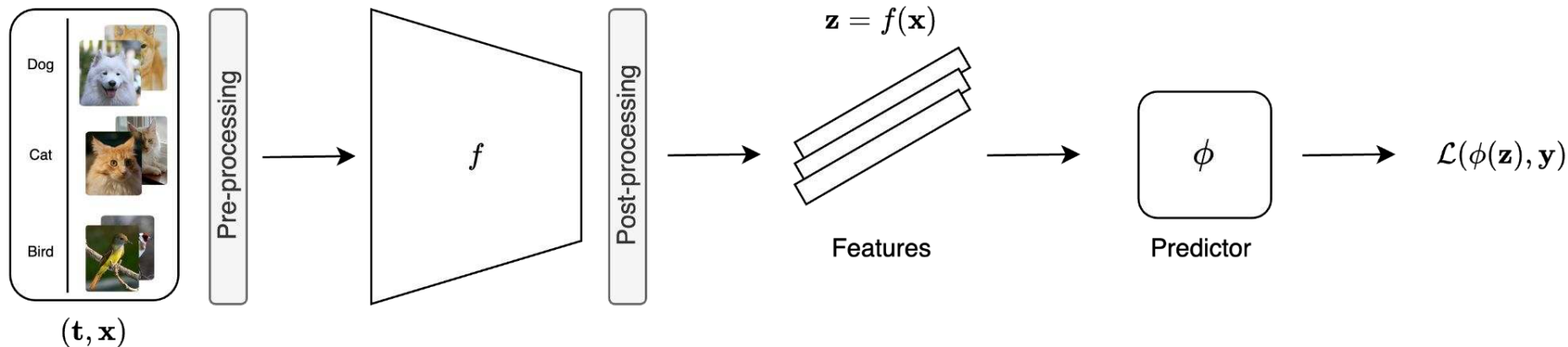
An overview of few-shot learning methods in image classification



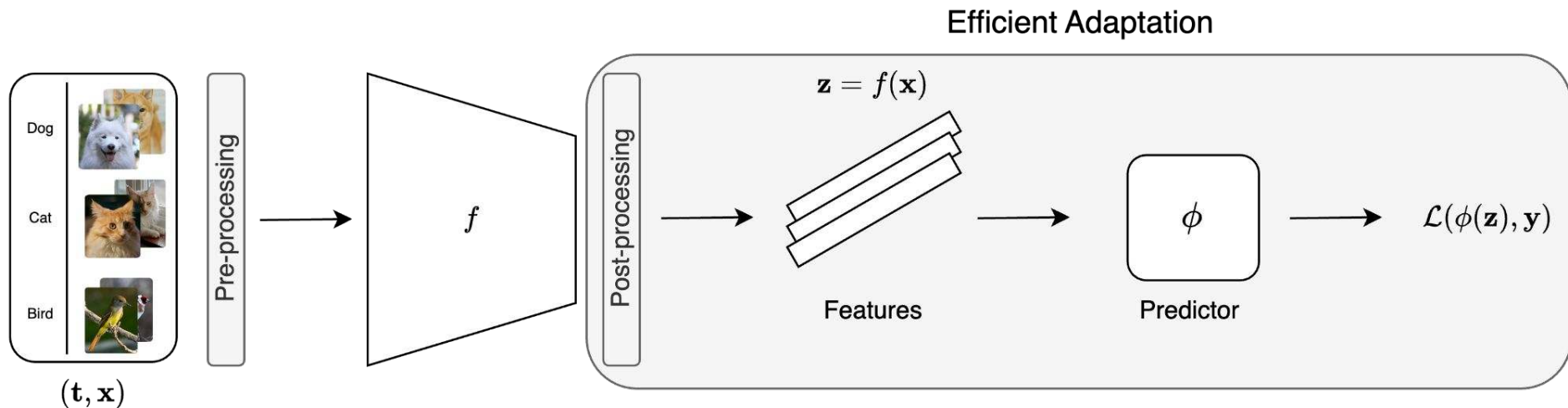
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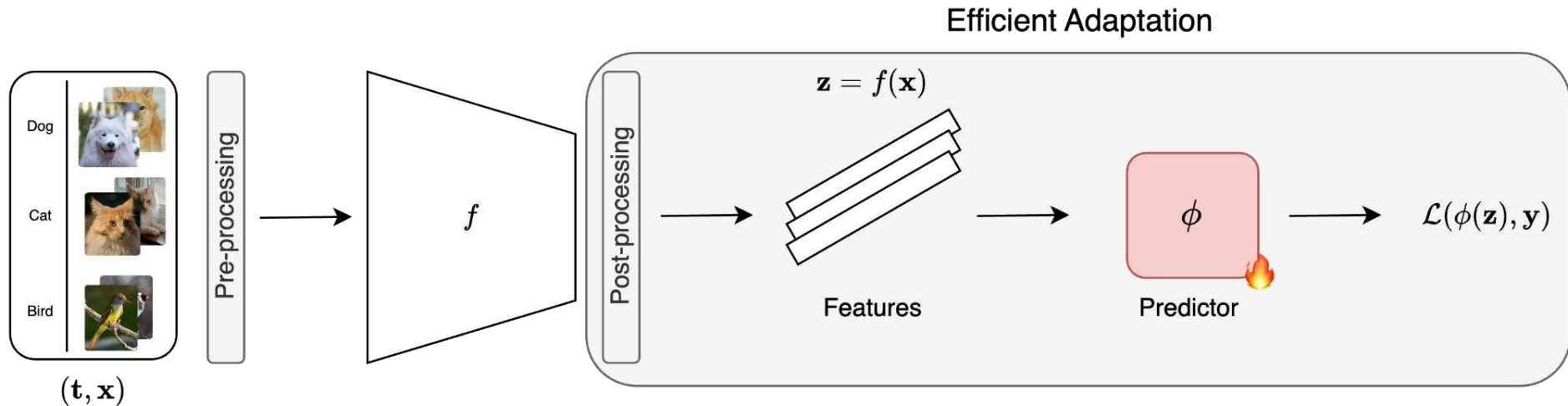
An overview of few-shot learning methods in image classification



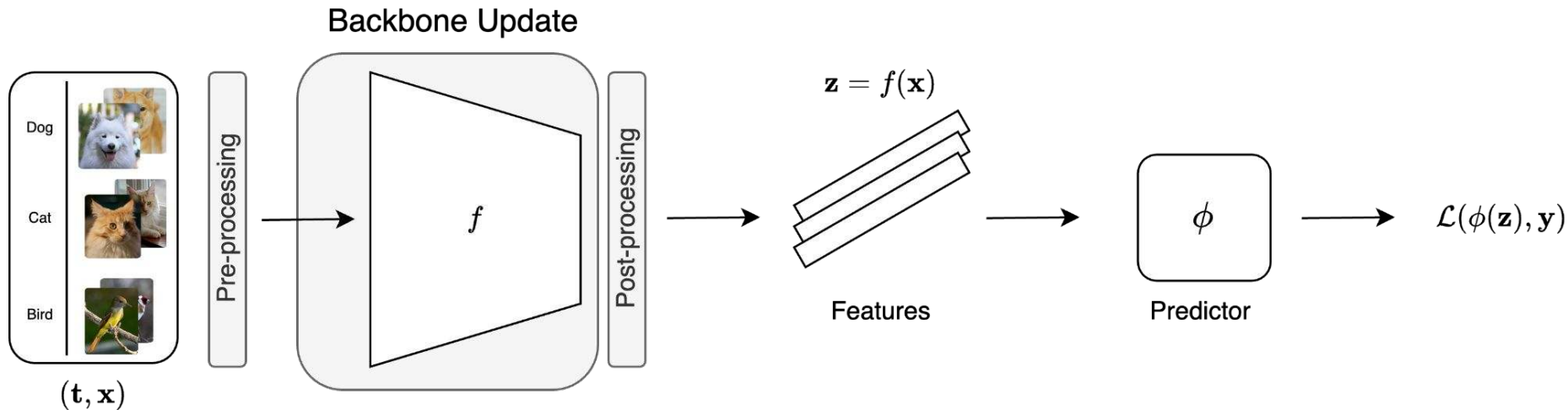
An overview of few-shot learning methods in image classification



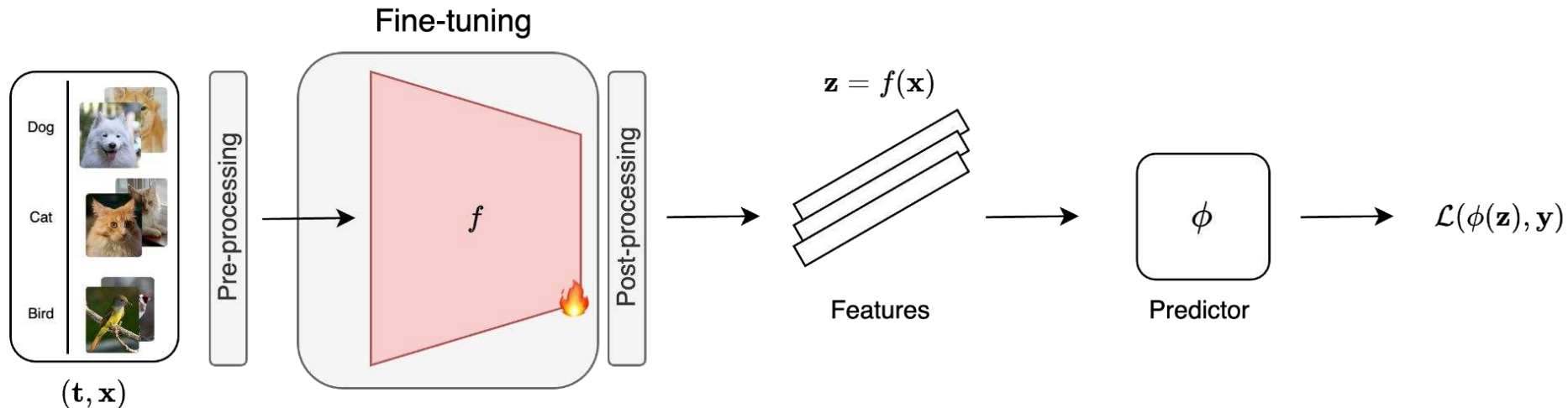
An overview of few-shot learning methods in image classification



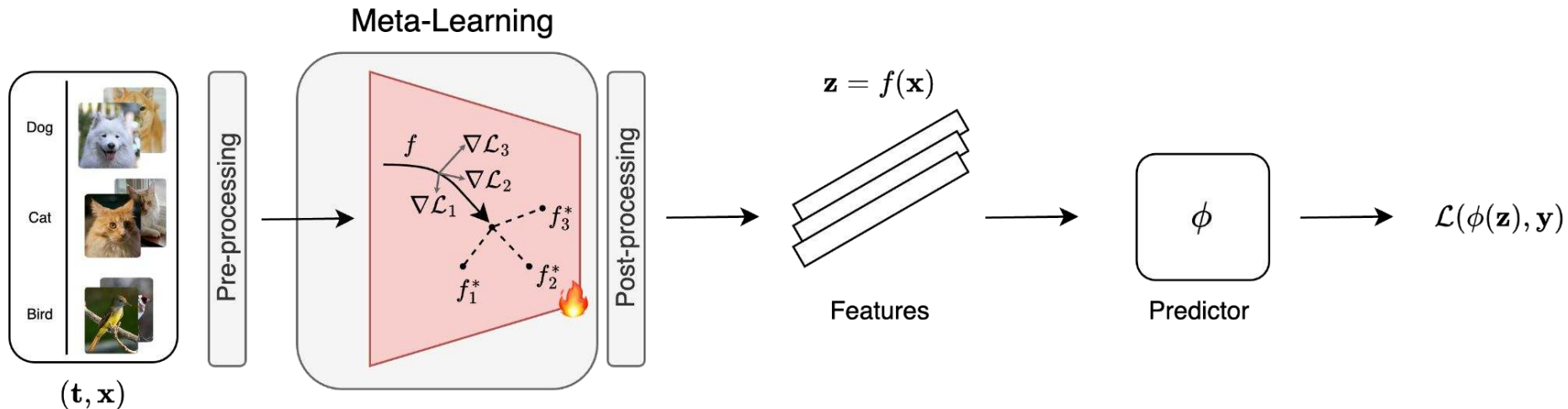
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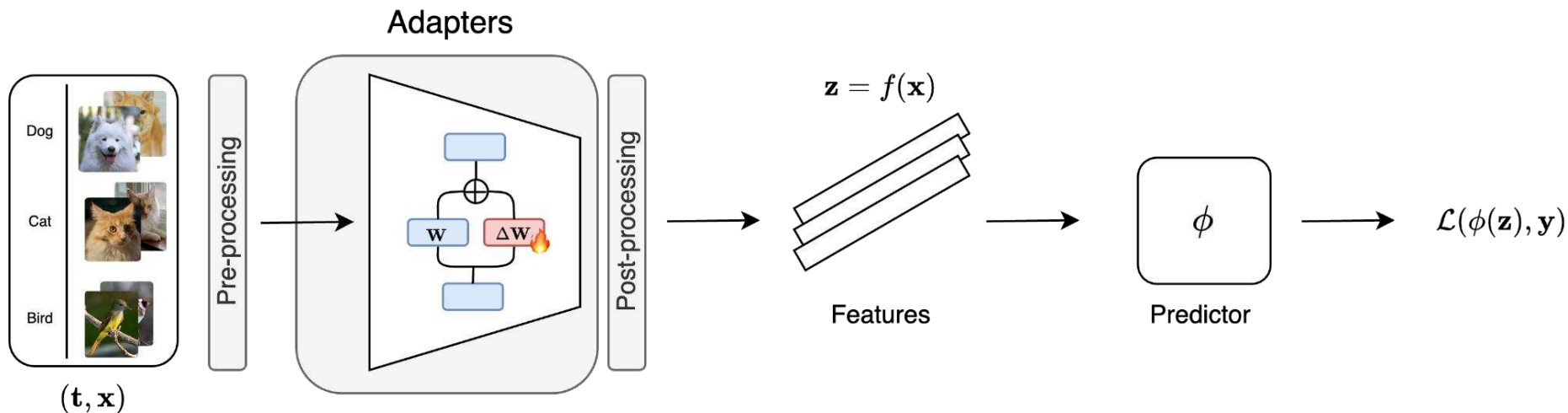
An overview of few-shot learning methods in image classification



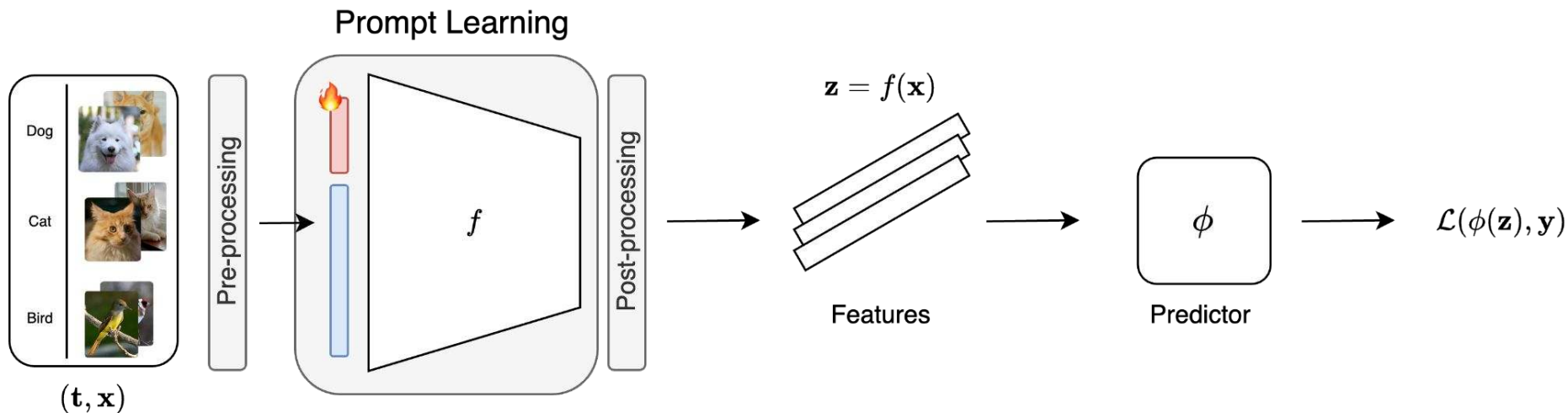
An overview of few-shot learning methods in image classification



An overview of few-shot learning methods in image classification

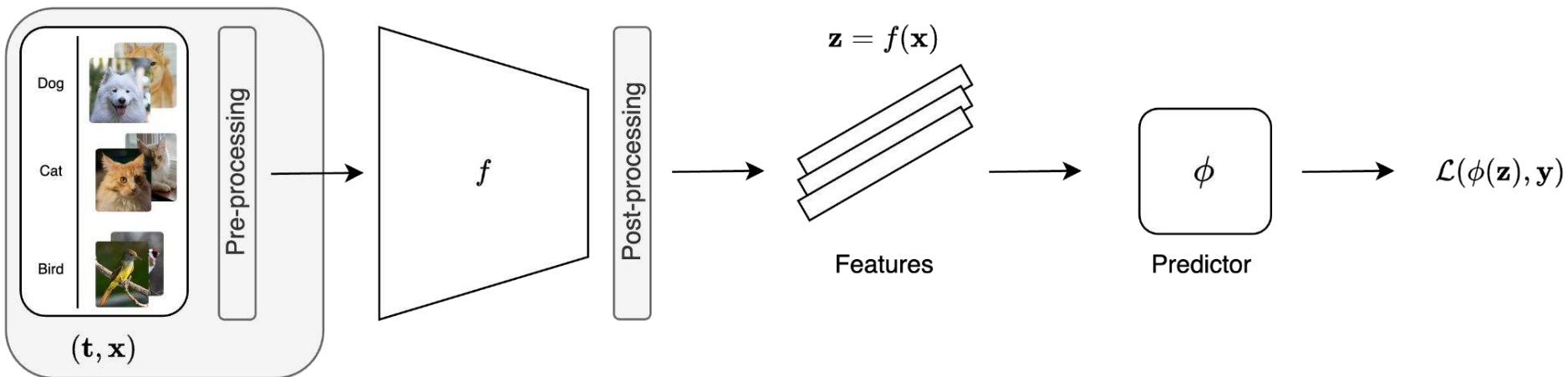


An overview of few-shot learning methods in image classification

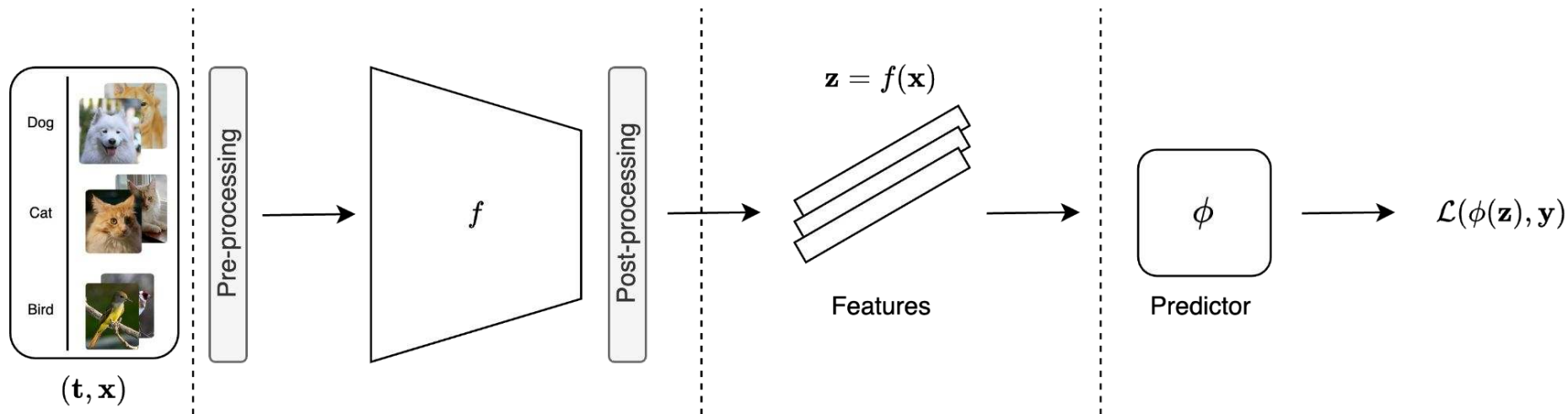


An overview of few-shot learning methods in image classification

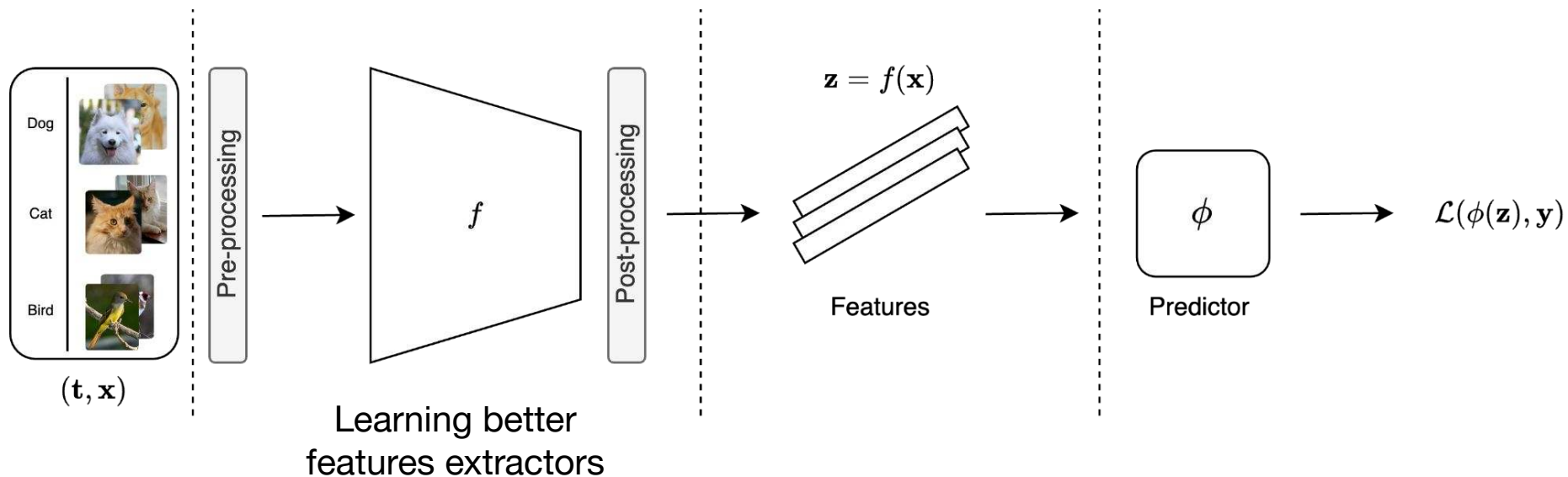
Data Augmentation



An overview of few-shot learning methods in image classification

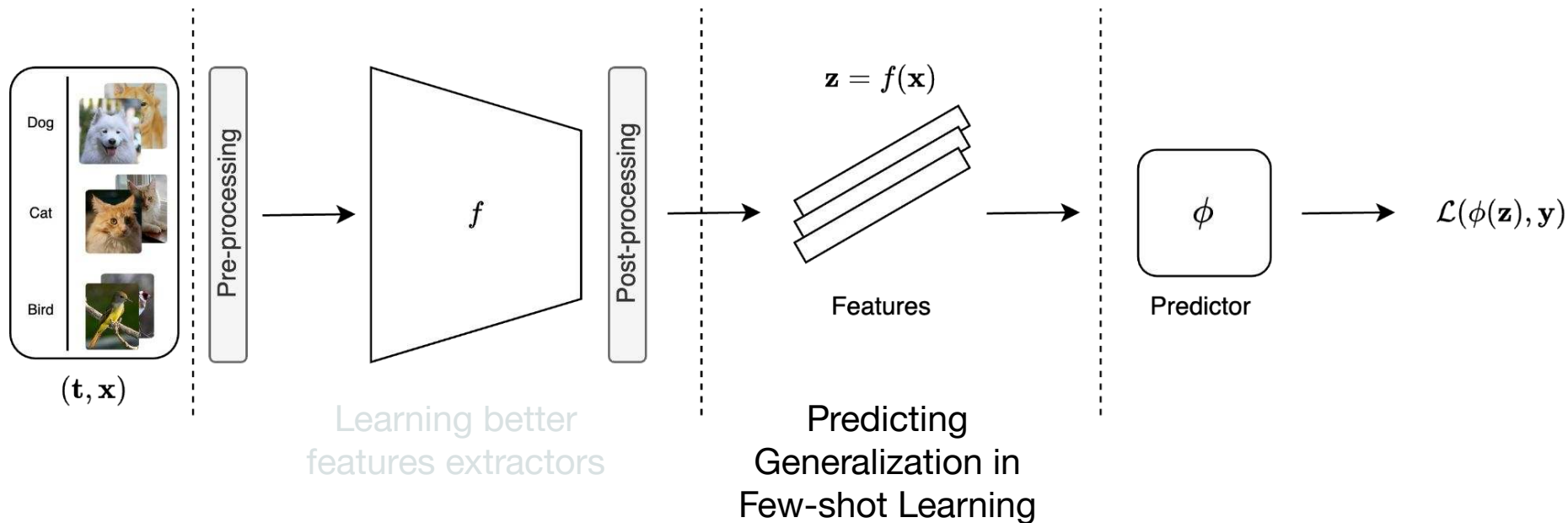


Contributions



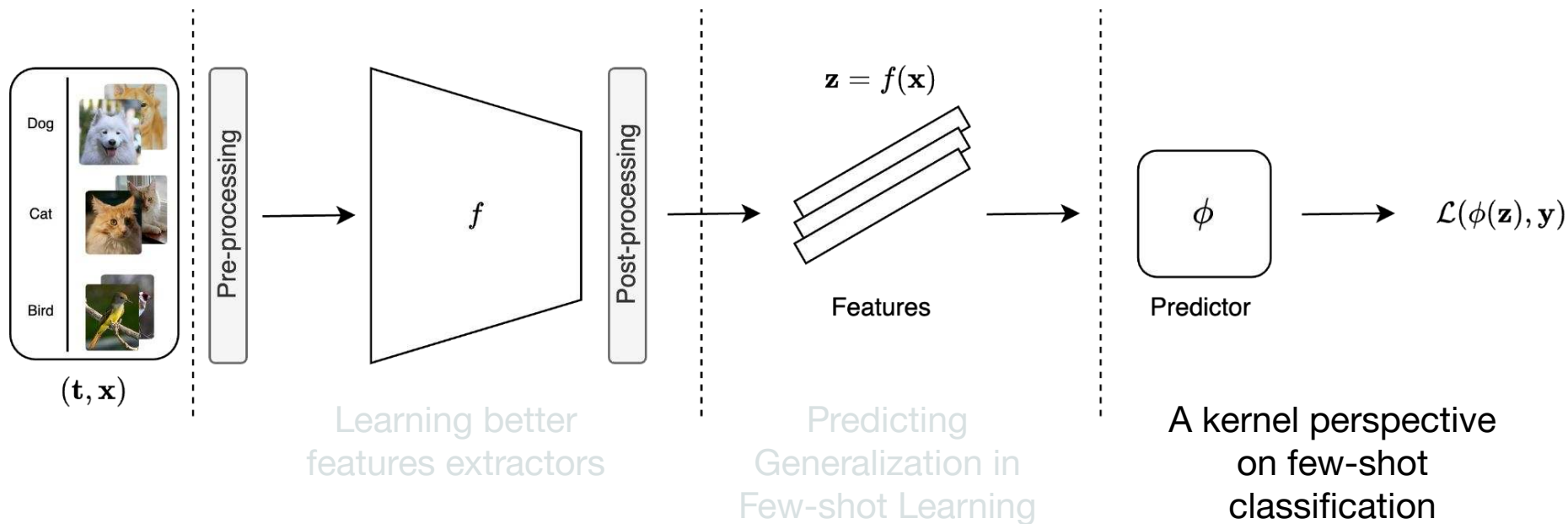
- Easy: Ensemble Augmented-Shot Y-shaped Learning: State-Of-The-Art Few-Shot Classification with Simple Components, Bendou et al. Journal of Imaging 2022.

Contributions



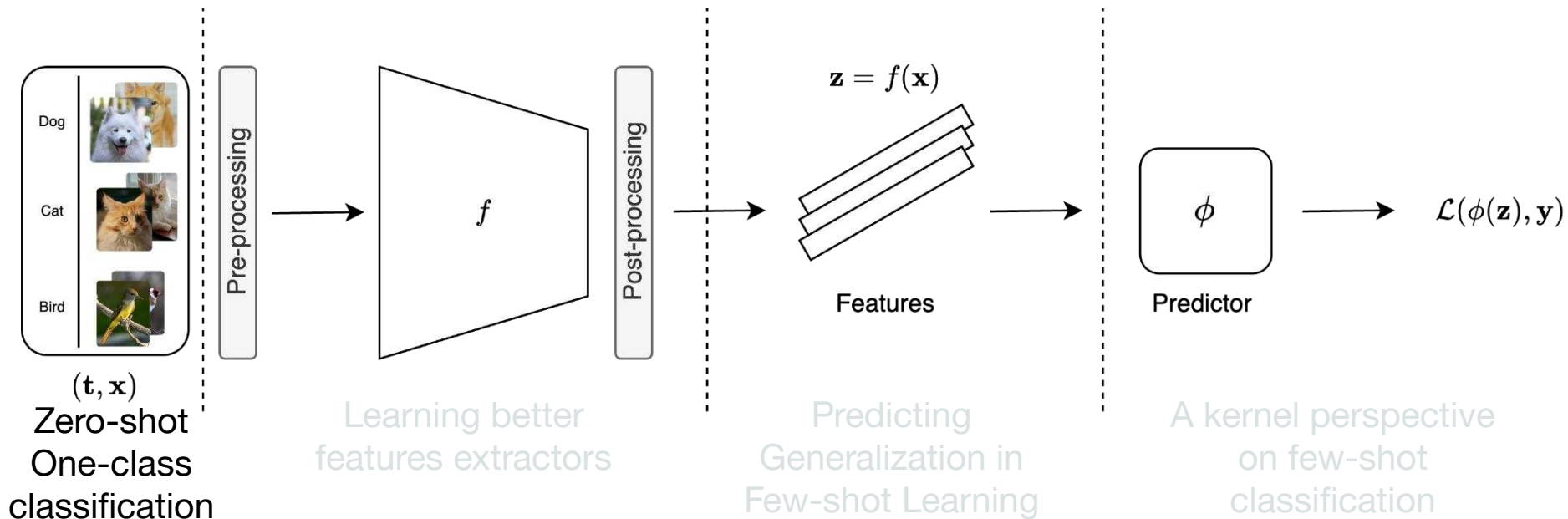
- A Statistical Model for Predicting Generalization in Few-Shot Classification, Bendou et al. IEEE EUSIPCO 2023.

Contributions



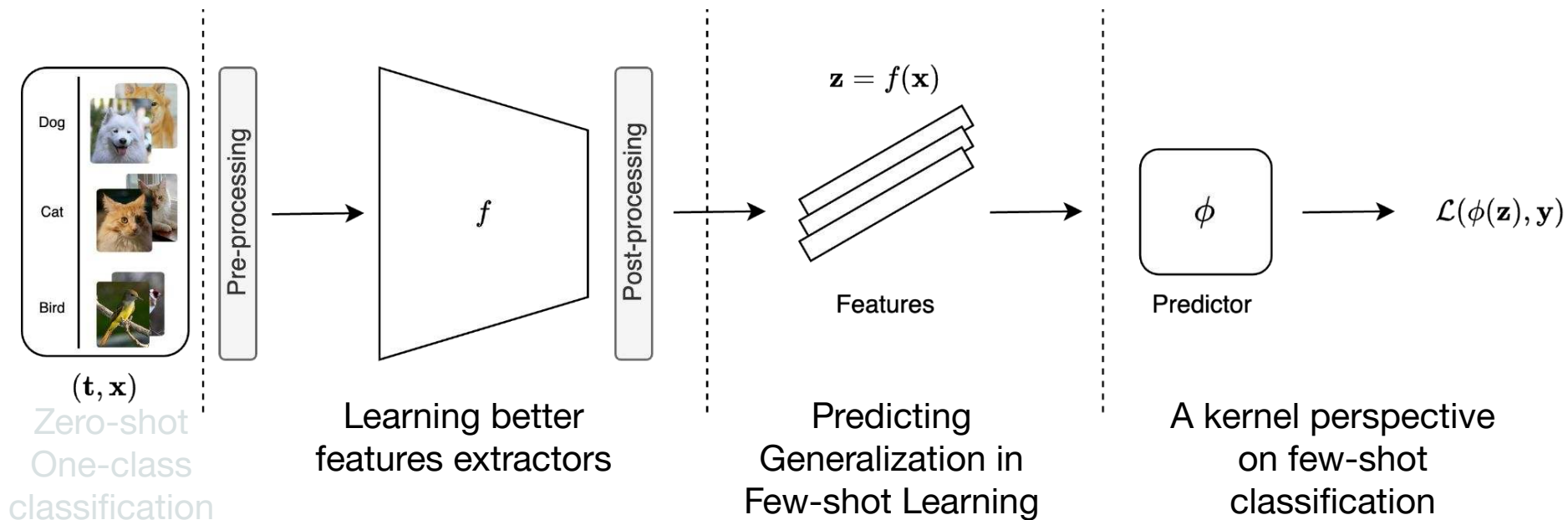
- A Kernel Perspective on Training-Free Few-Shot Adaptation of Large Vision-Language Models, Bendou et al. under review at CVPR 2025.

Contributions

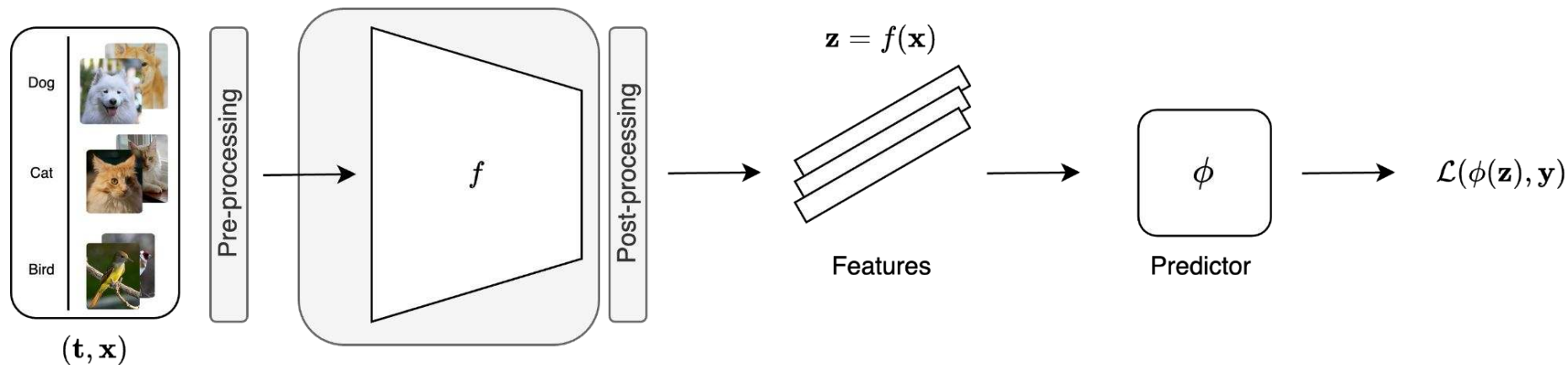


- LLM meets Vision-Language Models for Zero-Shot One-Class Classification, Bendou et al. ArXiv preprint 2024.

Contributions

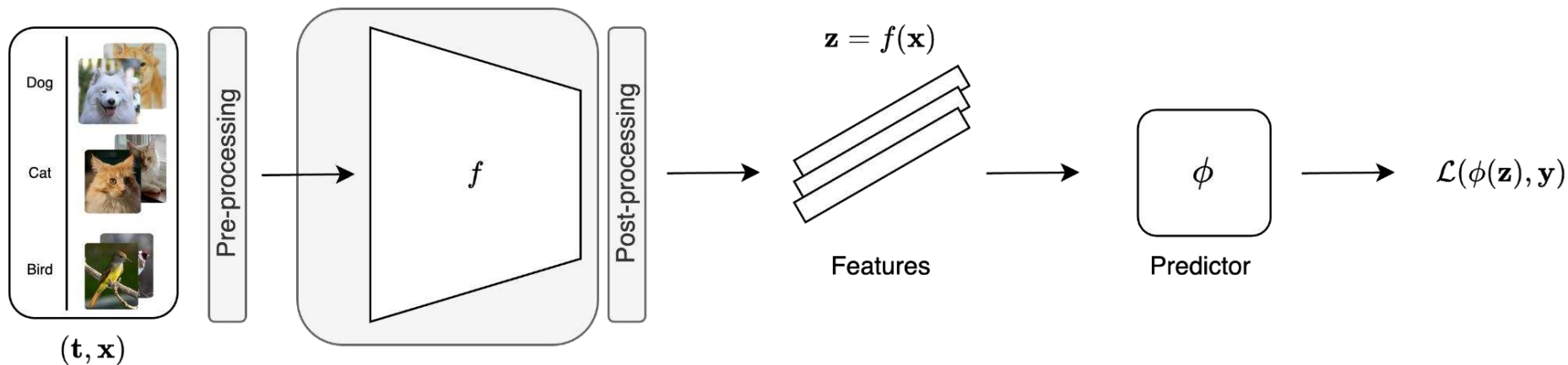


A strong baseline for few shot image classification

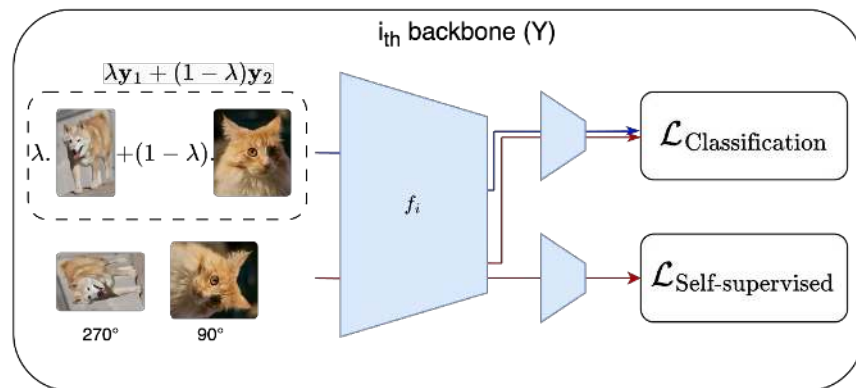


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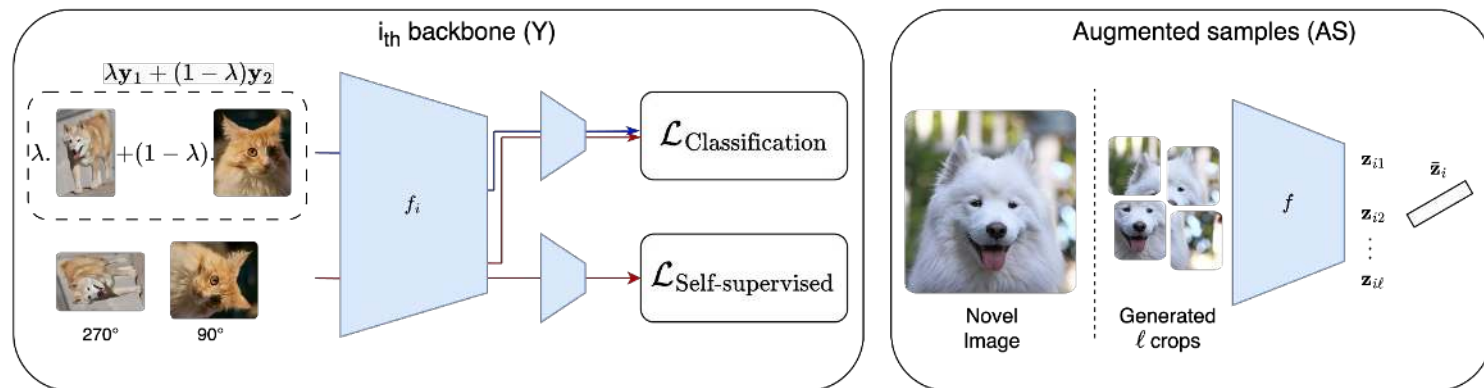
EASY - Ensemble Augmented-Shot Y-shaped Learning: State-Of-The-Art Few-Shot Classification with Simple Ingredients.



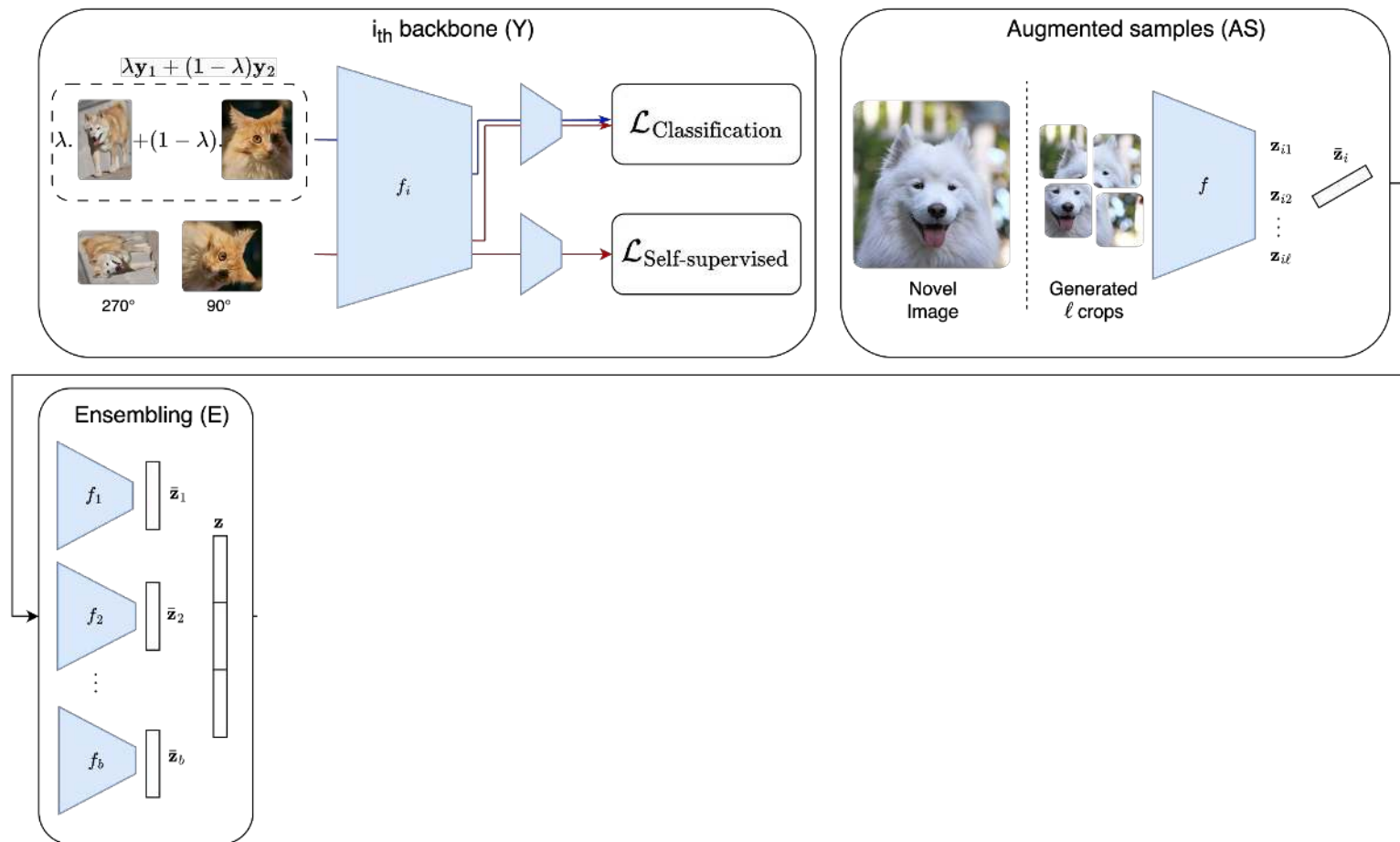
A strong baseline for few shot image classification



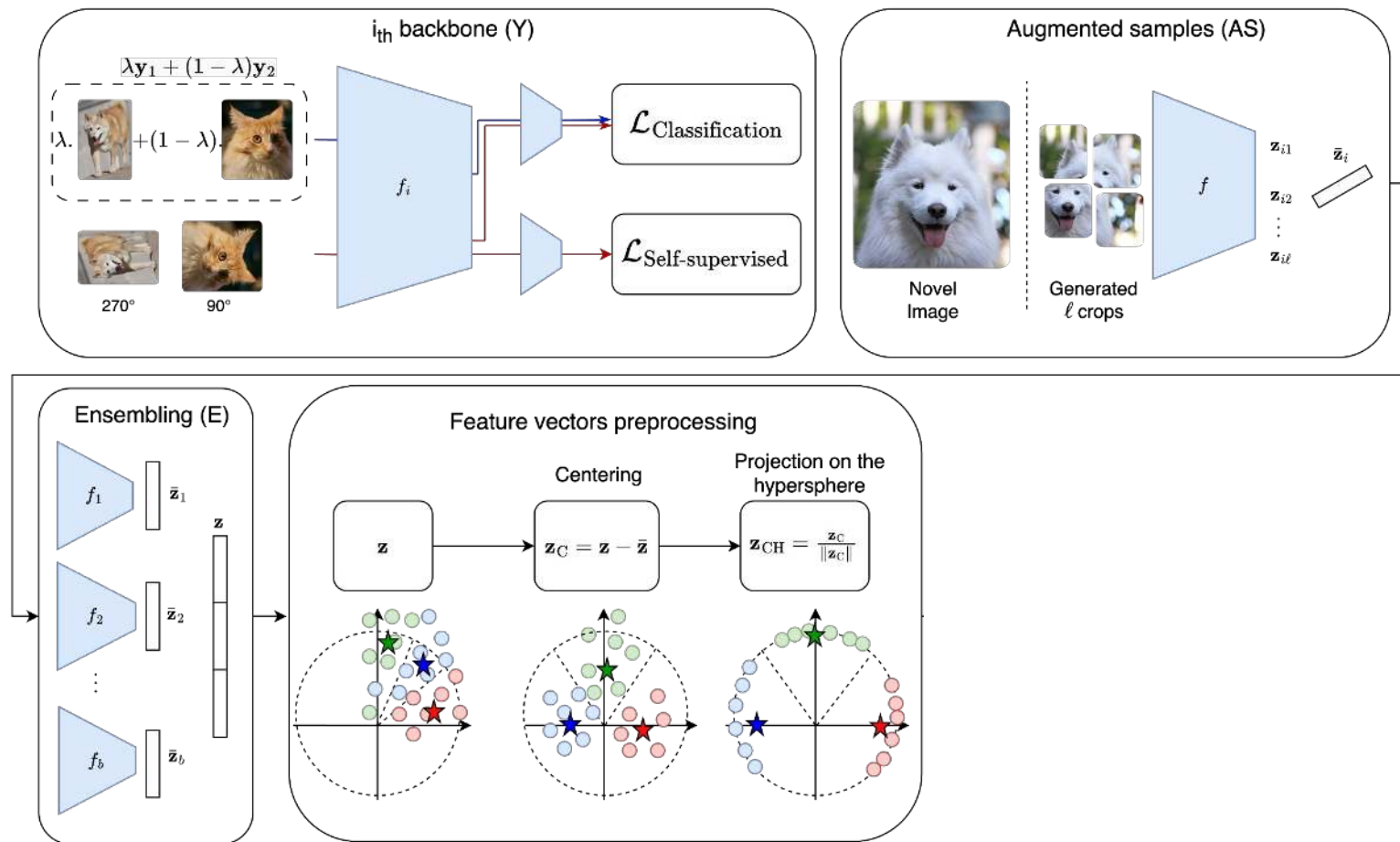
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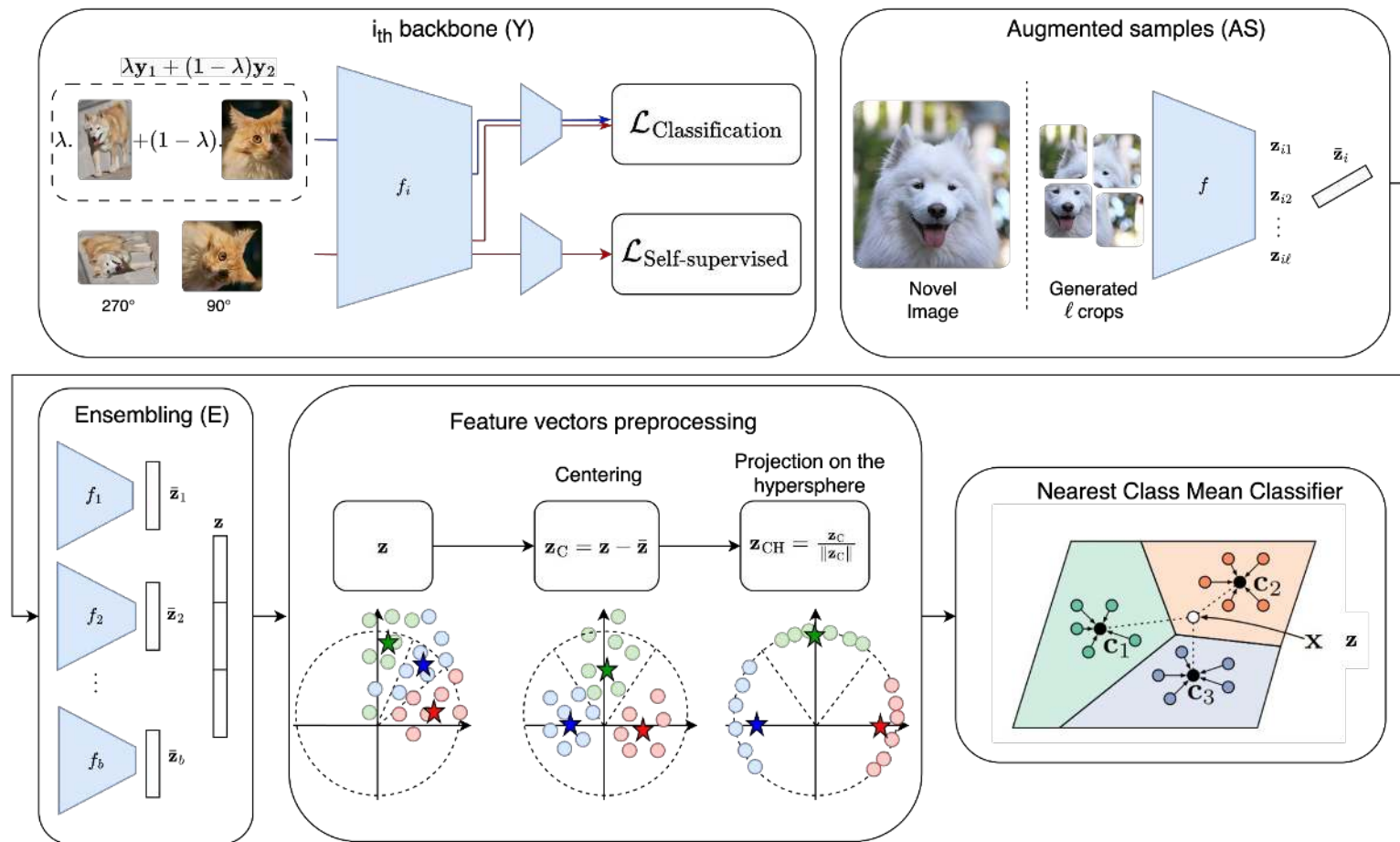
A strong baseline for few shot image classification



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A strong baseline for few shot image classification



A strong baseline for few shot image classification

Table 1: Results on minImageNet

Method	Year	backbone	#Params	1-shot	5-shot
SimpleShot	2019	ResNet12	12 M	62.85 ± 0.20	80.02 ± 0.14
Baseline++	2019	ResNet12	12 M	53.97 ± 0.79	75.90 ± 0.61
TADAM	2018	ResNet12	12 M	58.50 ± 0.30	76.70 ± 0.30
ProtoNet	2017	ResNet12	12 M	60.37 ± 0.83	78.02 ± 0.57
R2-D2 (+ens)	2020	ResNet12	12 M	64.79 ± 0.45	81.08 ± 0.32
FEAT	2020	ResNet12	12 M	66.78	82.05
CNL	2021	ResNet12	12 M	67.96 ± 0.98	83.36 ± 0.51
MERL	2020	ResNet12	12 M	67.40 ± 0.43	83.40 ± 0.28
Deep EMD v2	2020	ResNet12	12 M	68.77 ± 0.29	84.13 ± 0.53
PAL	2021	ResNet12	12 M	69.37 ± 0.64	84.40 ± 0.44
invariance-equivariance	2021	ResNet12	12 M	67.28 ± 0.80	84.78 ± 0.50
CSEI	2021	ResNet12	12 M	68.94 ± 0.28	85.07 ± 0.50
COSOC	2021	ResNet12	12 M	69.28 ± 0.49	85.16 ± 0.42
EASY (ours)	2022	2×ResNet12($\frac{1}{\sqrt{2}}$)	12 M	70.63 ± 0.20	86.28 ± 0.12
S2M2R	2020	WRN-28-10	36 M	64.93 ± 0.18	83.18 ± 0.11
LR + DC	2021	WRN-28-10	36 M	68.55 ± 0.55	82.88 ± 0.42
EASY (ours)	2022	3×ResNet12	36 M	71.75 ± 0.19	87.15 ± 0.12
State-of-the-art after the publication of EASY					
FewTURE	2022	ViT-Small	22 M	68.02 ± 0.88	84.51 ± 0.53
SMKD	2023	ViT-Small	21 M	74.28 ± 0.18	88.89 ± 0.09
MetaFormer-I	2024	ViT-Small	24.5 M	51.14 ± 0.71	68.33 ± 0.74

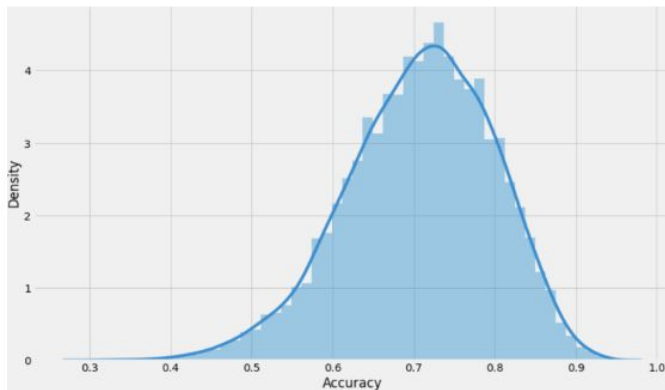
Table 2: Results on Tiered-ImageNet

Method	Year	backbone	#Params	1-shot	5-shot
SimpleShot	2019	ResNet12	12 M	69.09 ± 0.22	84.58 ± 0.16
ProtoNet	2017	ResNet12	12 M	65.65 ± 0.92	83.40 ± 0.65
FEAT	2018	ResNet12	12 M	70.80 ± 0.23	84.79 ± 0.16
PAL	2021	ResNet12	12 M	72.25 ± 0.72	86.95 ± 0.47
DeepEMD v2	2020	ResNet12	12M	74.29 ± 0.32	86.98 ± 0.60
MERL	2020	ResNet12	12 M	72.14 ± 0.51	87.01 ± 0.35
COSOC	2021	ResNet12	12 M	73.57 ± 0.43	87.57 ± 0.10
CNL	2021	ResNet12	12 M	73.42 ± 0.95	87.72 ± 0.75
invariance-equivariance	2021	ResNet12	12 M	72.21 ± 0.90	87.08 ± 0.58
CSEI	2021	ResNet12	12 M	73.76 ± 0.32	87.83 ± 0.59
ASY (ours)	2022	ResNet12	12 M	74.31 ± 0.22	87.86 ± 0.15
S2M2R	2020	WRN-28-10	36 M	73.71 ± 0.22	88.52 ± 0.14
EASY (ours)	2022	3×ResNet12	36 M	74.71 ± 0.22	88.33 ± 0.14
State-of-the-art after the publication of EASY					
FewTURE	2022	ViT-Small	22 M	72.96 ± 0.92	86.43 ± 0.67
SMKD	2023	ViT-Small	21 M	78.83 ± 0.20	91.21 ± 0.11
MetaFormer-I	2024	ViT-Small	24.5 M	79.05 ± 0.81	90.40 ± 0.53

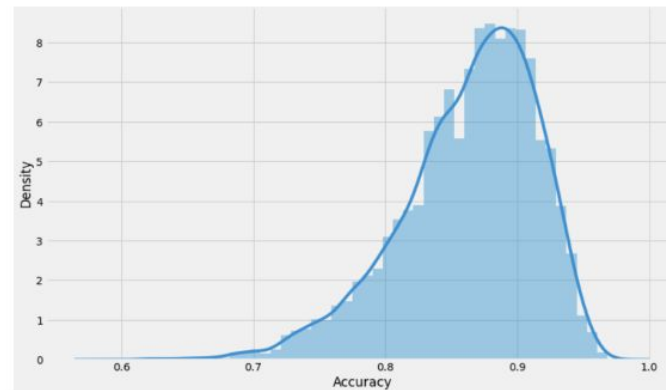
A strong baseline for few shot image classification

Table 3: Average Performance of EASY on 10000 few-shot problems

Dataset	1-shot 5-class	5-shot 5-class
MinilImageNet	70.63 ± 0.20	86.28 ± 0.12
TieredImageNet	74.31 ± 0.22	87.36 ± 0.15

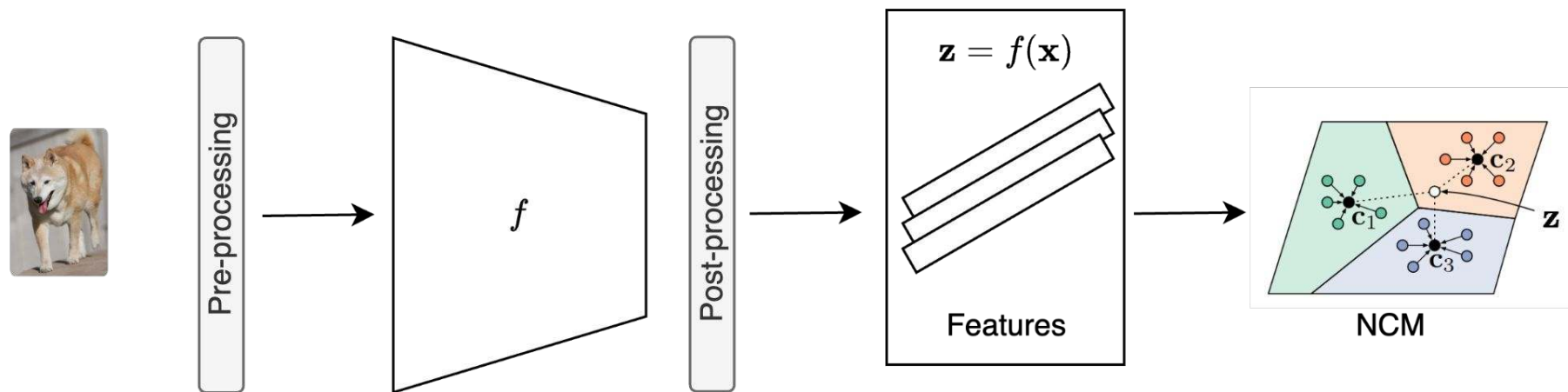


Histogram of accuracies on Mini-ImageNet

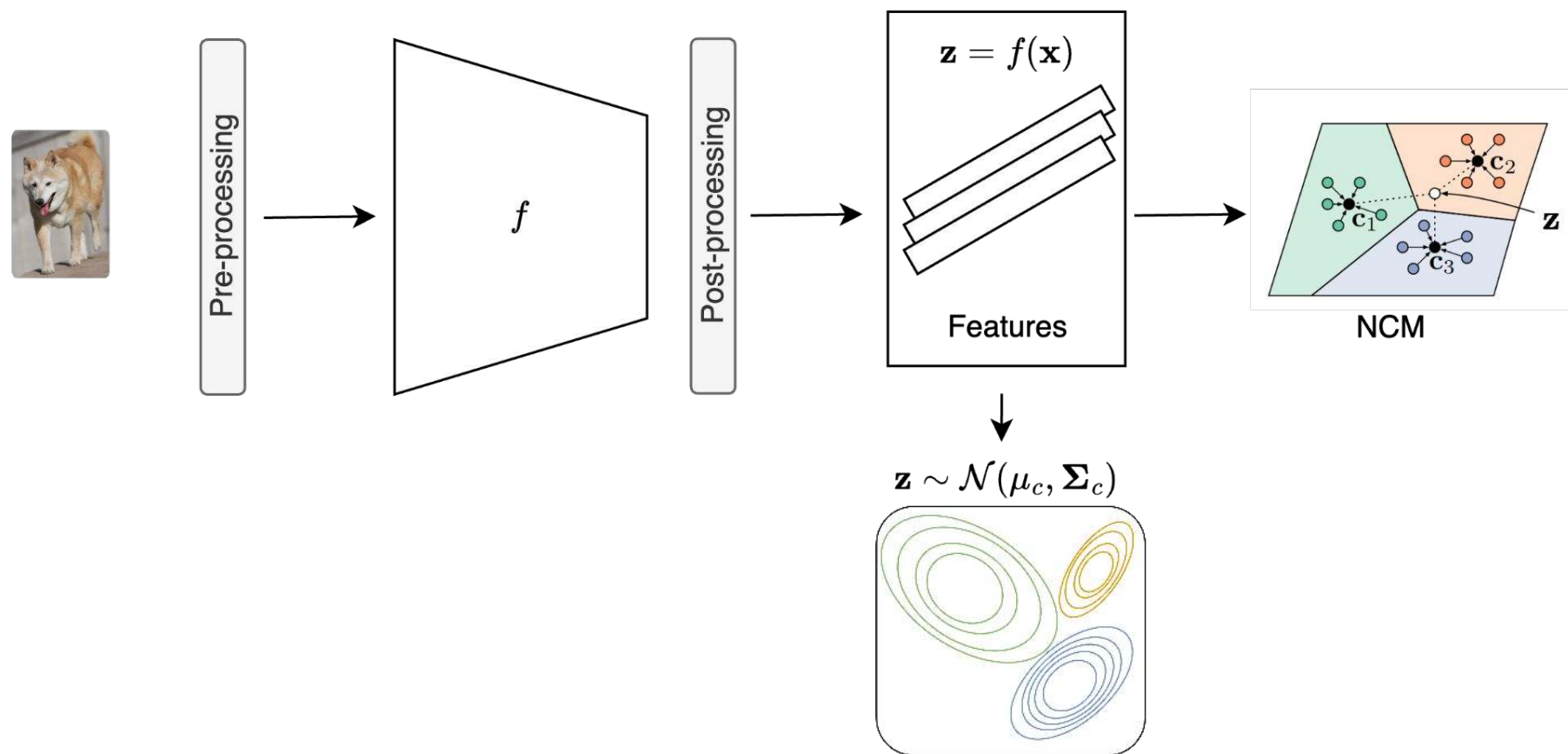


Histogram of accuracies on Tiered-ImageNet

A Statistical Model for Predicting Generalization in Few-Shot Classification



A Statistical Model for Predicting Generalization in Few-Shot Classification

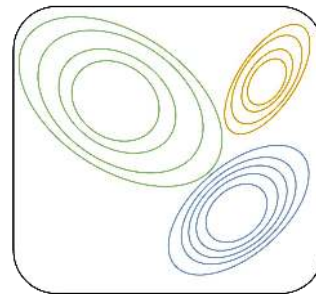


A Statistical Model for Predicting Generalization in Few-Shot Classification

Definition 1

The class-conditional densities are:

$$p(\mathbf{z}|y = c) = \mathcal{N}(\mathbf{z}|\mu_c, \Sigma_c)$$

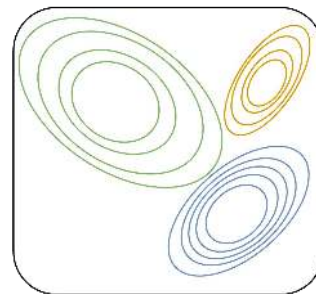


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Definition 1

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$$p(\mathbf{z}|y = c) = \mathcal{N}(\mathbf{z}|\mu_c, \Sigma_c)$$



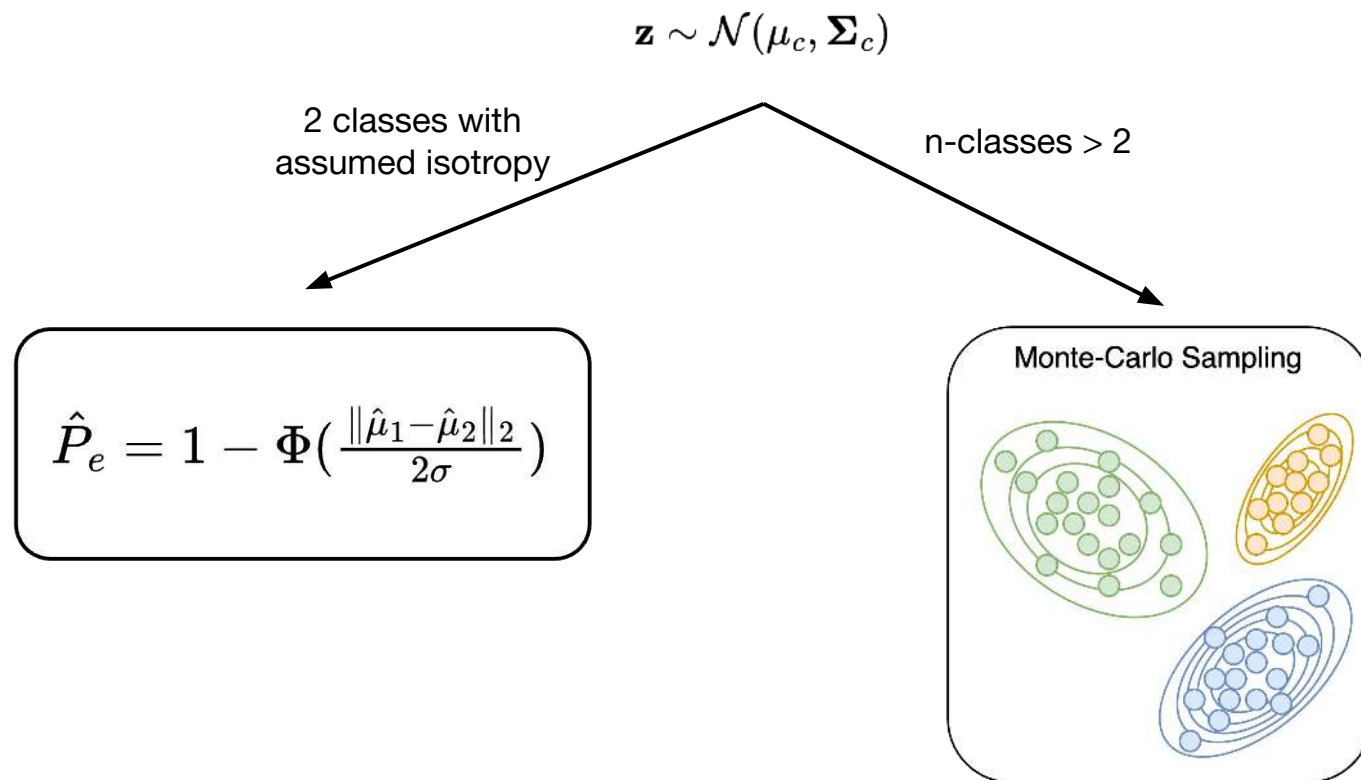
Definition 2

Let $R_c = \{\mathbf{z} \mid C(\mathbf{z}) = c\}$ be the decision region for class c using the classifier C and $R = \bigcup_{c \in \mathcal{Y}} R_c$.

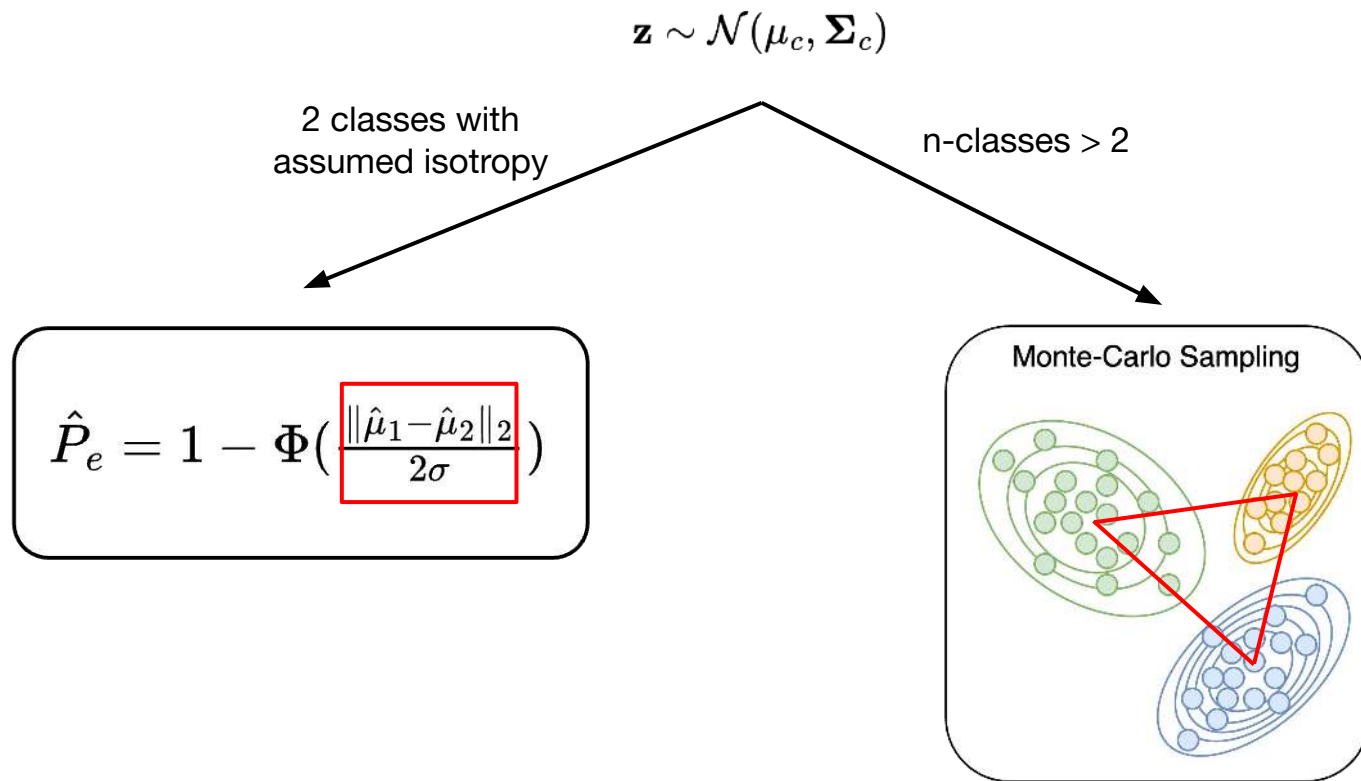
The probability error of a classifier C is defined as:

$$P_e = \sum_{c \in \mathcal{Y}} \int_{R \setminus R_c} p(\mathbf{z} \mid y = c) p(y = c) d\mathbf{z}$$

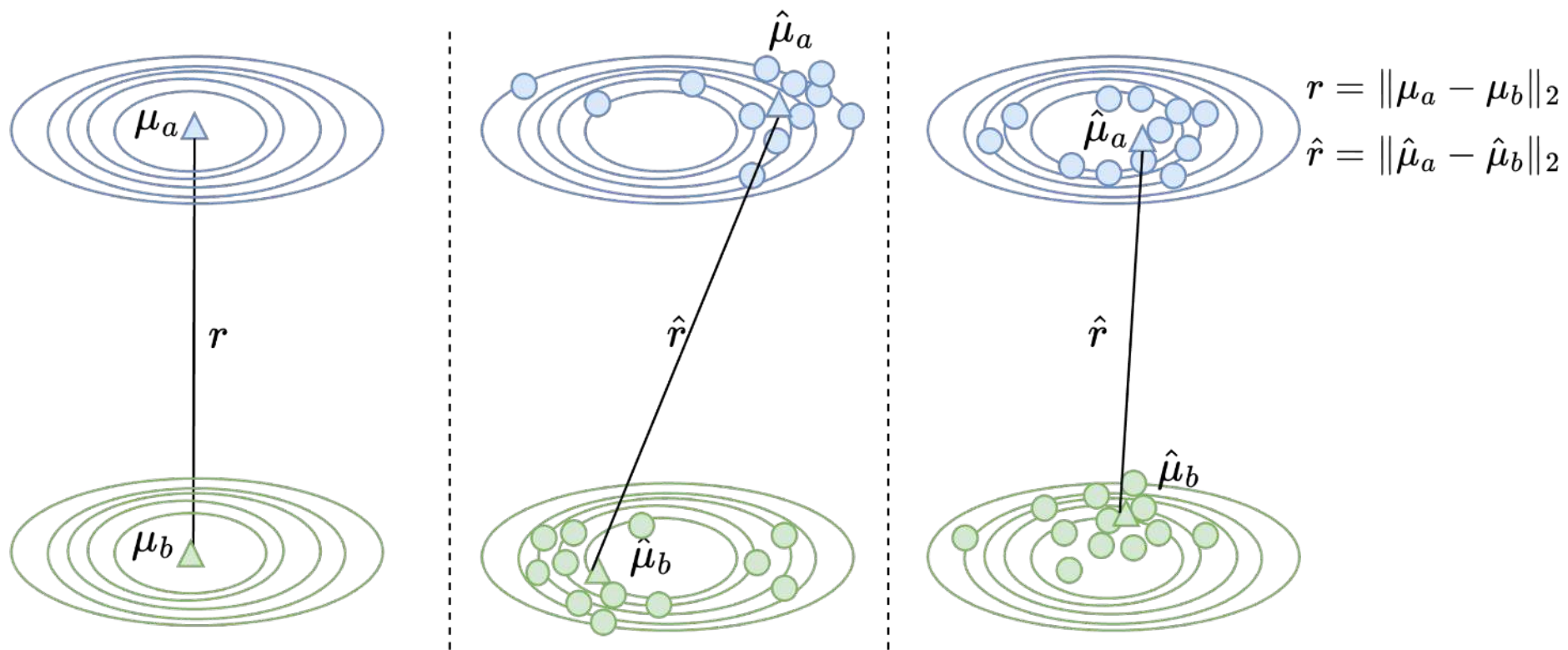
A Statistical Model for Predicting Generalization in Few-Shot Classification



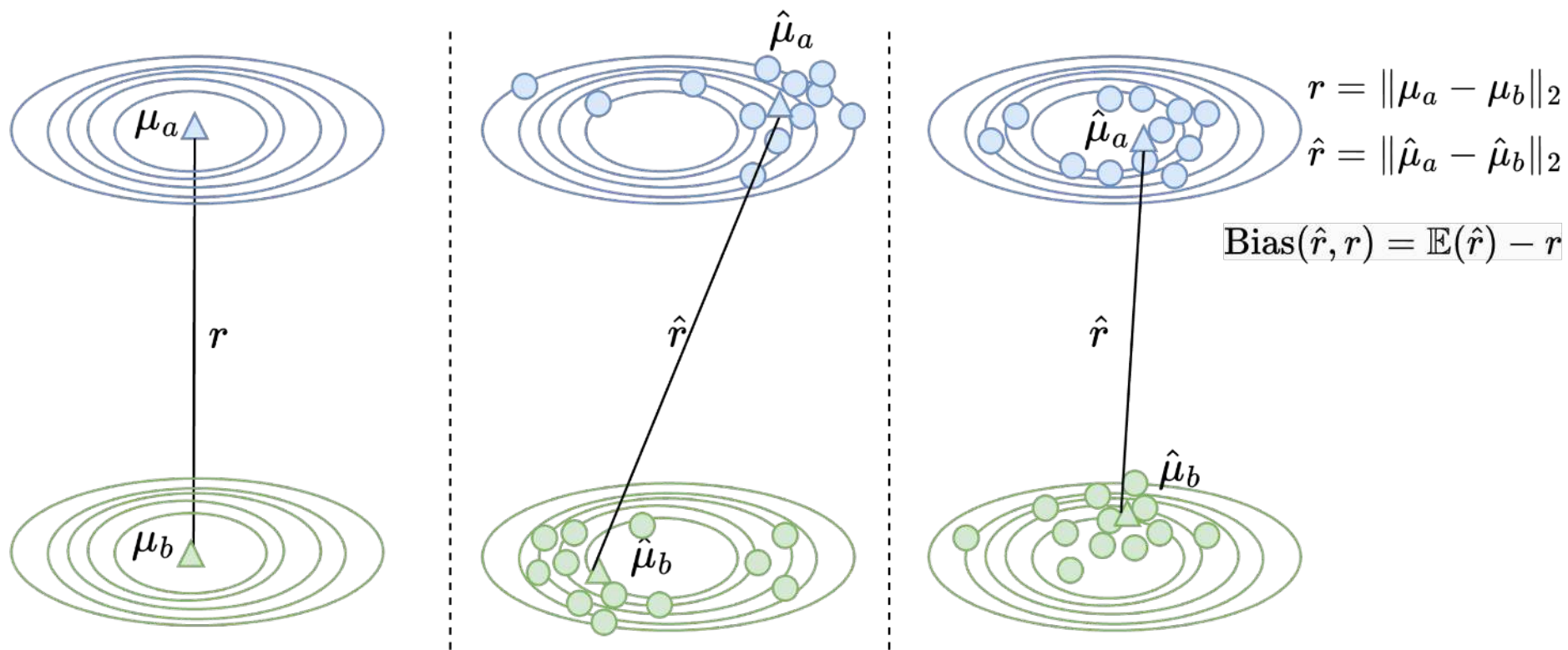
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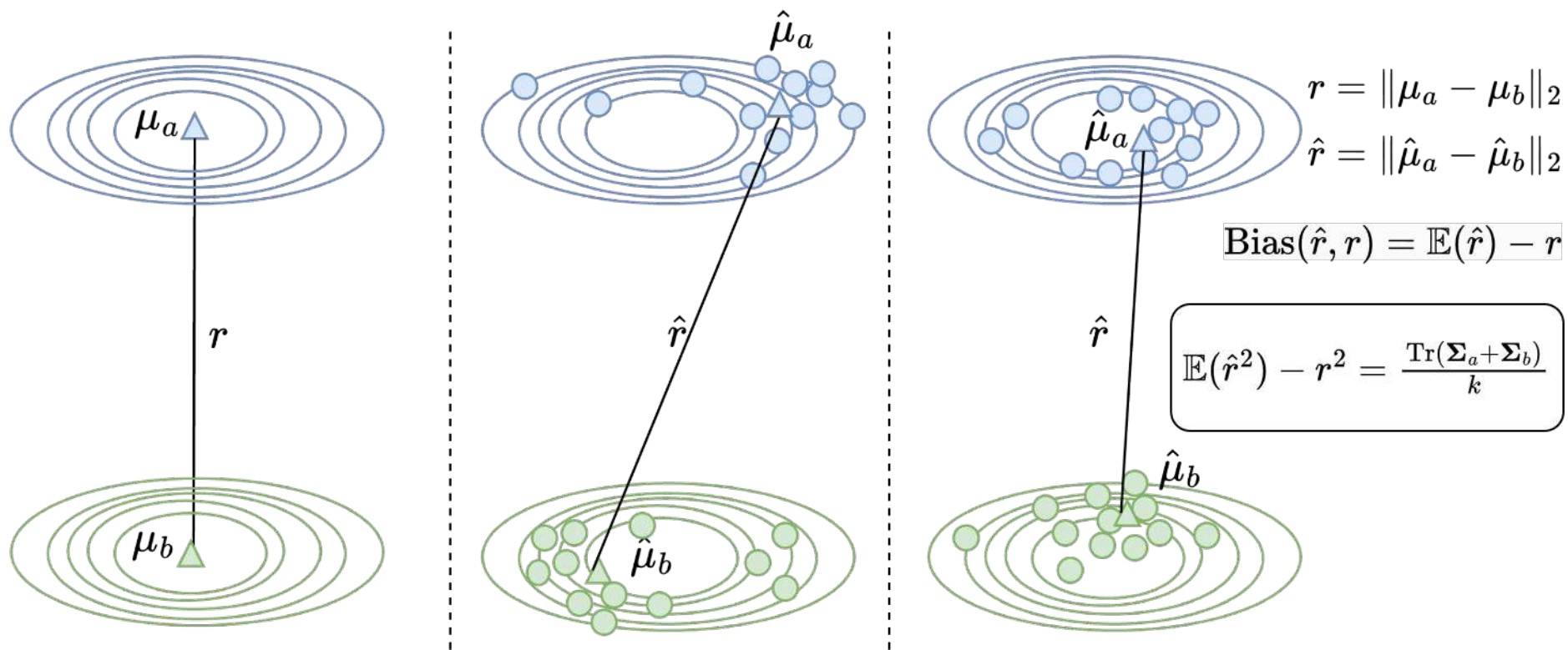
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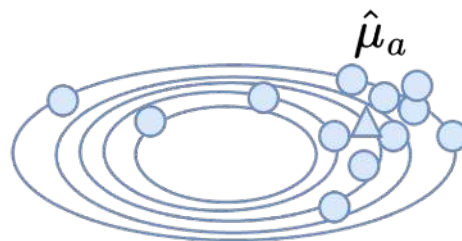
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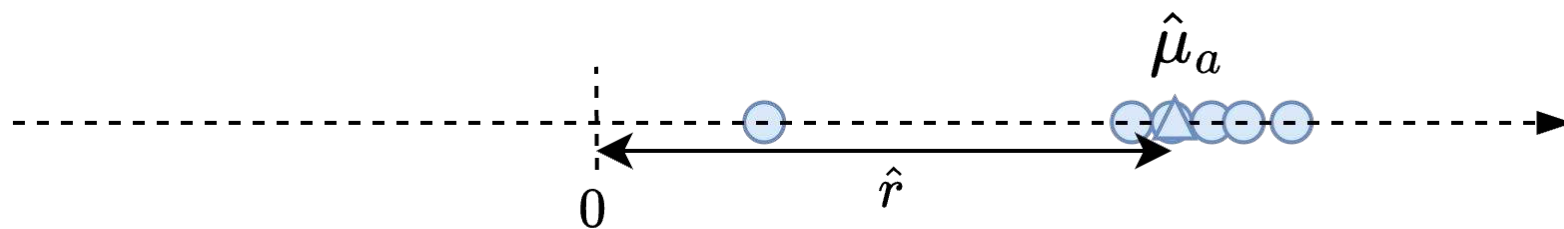
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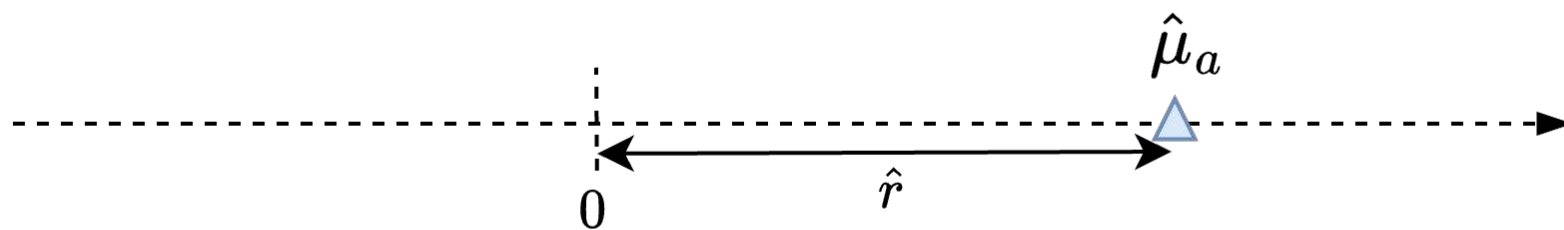
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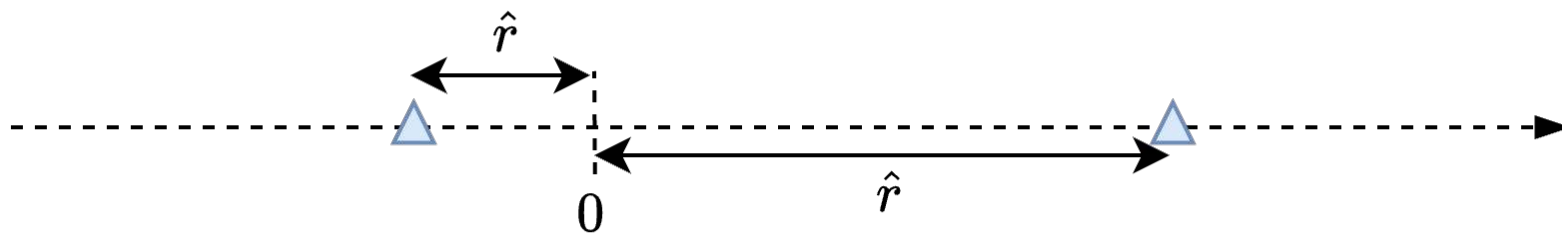
A Statistical Model for Predicting Generalization in Few-Shot Classification



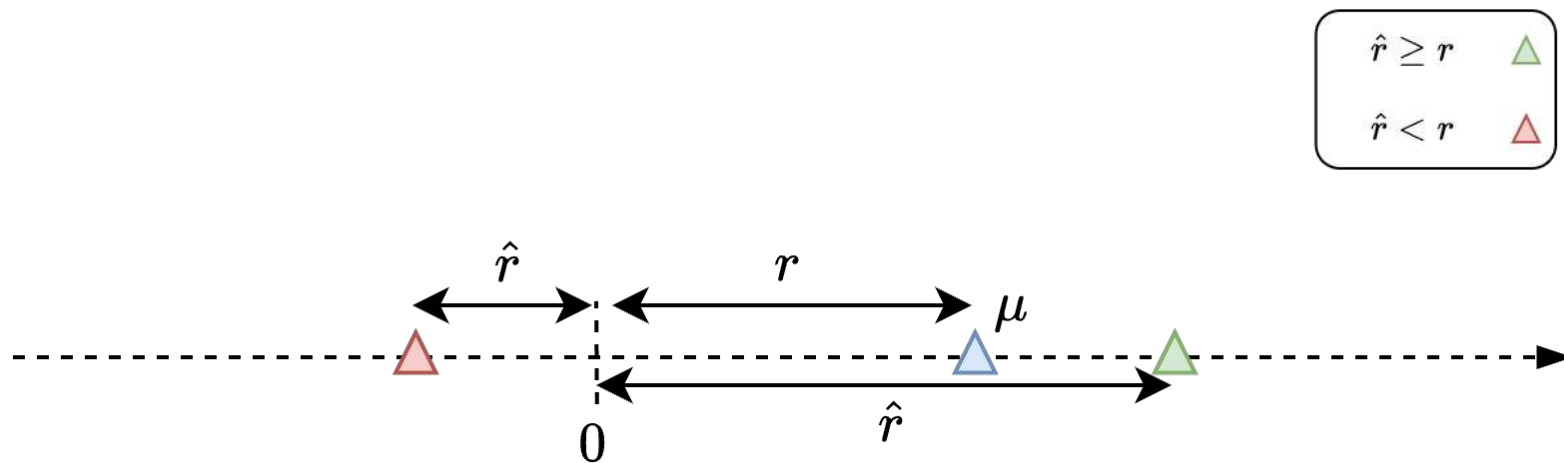
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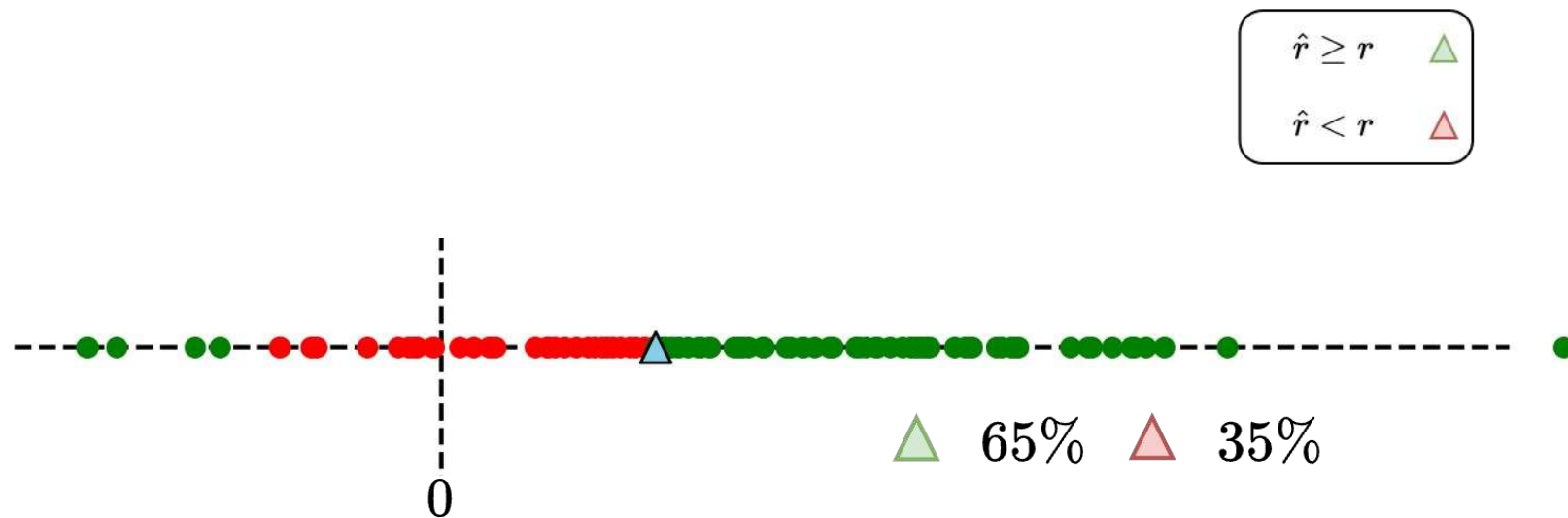
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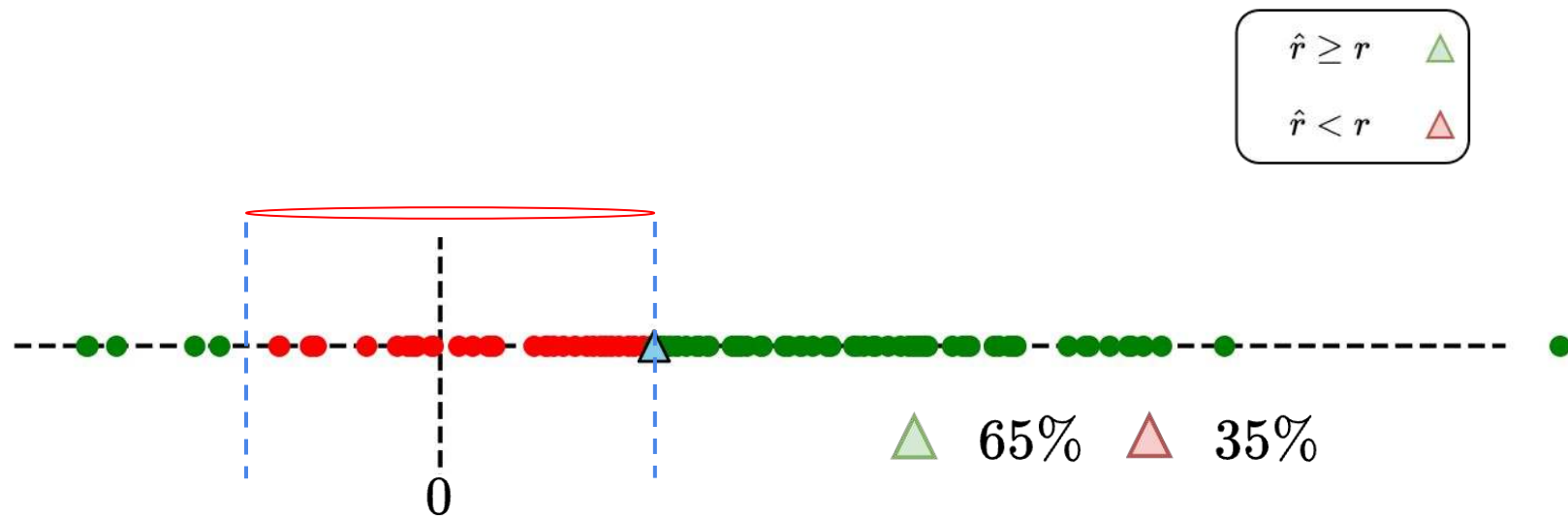
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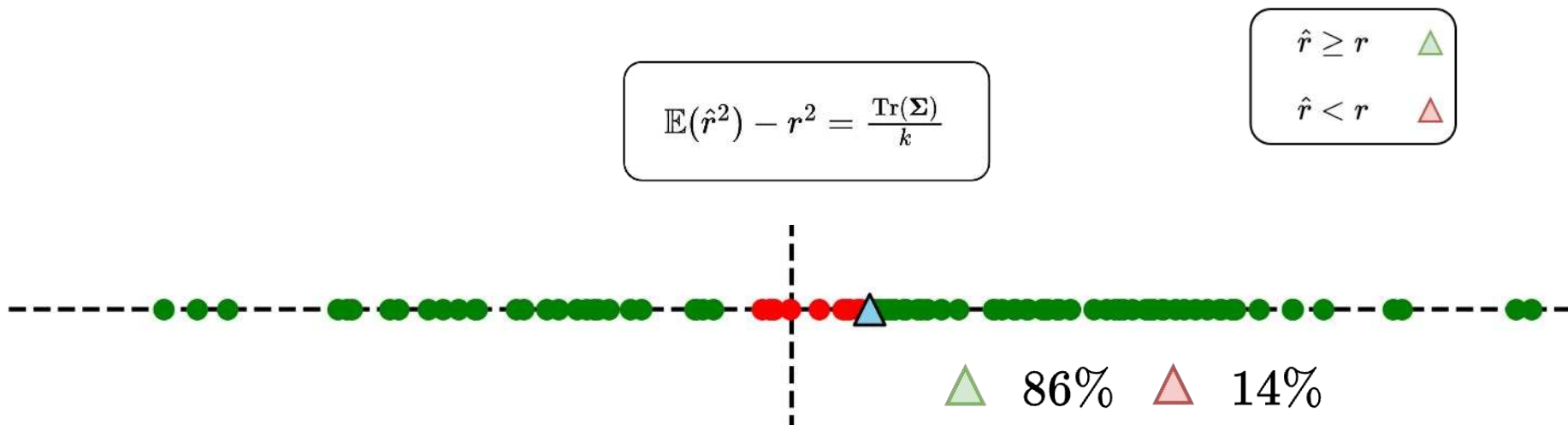
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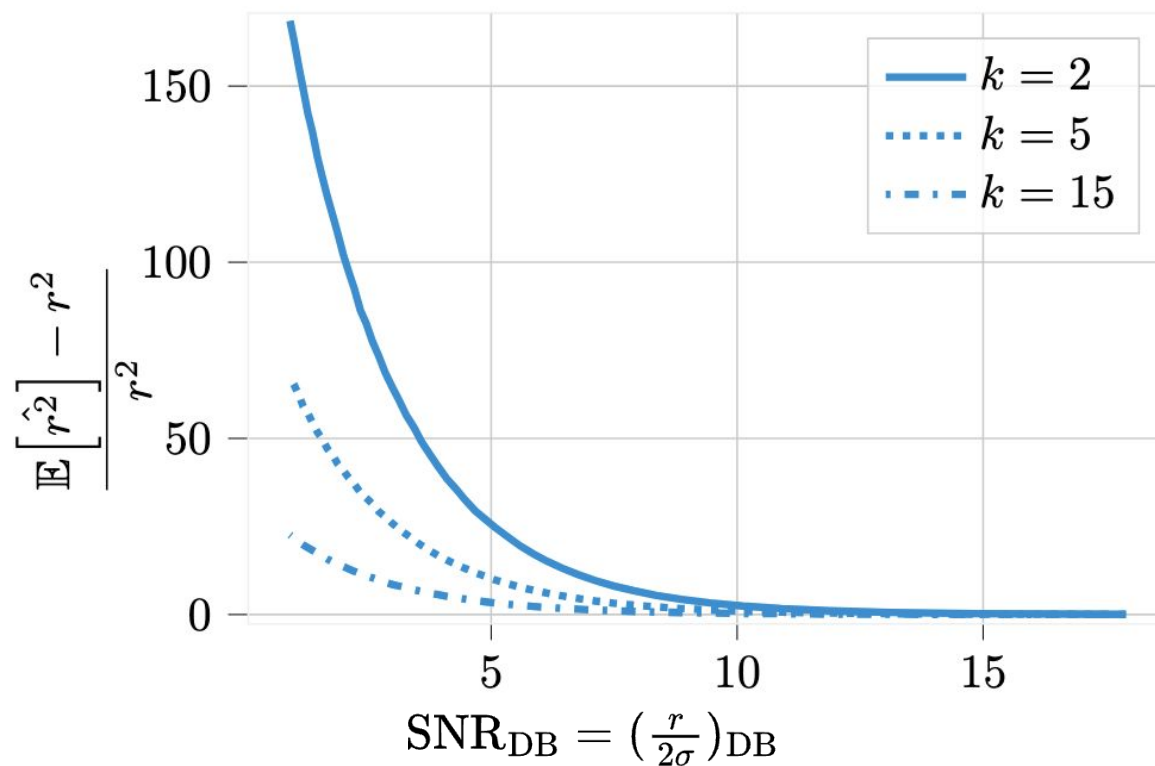
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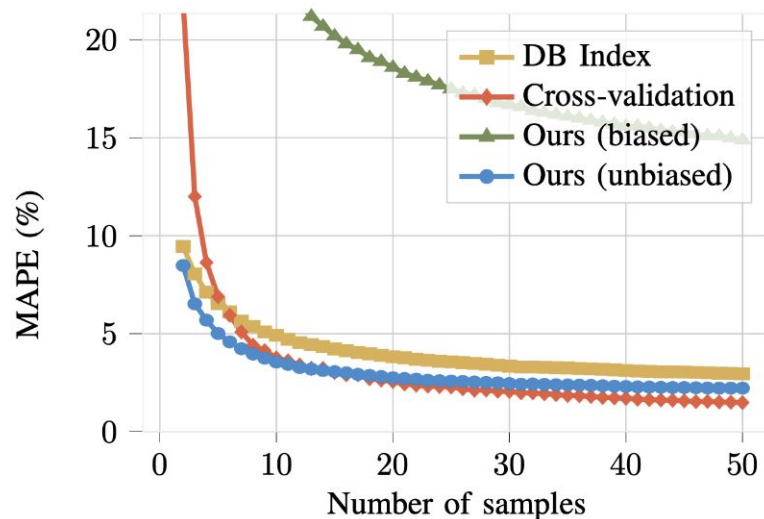
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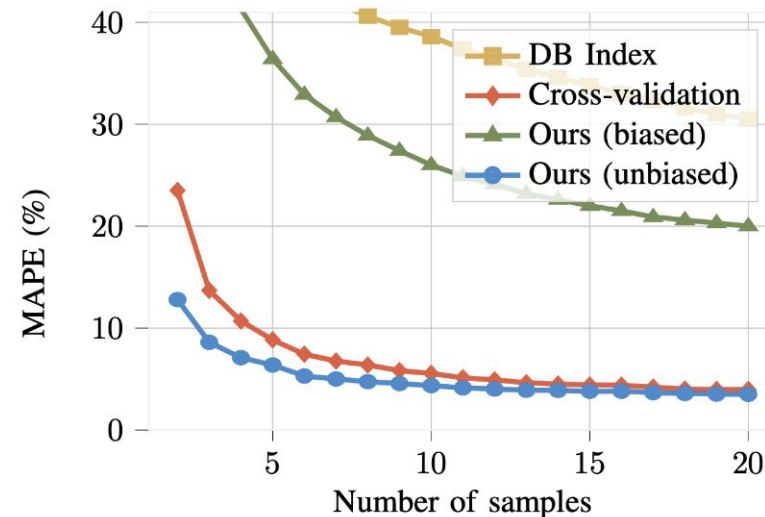
Bias with respect to Signal-to-Noise Ratio

$$\mathbb{E}(\hat{r}^2) - r^2 = \frac{\text{Tr}(\Sigma_a + \Sigma_b)}{k}$$

A Statistical Model for Predicting Generalization in Few-Shot Classification



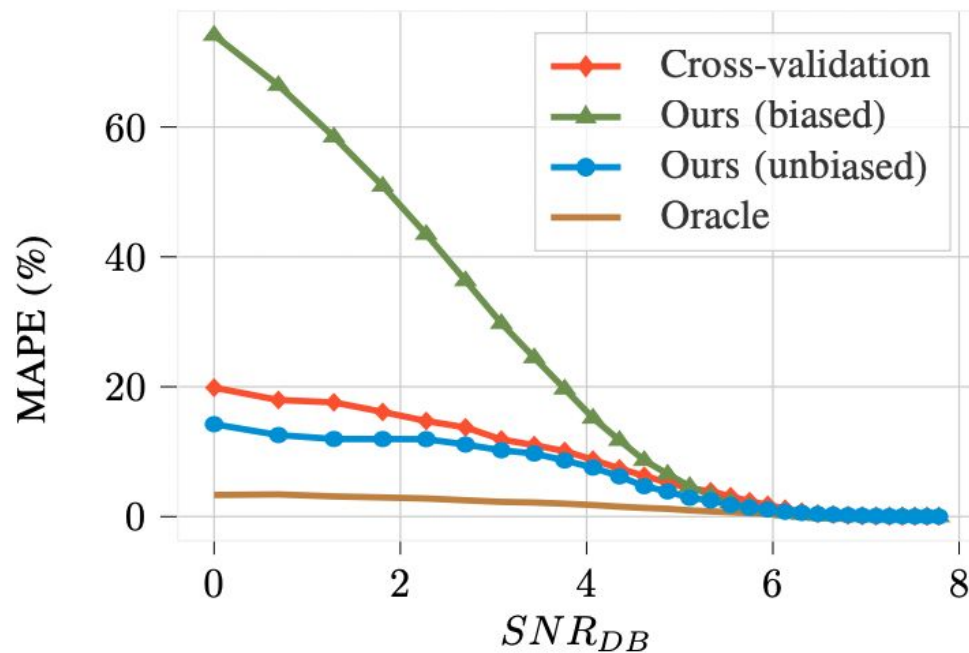
(a) Tiered-ImageNet



(b) VGG-Flower

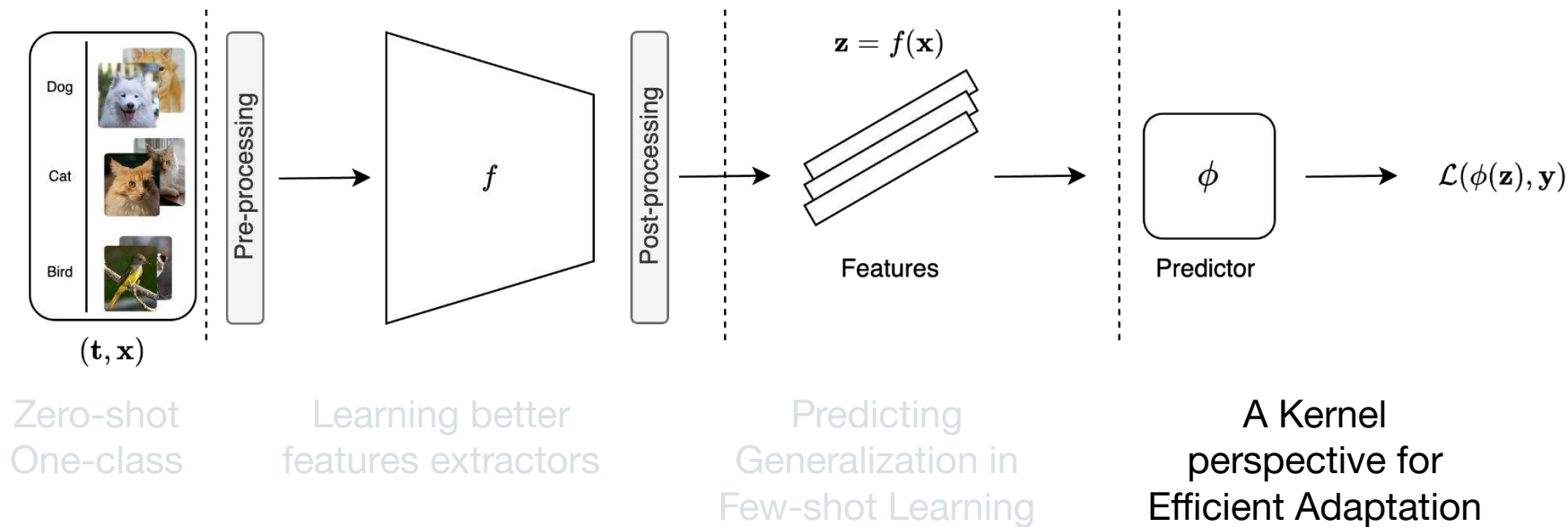
*Mean-Absolute-Percentage Error using 1k 5-class few-shot problems.
Figure (a) is in-domain and Figure (b) is cross-domain.*

A Statistical Model for Predicting Generalization in Few-Shot Classification

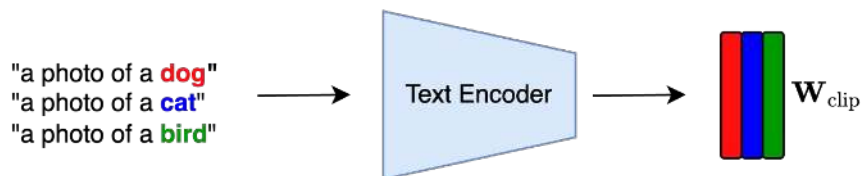


Prediction error on 10k binary 10-shot problems generated from isotropic Gaussian distributions with different SNR values. The oracle uses our approach with the true parameter distributions

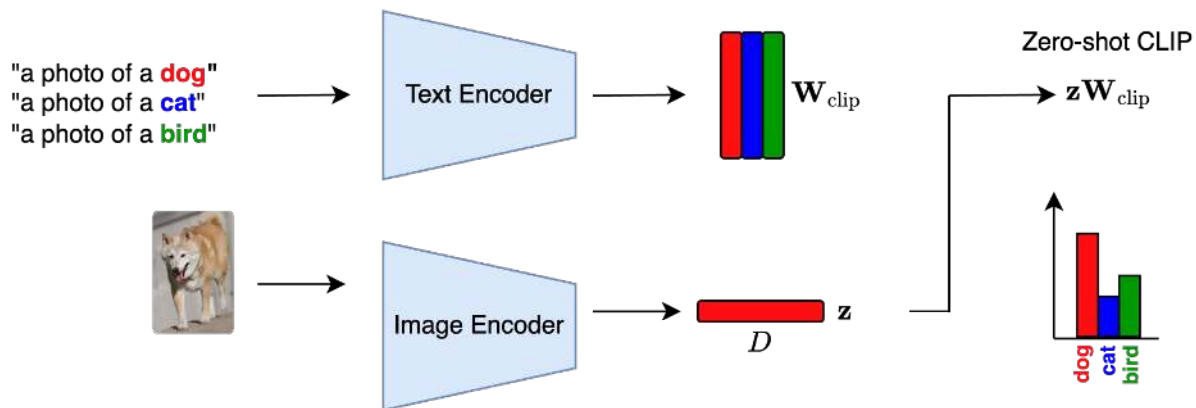
A Kernel Perspective on Training-Free Few-Shot Adaptation of Large Vision-Language Models



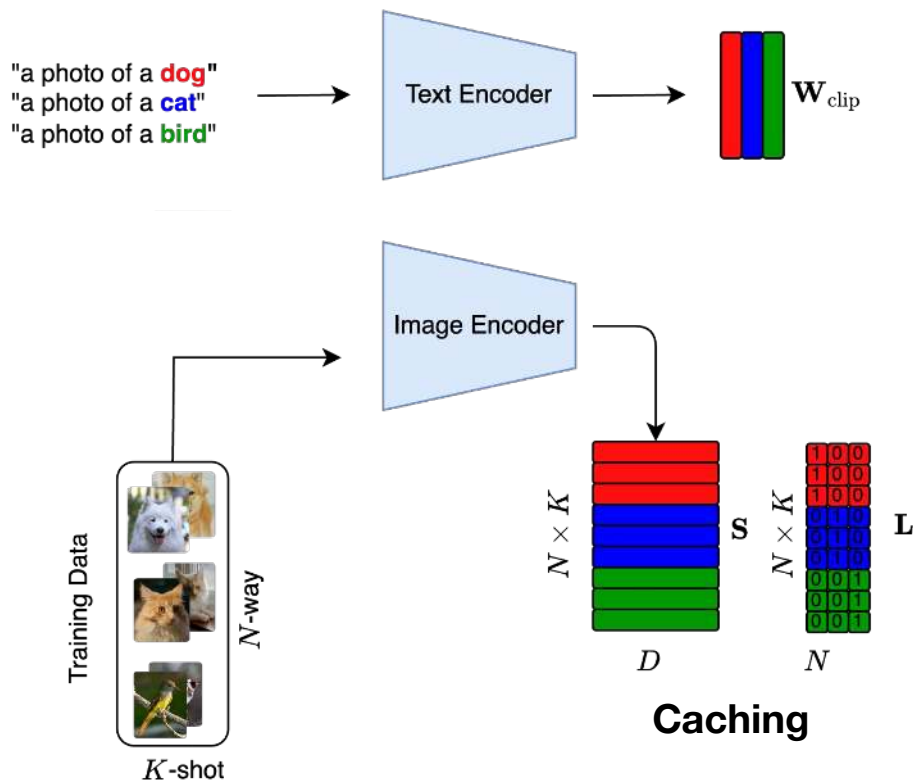
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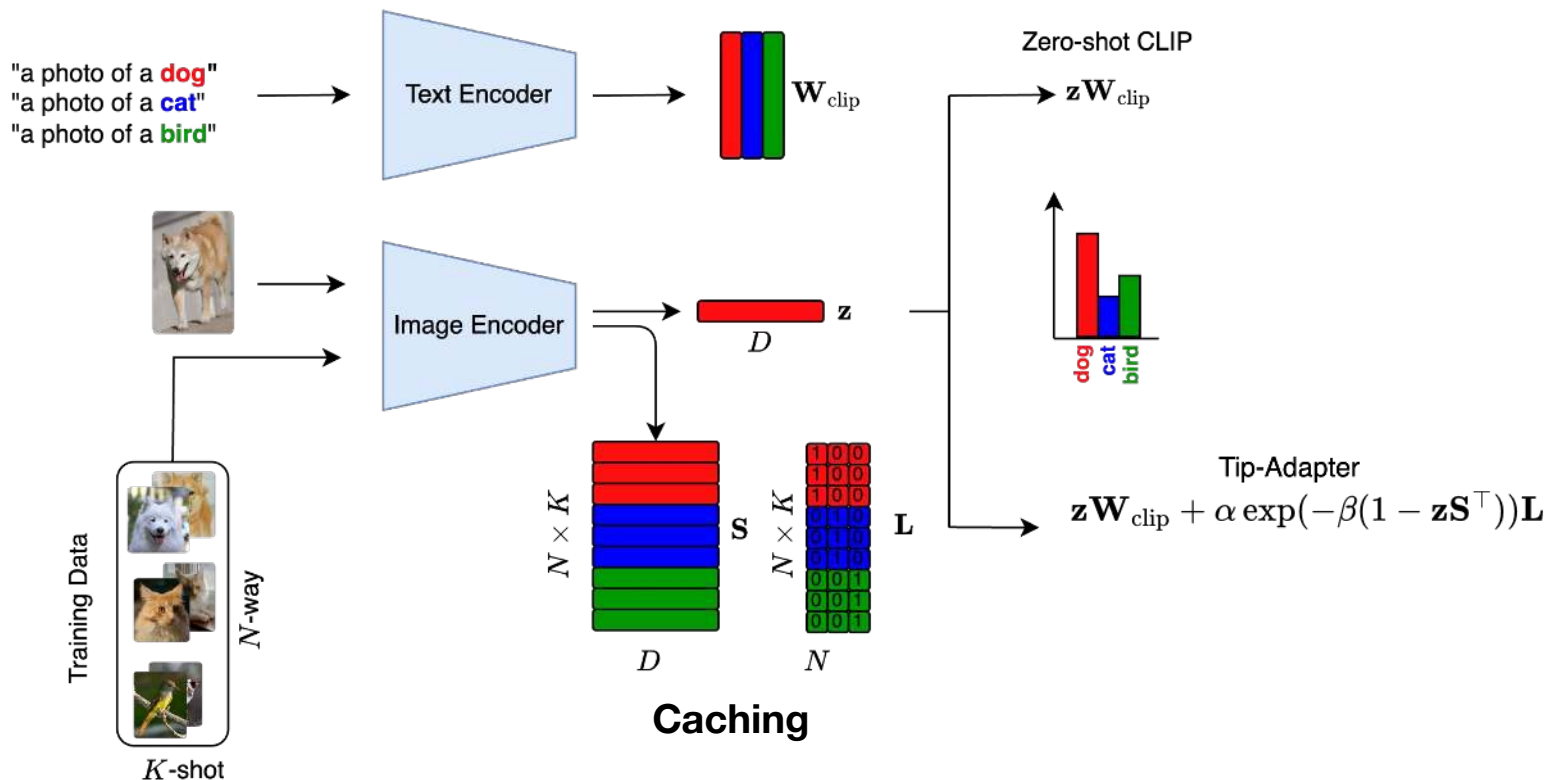
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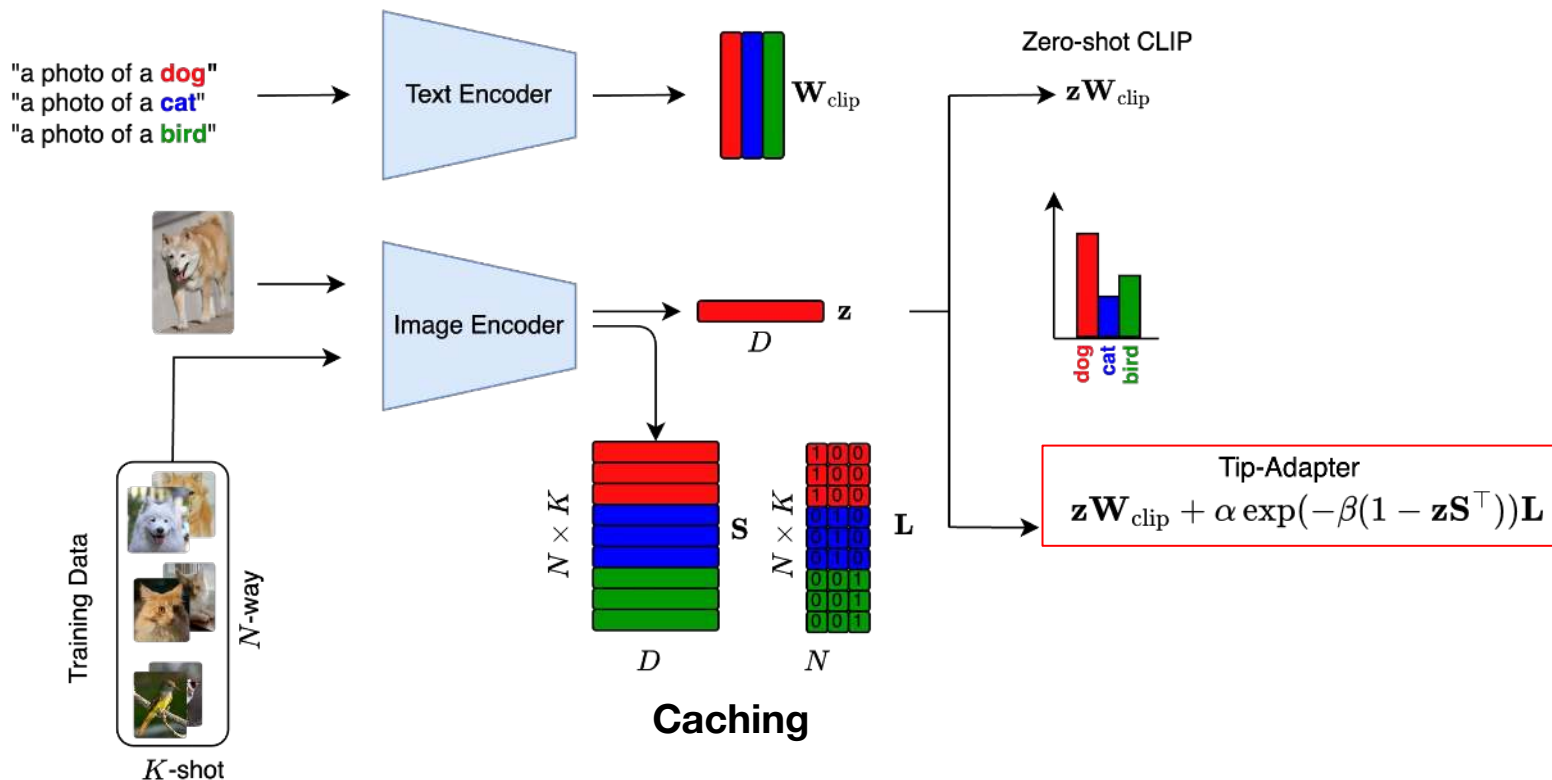
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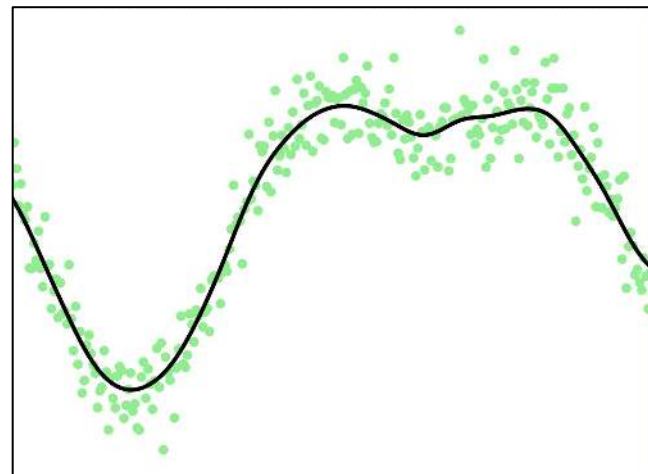
$$\phi_{\text{Tip}}(\mathbf{z}) = f_{\text{clip}}(\mathbf{z}) + \alpha \exp(-\beta (1 - \mathbf{z}\mathbf{S}^\top))\mathbf{L}$$

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$$\phi_{\text{Tip}}(\mathbf{z}) = f_{\text{clip}}(\mathbf{z}) + \alpha \sum_{i=1}^{NK} \exp\left(-\frac{\beta}{2} \|\mathbf{S}_i - \mathbf{z}\|_2^2\right) \mathbf{L}_i$$

$$\phi_{\text{Tip}}(\mathbf{z}) = f_{\text{clip}}(\mathbf{z}) + \alpha \exp(-\beta(1 - \mathbf{z}\mathbf{S}^\top))\mathbf{L}$$

$$\phi_{\text{Tip}}(\mathbf{z}) = f_{\text{clip}}(\mathbf{z}) + \alpha \sum_{i=1}^{NK} \exp\left(-\frac{\beta}{2}\|\mathbf{S}_i - \mathbf{z}\|_2^2\right)\mathbf{L}_i$$

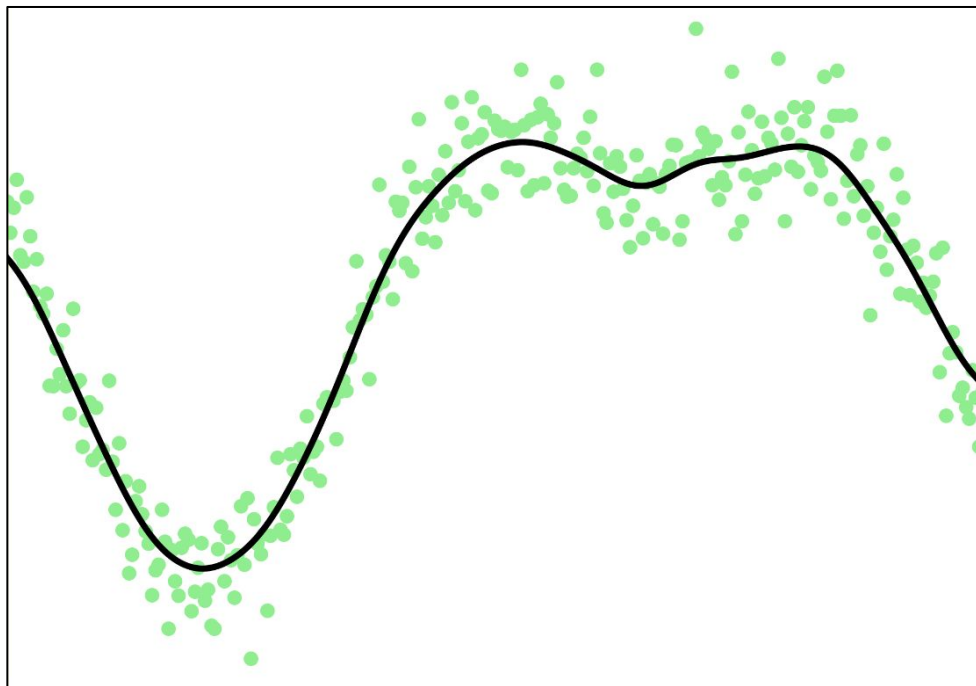


$$\phi(\mathbf{z}) = \frac{\sum_{i=1}^{NK} k_{\beta}(d(\mathbf{z}, \mathbf{S}_i))\mathbf{L}_i}{\sum_{i=1}^{NK} k_{\beta}(d(\mathbf{z}, \mathbf{S}_i))}$$

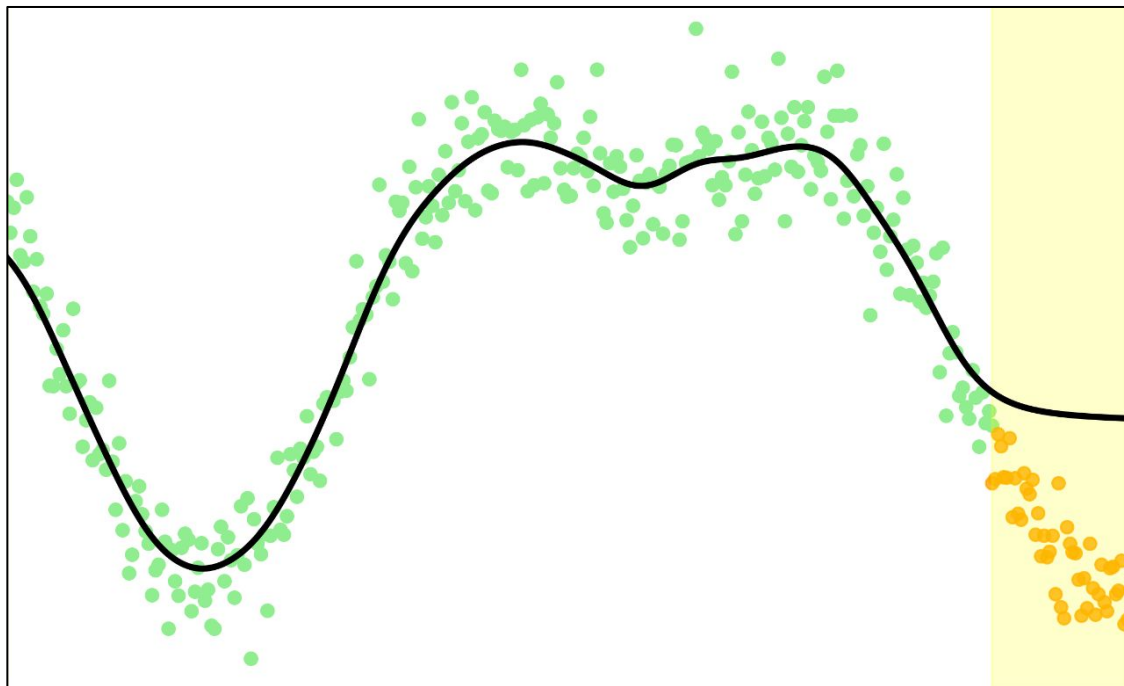
where $k_{\beta}(u) = \left(\frac{\beta}{2}\right)^D \exp(-\frac{\beta}{2}u)$

Nadaraya-Watson Estimator

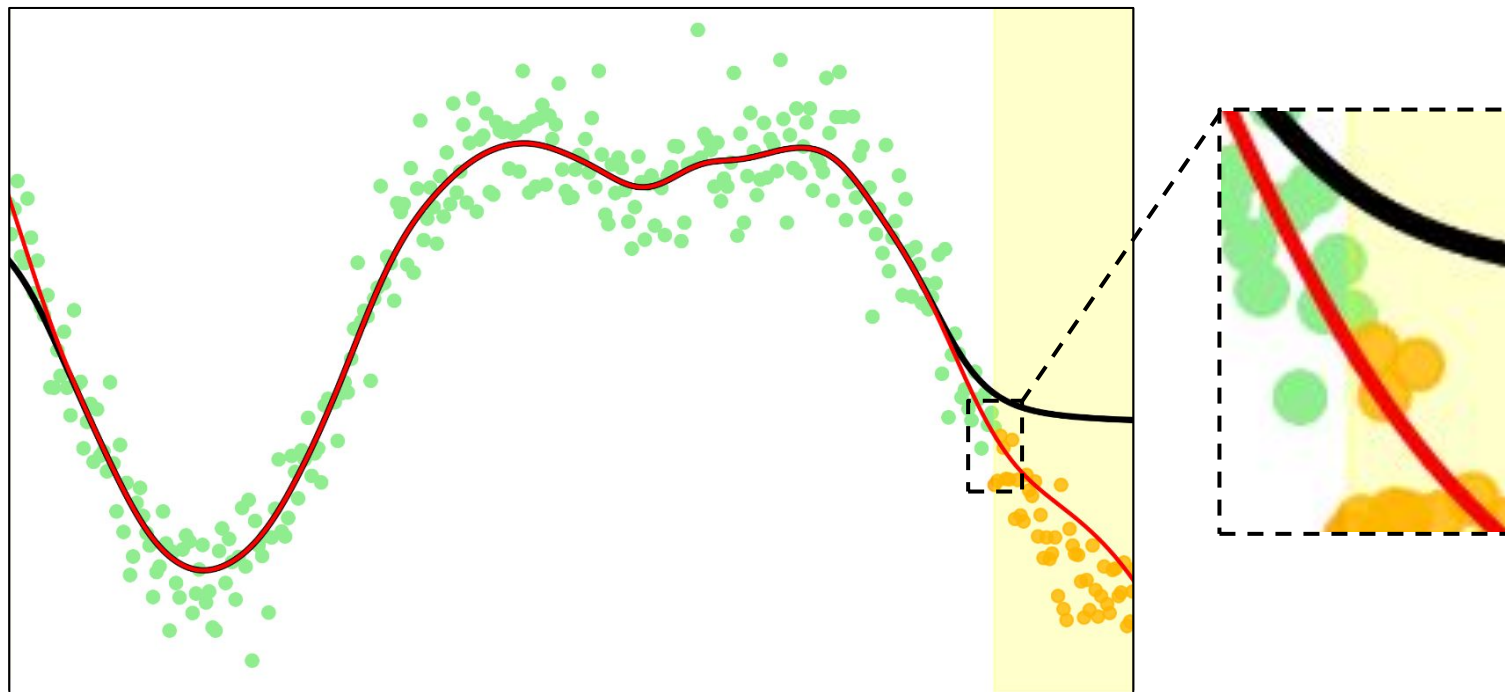
A Kernel Perspective on Training-Free Few-Shot Adaptation of Large Vision-Language Models



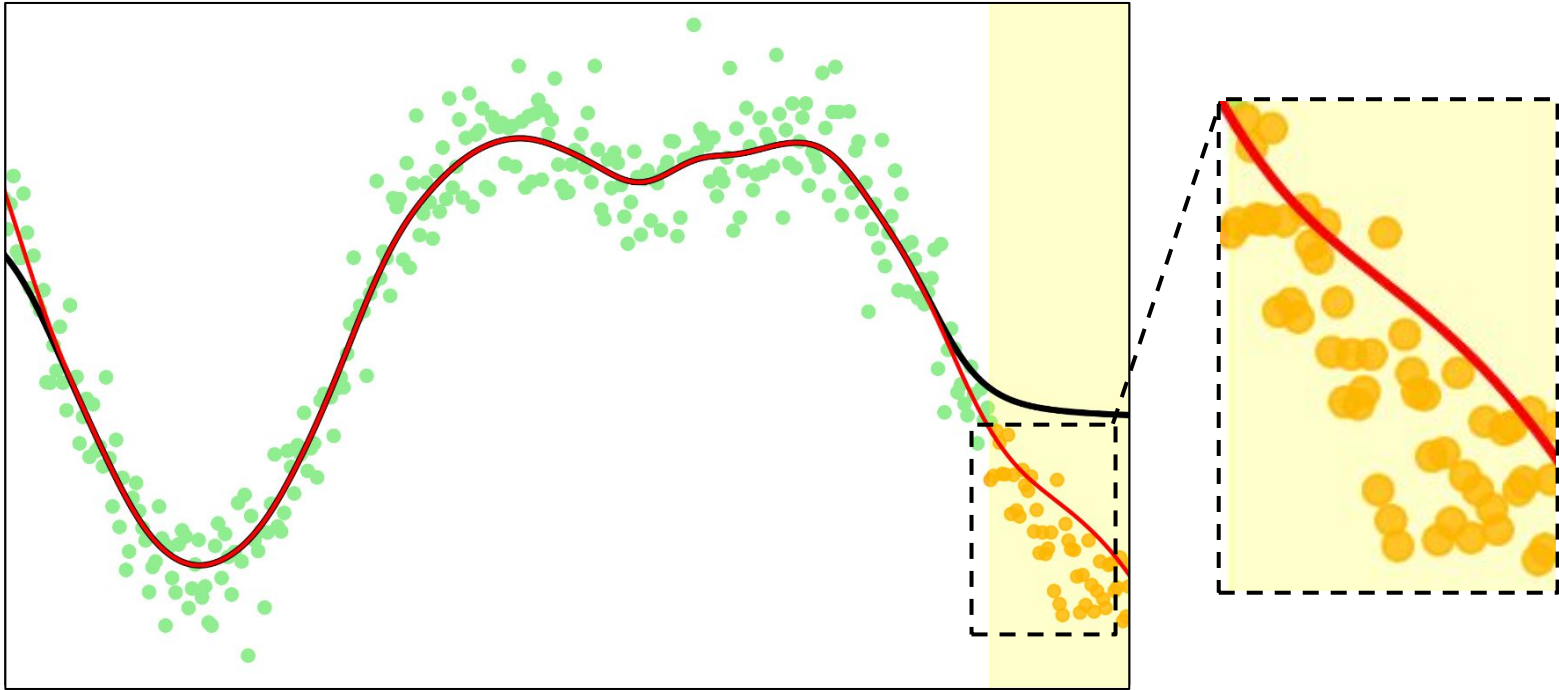
A Kernel Perspective on Training-Free Few-Shot Adaptation of Large Vision-Language Models



A Kernel Perspective on Training-Free Few-Shot Adaptation of Large Vision-Language Models



A Kernel Perspective on Training-Free Few-Shot Adaptation of Large Vision-Language Models



1. $\phi \in \mathcal{H}$ Reproducing Kernel Hilbert Space

1. $\phi \in \mathcal{H}$ Reproducing Kernel Hilbert Space

$$2. \min_{\phi \in \mathcal{H}} \sum_{i=1}^{NK} \|\phi(\mathbf{S}_i) - \mathbf{L}_i\|_2^2 + \underbrace{\lambda \|\phi - f_{\text{clip}}\|_{\mathcal{H}}^2}_{\text{Proximal Regularization}}$$

ProKeR: **P**roximal **K**ernel ridge **R**egression

$$\phi(\mathbf{z}) = f_{\text{clip}}(\mathbf{z}) + \sum_{i=1}^{NK} k(\mathbf{S}_i, \mathbf{z}) \gamma_i$$

where $\gamma_i = (\mathbf{I} + \frac{1}{\lambda} k_{\beta}(\mathbf{S}, \mathbf{S}))^{-1} (\mathbf{L}_i - f_{\text{clip}}(\mathbf{S}_i))$

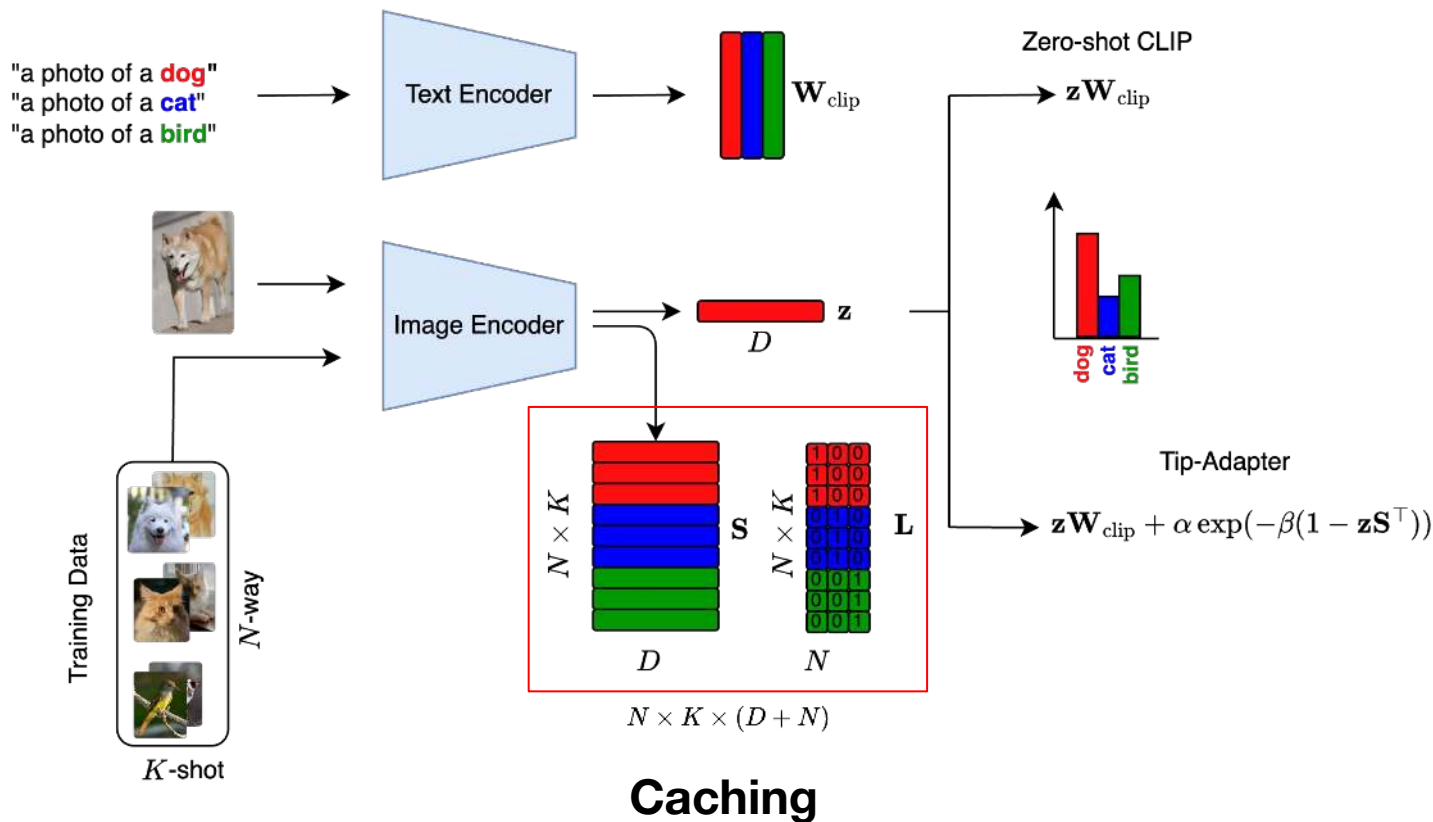
Performance on 11 classification datasets (CoOp's Benchmark).

Shots	1	2	4	8	16	Average
Tip-Adapter	62.83	64.84	66.50	68.64	70.02	66.56
APE	64.43	66.48	68.66	70.75	72.81	68.62
GDA	62.19	66.19	69.77	73.30	76.04	69.49
ProKeR (Polynomial) (ours)	63.65	66.65	69.61	73.29	76.61	69.96
ProKeR (RBF) (ours)	64.01	67.31	70.42	73.85	76.75	70.46

Performance on 11 classification datasets. Hyperparameters are transferred from ImageNet.

Shots	1	2	4	8	16	Average
Tip-Adapter	58.86	60.33	61.49	63.15	64.61	61.68
APE	60.09	62.33	65.36	67.95	69.89	65.12
GDA	57.49	63.43	66.68	72.46	75.12	67.03
ProKeR (Polynomial) (ours)	62.39	65.59	68.05	71.81	75.82	68.72
ProKeR (RBF) (ours)	63.13	66.31	68.64	72.15	75.12	69.07

A Kernel Perspective on Training-Free Few-Shot Adaptation of Large Vision-Language Models



Random Fourier Features (RFF) [1]

$$k(\mathbf{z}, \mathbf{z}') \approx \psi(\mathbf{z})\psi(\mathbf{z}') \text{ where } \psi(\mathbf{z}) = \exp(i\mathbf{w}\mathbf{z})$$

[1] Rahimi A, Recht B. Random features for large-scale kernel machines. Advances in neural information processing systems 2007.

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$$\phi(\mathbf{z}) = f_{\text{clip}}(\mathbf{z}) + \sum_{i=1}^{NK} \psi(\mathbf{z})\psi(\mathbf{S}_i)\gamma_i$$

[1] Rahimi A, Recht B. Random features for large-scale kernel machines. Advances in neural information processing systems 2007.

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$$\phi(\mathbf{z}) = f_{\text{clip}}(\mathbf{z}) + \sum_{i=1}^{NK} \psi(\mathbf{z})\psi(\mathbf{S}_i)\gamma_i$$

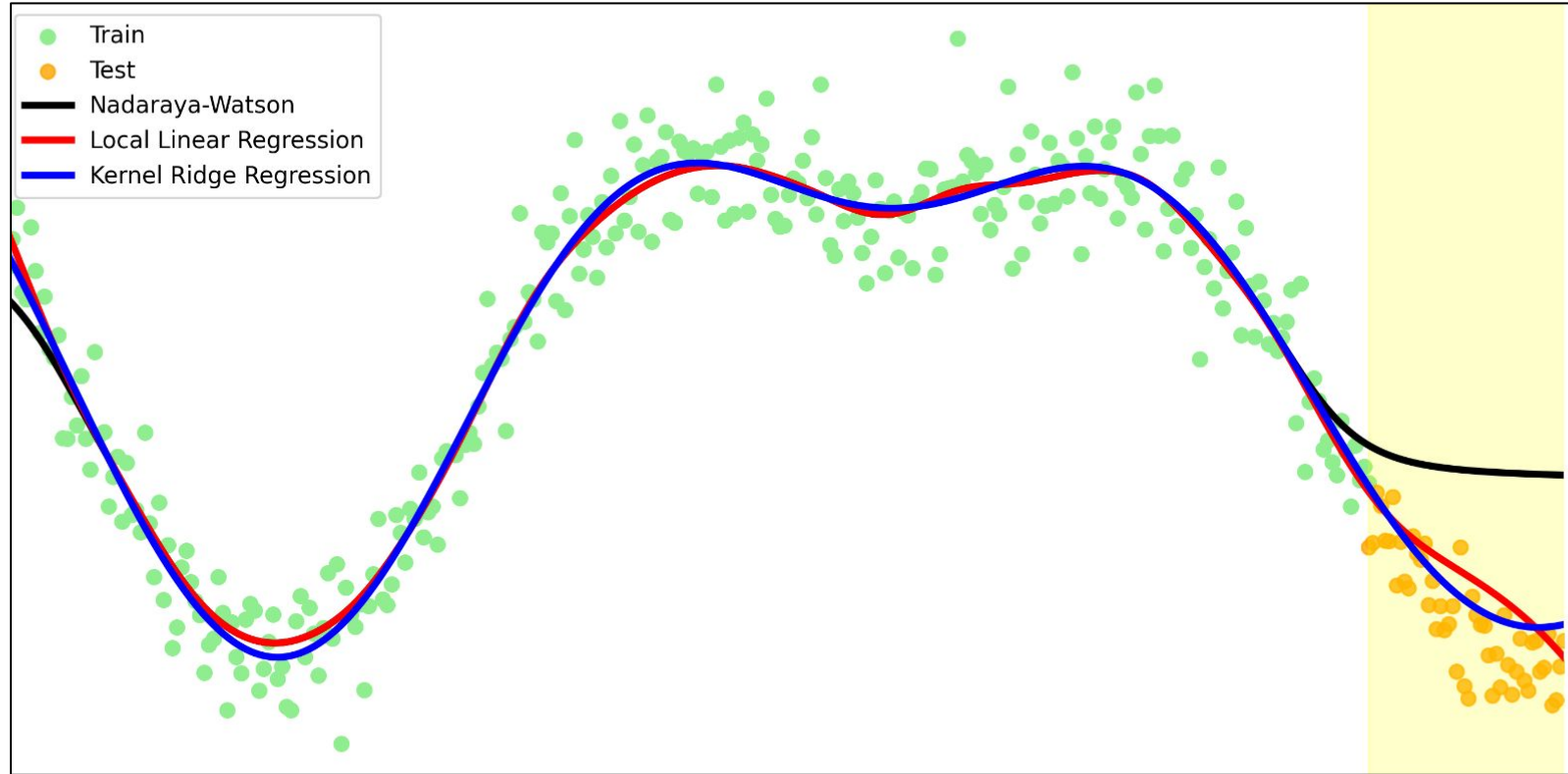
$$\phi(\mathbf{z}) = f_{\text{clip}}(\mathbf{z}) + \psi(\mathbf{z})[\Psi_1, \dots, \Psi_N]$$

[1] Rahimi A, Recht B. Random features for large-scale kernel machines. Advances in neural information processing systems 2007.

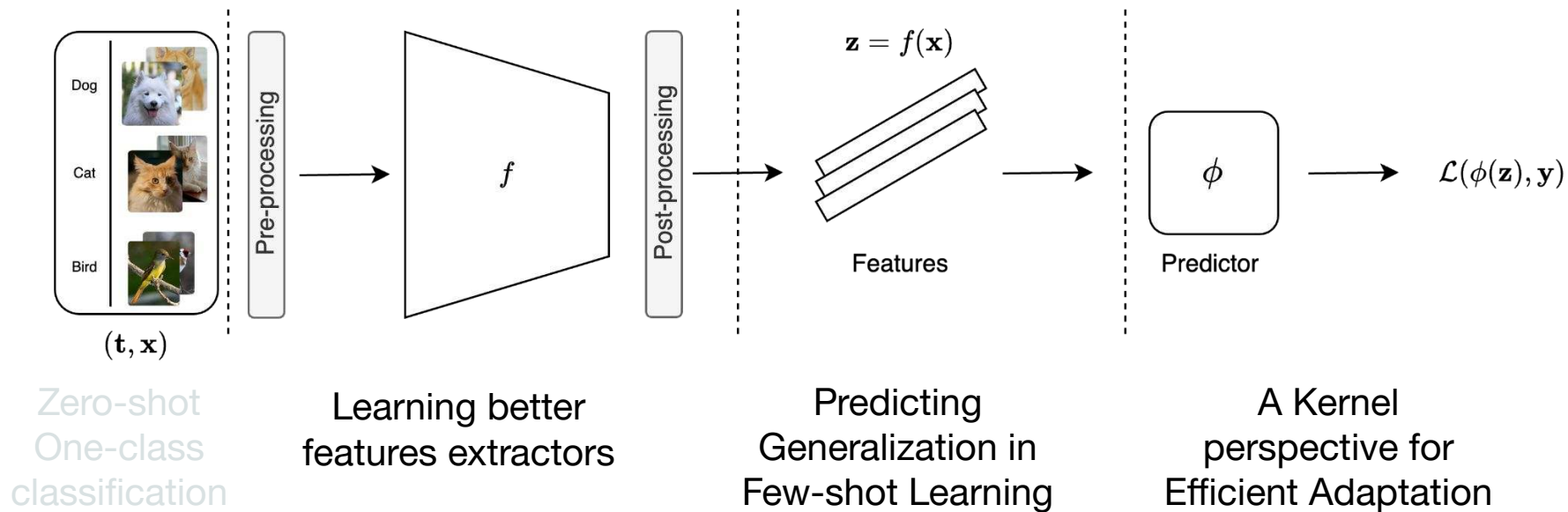
Performance on 11 classification datasets (CoOp's benchmark)

Shots	1	2	4	8	16	Average
GDA	62.19	66.19	69.77	73.30	76.04	69.49
Tip-Adapter	62.83	64.84	66.50	68.64	70.02	66.56
Tip-Adapter with RFF	62.77	64.58	66.29	68.15	69.60	66.27
APE	64.43	66.48	68.66	70.75	72.81	68.62
APE with RFF	64.16	66.11	68.76	70.37	72.02	68.28
ProKeR (ours)	64.01	67.31	70.42	73.85	76.75	70.46
ProKeR with RFF (ours)	64.01	67.12	70.28	73.61	76.44	70.29

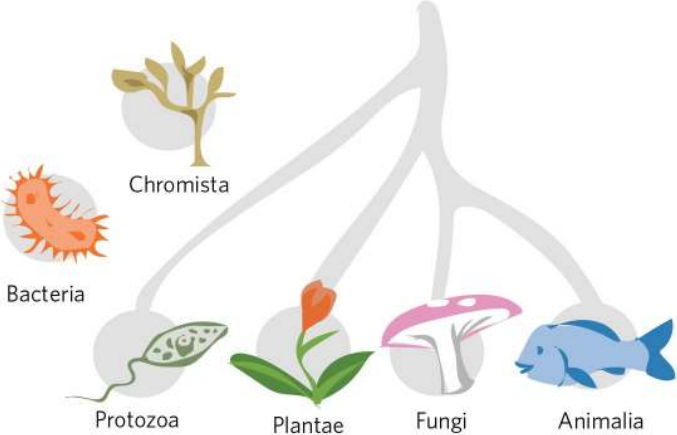
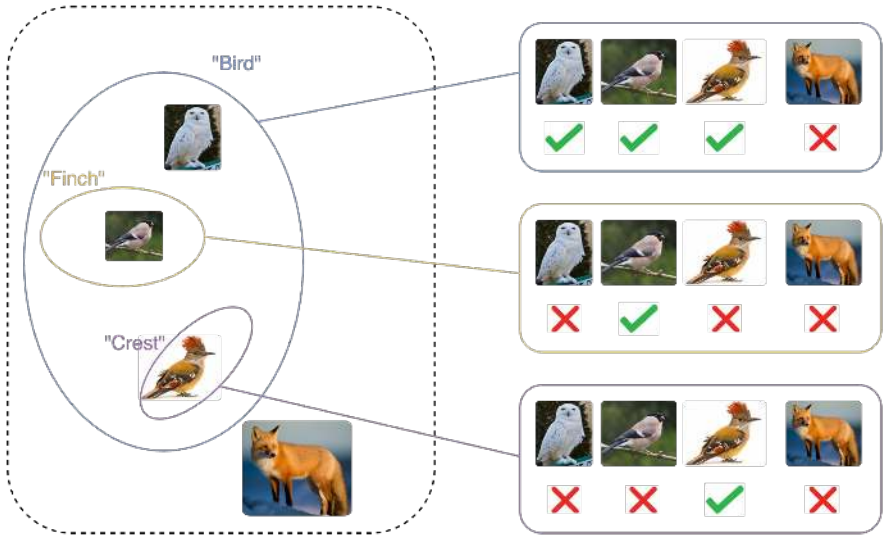
A Kernel Perspective on Training-Free Few-Shot Adaptation of Large Vision-Language Models



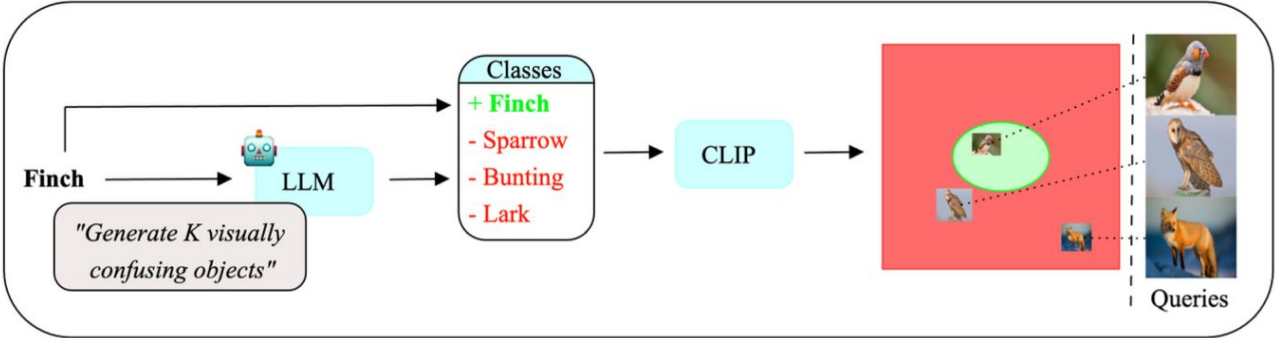
Summary



Zero-shot One-class Classification



inaturalist taxonomy



Method	F1 score
CLIPN	62.74
ZOC	78.16
Ours	79.60

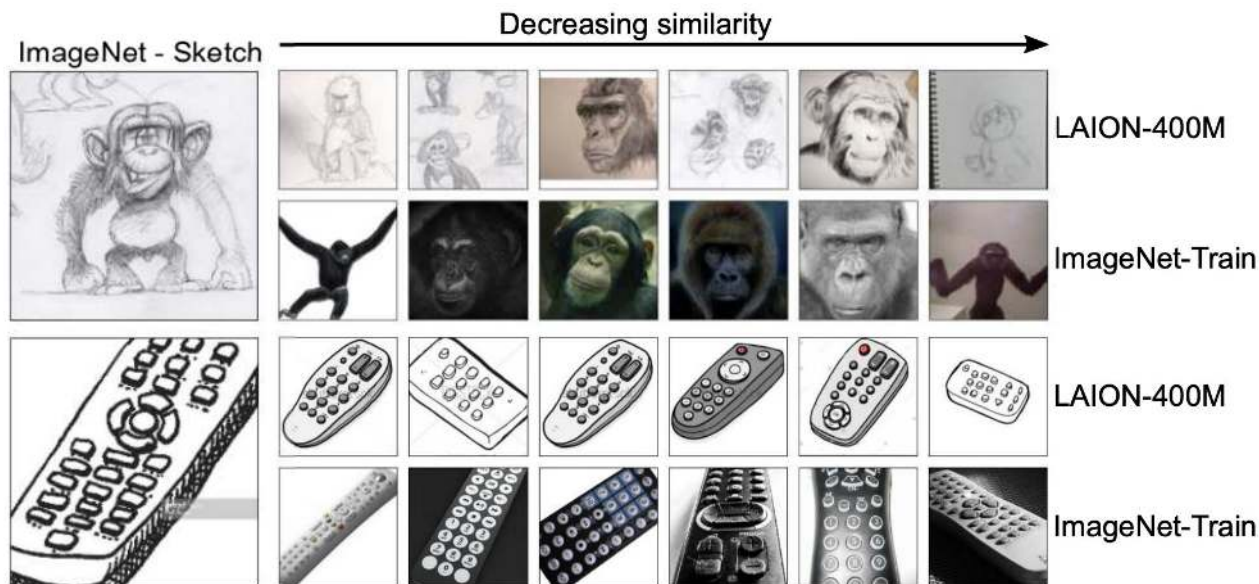
Other contributions

- A Simplex-Based Approach in Grets2023, First-author.
- FICUS: Few-shot image classification with unsupervised segmentation in IEEE EUSIPCO 2024, Co-author.
- Few and Fewer: Learning Better from Few Examples Using Fewer Base Classes in ArXiv 2023, Co-author.



Perspectives

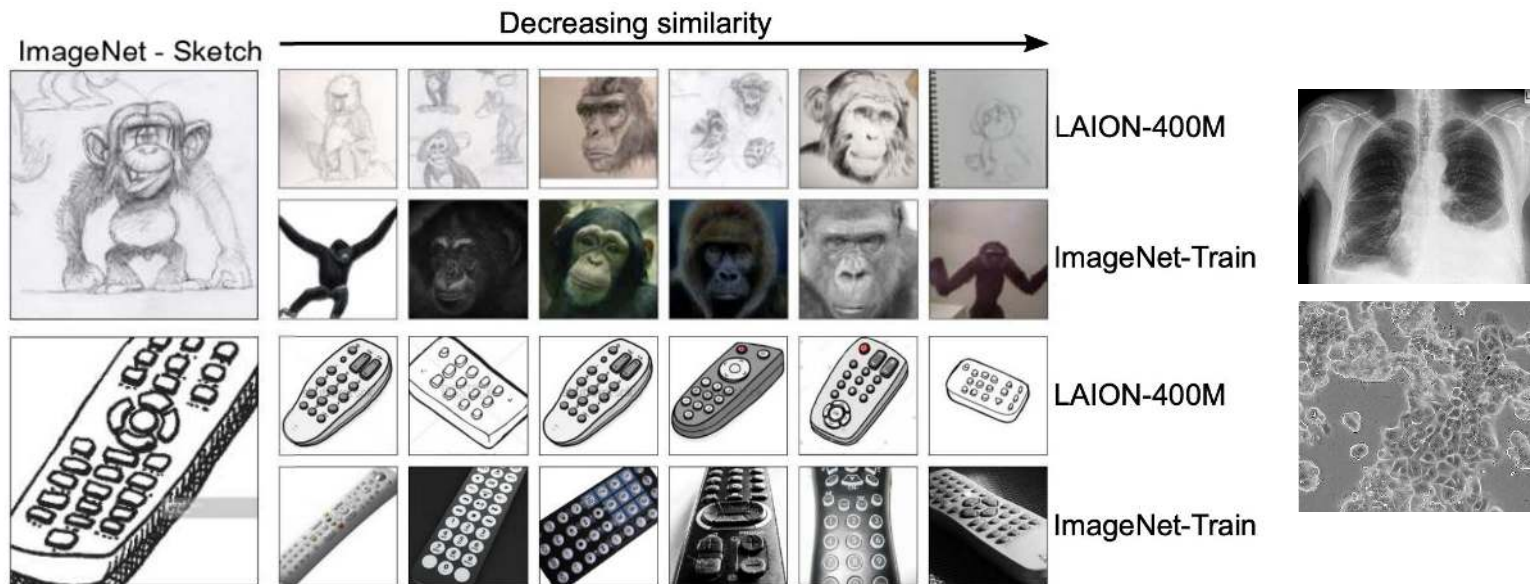
Task adaptation instead of learning.



Mayilvahanan et al., Does CLIP's generalization performance mainly stem from high train-test similarity?
ICLR 2024.

Perspectives

Task adaptation instead of learning.

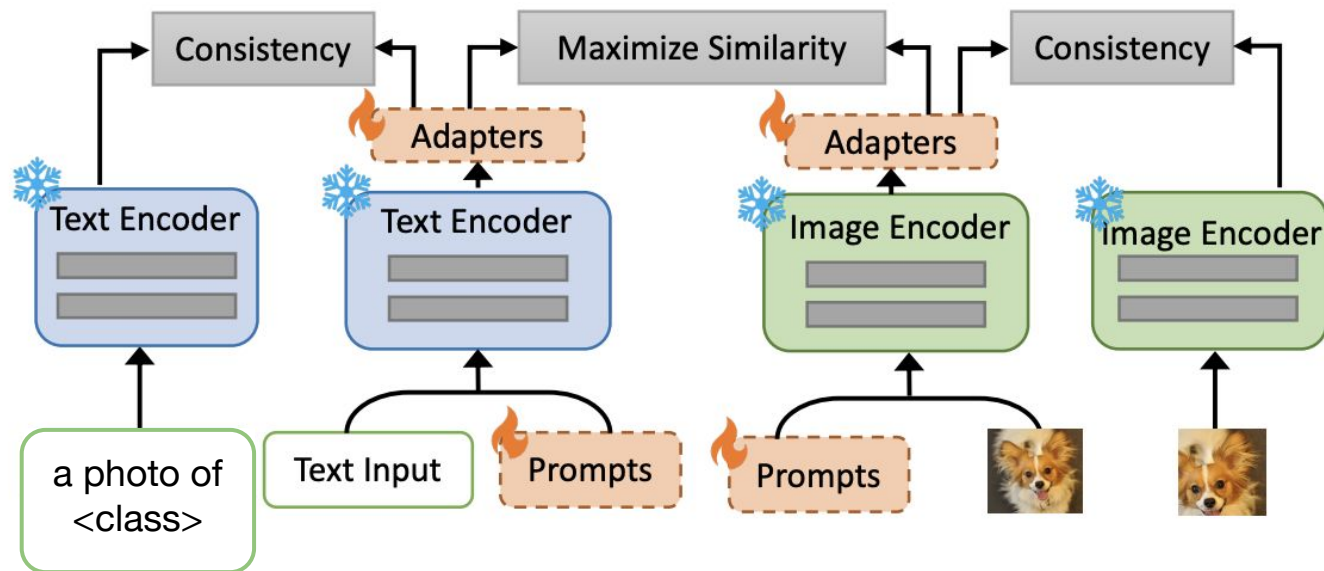


Mayilvahanan et al., Does CLIP's generalization performance mainly stem from high train-test similarity?
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Perspectives

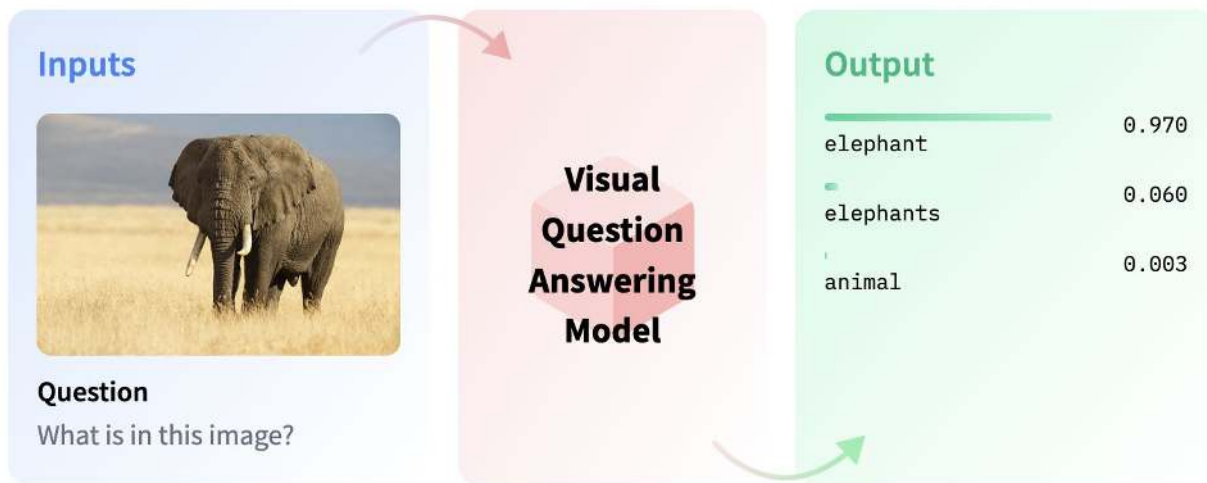
	Base	Novel	HM
CLIP	72.08	77.80	74.83
CoOp	97.60	59.67	74.06
Co-CoOp	94.87	71.75	81.71
ProGrad	95.54	71.87	82.03
KgCoOp	95.00	74.73	83.65
MaPLe	95.92	72.46	82.56
PromptSRC	98.07	76.50	85.95
CoPrompt	97.27	76.60	85.71

Performance on Flowers Dataset



Consistency-guided Prompt Learning for Vision-Language Models. ICLR 2024

Perspectives



<https://huggingface.co/tasks/visual-question-answering>

Perspectives

Concept: **night snake**

Definition: a **small light brown or beige colored snake**.

What is the species name of the animal in the photo?

European adder

garter snake

Generate a photo of a **night snake**

DALL-E 3

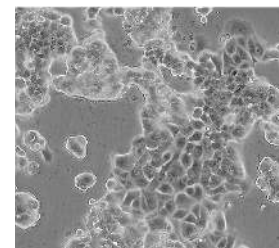
SD-XL



Nothofagus cunninghamii

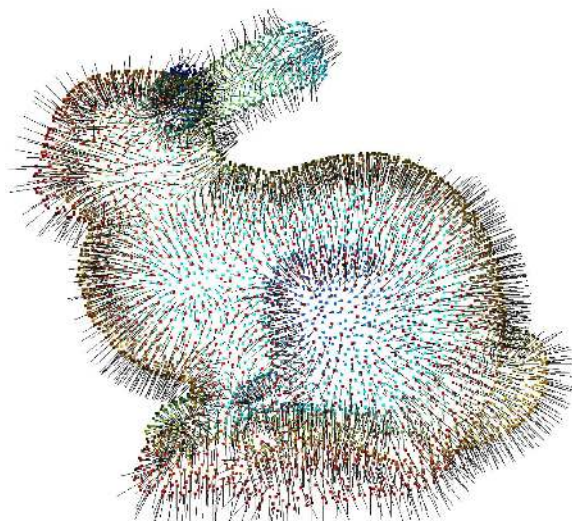


Atelectasis



MCF-7

The use of Foundation models for few shot learning in other tasks.



Additional slides

Table 2.6 – 1-shot and 5-shot accuracy of state-of-the-art methods and proposed solution on **MiniImageNet** in **transductive** setting. We include in gray state-of-the-art after the publication of our contribution EASY.

Method	Year	backbone	#Params	1-shot	5-shot
TIM-GD [97]	2020	ResNet12	12 M	73.90	85.00
ODC [98]	2021	ResNet12	12 M	77.20 ± 0.36	87.11 ± 0.42
PEM _n E-BMS* [81]	2021	ResNet12	12 M	80.56 ± 0.27	87.98 ± 0.14
SSR [99]	2021	ResNet12	12 M	68.10 ± 0.60	76.90 ± 0.40
iLPC [100]	2020	ResNet12	12 M	69.79 ± 0.99	79.82 ± 0.55
EPNet [80]	2020	ResNet12	12 M	66.50 ± 0.89	81.60 ± 0.60
DPGN [101]	2020	ResNet12	12 M	67.77 ± 0.32	84.60 ± 0.43
ECKPN [102]	2021	ResNet12	12 M	70.48 ± 0.38	85.42 ± 0.46
Rot+KD+POODLE [103]	2021	ResNet12	12 M	77.56	85.81
EASY (ours)	2022	$2 \times \text{ResNet12}(\frac{1}{\sqrt{2}})$	12 M	82.31 ± 0.24	88.57 ± 0.12
SSR [99]	2021	WRN-28-10	36 M	72.40 ± 0.60	80.20 ± 0.40
fine-tuning(train+val) [104]	2019	WRN-28-10	36 M	68.11 ± 0.69	80.36 ± 0.50
	2020	WRN-28-10	36 M	71.40	81.20
SIB+E ³ BM [105]	2020	WRN-28-10	36 M	71.40	81.20
LR+DC [69]	2021	WRN-28-10	36 M	68.57 ± 0.55	82.88 ± 0.42
EPNet [80]	2020	WRN-28-10	36 M	70.74 ± 0.85	84.34 ± 0.53
TIM-GD [97]	2020	WRN-28-10	36 M	77.80	87.40
PT+MAP [85]	2021	WRN-28-10	36 M	82.92 ± 0.26	88.82 ± 0.13
iLPC [100]	2020	WRN-28-10	36 M	83.05 ± 0.79	88.82 ± 0.42
ODC [98]	2021	WRN-28-10	36 M	80.64 ± 0.34	89.39 ± 0.39
PEM _n E-BMS* [81]	2021	WRN-28-10	36 M	83.35 ± 0.25	89.53 ± 0.13
EASY (ours)	2022	$3 \times \text{ResNet12}$	36 M	84.04 ± 0.23	89.14 ± 0.11
State-of-the-art after the publication of EASY					
protoLP [106]	2023	WRN-28-10	36 M	84.32 ± 0.21	90.02 ± 0.12
BECLR [107]	2024	RN50	21M	80.57 ± 0.57	87.82 ± 0.29
MetaFormer-A [61]	2024	ViT-Small	24.5 M	84.78 ± 0.79	91.39 ± 0.42

*Comparison with fine-tuned methods for **16** shots. Hyperparameters are transferred from ImageNet.*

Method	Average
Tip-Adapter-F	71.40
CrossModallP	73.65
TaskRes	74.42
APE-T	68.28
CLAP	74.57
ProKeR (ours)	75.11

*Comparison with fine-tuned methods for **16** shots. Hyperparameters are transferred from ImageNet.*

Method	Average
Tip-Adapter-F	71.40
CrossModallP	73.65
TaskRes	74.42
APE-T	68.28
CLAP	74.57
ProKeR (ours)	75.11

$$\min_{\mathcal{W}} \sum_{m=1}^M \mathcal{H}(\mathbf{y}^{(m)}, \hat{\mathbf{y}}^{(m)}) + \sum_{m=1}^M \sum_{c=1}^C \Lambda_{mc} \|\mathbf{t}_c - \mathbf{w}_c^{(m)}\|_2^2.$$

$$\min_{\mathbf{W}} \mathcal{L}_{\text{CLAP}}(\phi_{\mathbf{W}}, \mathbf{S}, \mathbf{L})$$

$$\text{where } \phi_{\mathbf{W}}(\mathbf{x}) = \mathbf{x}\mathbf{W} + \sum_{i=1}^{NK} k_{\beta}(\mathbf{S}_i, \mathbf{W})\gamma_i$$

$$\text{and } \gamma = (\mathbf{I} + \frac{1}{\lambda} k_{\beta}(\mathbf{S}, \mathbf{S}))^{-1}(\mathbf{L} - \mathbf{S}\mathbf{W})$$

Comparison with training-based methods using 16 shots. Hyperparameters are transferred from ImageNet

Method	Average
Tip-Adapter-F	71.40
CrossModalLP	73.65
TaskRes	74.42
APE-T	68.28
CLAP	74.57
ProKeR (ours)	75.11
ProKeR + CLAP (ours)	76.84

$$\phi(\mathbf{x}) = \arg \min_{\mathbf{q}} \frac{1}{NK} \sum_{i=1}^{NK} k_{\beta}(d(\mathbf{x}, \mathbf{S}_i)) s(\mathbf{q}, \mathbf{L}_i) + \mathcal{R}_{\text{clip}}$$

Nadaraya-Watson objective

$$\phi(\mathbf{x}) = \tilde{\mathbf{x}}\mathbf{A} \quad \min_{\mathbf{A}} \frac{1}{NK} \sum_{i=1}^{NK} k_{\beta}(d(\mathbf{x}, \mathbf{S}_i)) s(\tilde{\mathbf{S}}_i\mathbf{A}, \mathbf{L}_i) + \mathcal{R}_{\text{clip}}$$

Local Linear Regression objective

$$R(\mathbf{x}, \phi(\mathbf{x})) = \mathbb{E}_{Y|X}[s(Y, \phi(X)) + \mathcal{R}_{\text{clip}} \mid X = \mathbf{x}]$$

Pointwise risk

$$\phi(\mathbf{x}) = \arg \min_{\mathbf{q} \in M} R(\mathbf{x}, \mathbf{q}) = \arg \min_{\mathbf{q} \in M} \int_M s(\mathbf{y}, \mathbf{q}) d\mu_{\mathbf{x}}(\mathbf{y}) + \mathcal{R}_{\text{clip}}$$

$$\phi(\mathbf{x}) = \frac{\lambda NK}{\lambda NK + \mathcal{Z}(\mathbf{x})} f_{\text{clip}}(\mathbf{x}) + \frac{1}{\lambda NK + \mathcal{Z}(\mathbf{x})} \sum_{i=1}^{NK} k_{\beta}(d(\mathbf{x}, \mathbf{S}_i)) \mathbf{L}_i$$

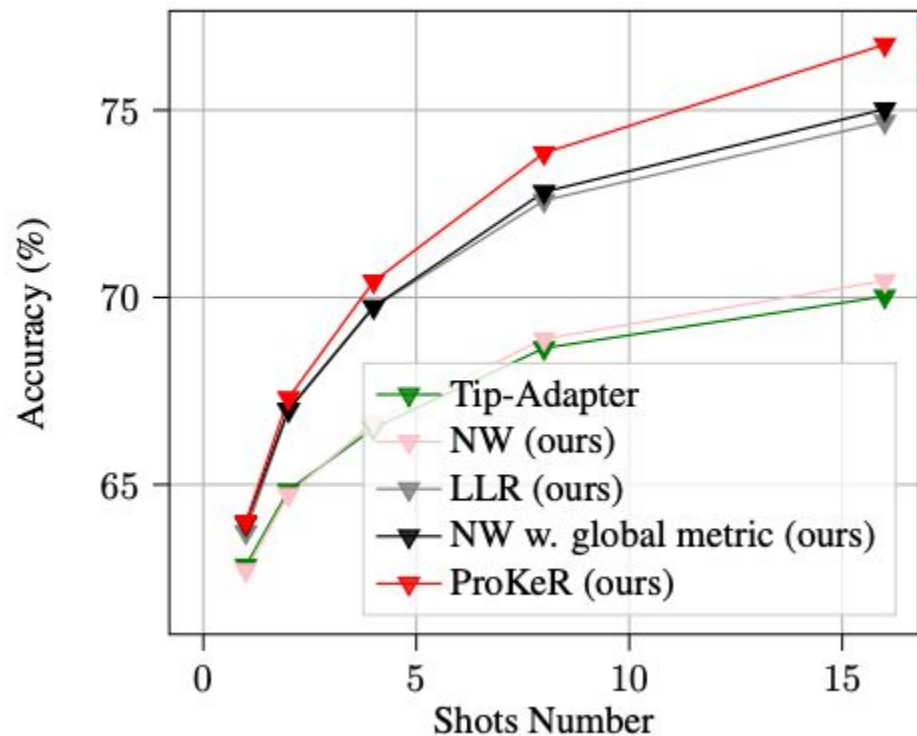
$$\text{where } \mathcal{Z}(\mathbf{x}) = \sum_{i=1}^{NK} k_{\beta}(d(\mathbf{x}, \mathbf{S}_i))$$

$$(\mathbf{K}_\beta(\mathbf{x}, \mathbf{x}'))_{j,j'} = k_\beta(\mathbf{x}, \mathbf{x}')\mathbf{B}$$

Separable kernels

Additional slides

Average over 11 datasets



$$d(\mathbf{x}, \mathbf{S}_i) = \|\mathbf{x} - \mathbf{S}_i\|_{\hat{\Lambda}}$$

Kernel	$k(\mathbf{x}, \mathbf{x}')$	Accuracy
Linear	$\mathbf{x}\mathbf{x}'^\top$	72.34
Epanechnikov	$\frac{3}{4} (1 - \ \mathbf{x} - \mathbf{y}'\ _2^2)$	74.43
Polynomial	$(\mathbf{x}\mathbf{y}'^\top)^2$	76.61
RBF	$\exp(-\frac{\beta}{2} \ \mathbf{x} - \mathbf{y}'\ _2^2)$	76.75

Additional slides

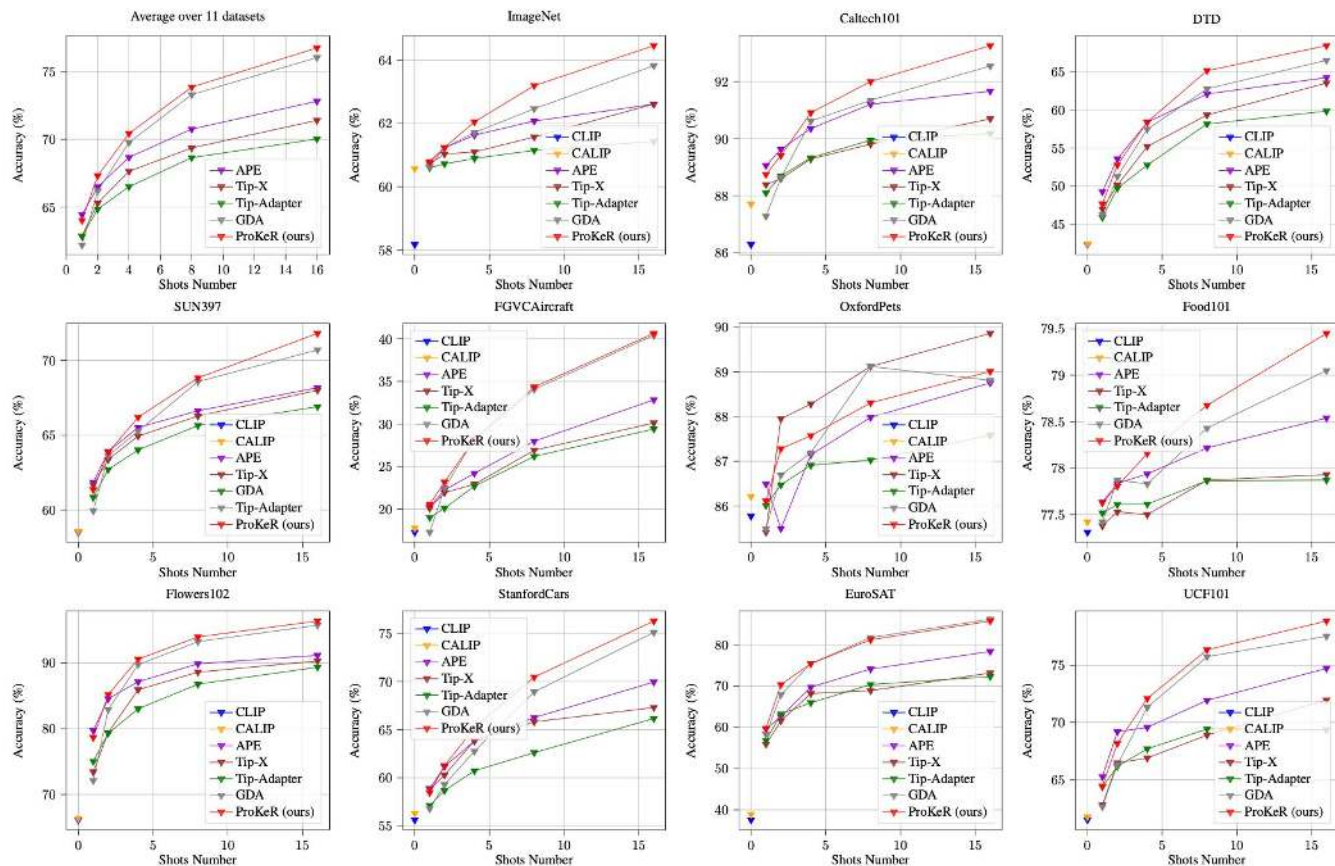


Table 4: **Robustness to distribution shift of different methods for 16 shots.**

Datasets	Source	Target				Average
	ImageNet [9]	-V2 [44]	-Sketch [48]	-A [20]	-R [19]	
Zero-Shot CLIP [41]	60.33	53.27	35.44	23.61	60.42	46.61
Tip-Adapter [62]	61.43	54.13	35.71	23.63	60.41	47.06
APE [67]	62.60	54.93	35.41	22.95	59.90	47.15
GDA [55]	63.82	55.35	34.32	19.53	55.56	45.71
ProKeR (Polynomial) (Ours)	64.66	56.11	36.08	23.27	60.55	48.13
ProKeR (Ours)	64.45	56.02	36.08	23.37	60.59	48.10

Kernel	$k(\mathbf{x}, \mathbf{x}')$	Accuracy
Linear	$\mathbf{x}\mathbf{x}'^\top$	72.34
Epanechnikov	$\frac{3}{4} (1 - \ \mathbf{x} - \mathbf{y}'\ _2^2)$	74.43
Polynomial	$(\mathbf{x}\mathbf{y}'^\top)^2$	76.61
RBF	$\exp(-\frac{\beta}{2} \ \mathbf{x} - \mathbf{y}'\ _2^2)$	76.75

Table 5: Kernel Ablation for 16 shots on 11 datasets on CoOp's benchmark.

Table 6: **Average performance on 16-shot ImageNet with different backbones.**

Models	ResNet-50	ResNet-101	ViT-B/32	ViT-B/16
Zero-shot CLIP [41]	60.33	62.53	63.80	68.73
Tip-Adapter [62]	61.43	64.08	65.18	70.25
APE [67]	62.60	65.61	66.31	71.37
ProKeR (Ours)	64.45	67.39	68.12	73.25

Additional slides

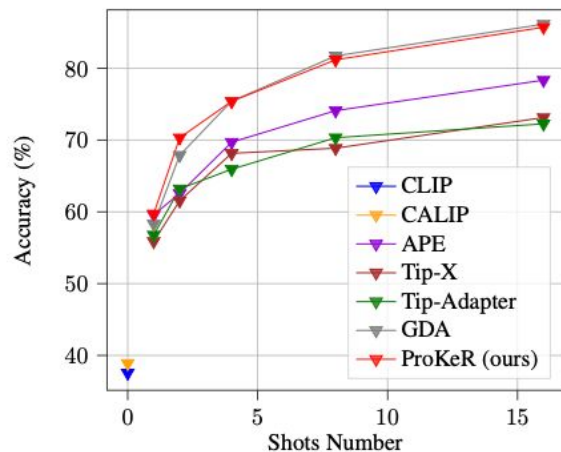
Table 7: **Running times on ImageNet for 16 shots.** All experiments are performed on a RTX A6000 GPU. For each method we report the memory requirements. D_1 is the inner dimension of the MLP used in Clip-Adapter [12] and T is the number of text tokens in CoOp [66]. R is the number of Fourier features used to approximate the RBF kernel.

Methods	Overall Time	Train	Memory requirements
CoOp[66]	$\sim 17\text{h}$	✓	$N \times T \times D$
Clip-Adapter[12]	$\sim 40\text{min}$	✓	$(2N + D_1) \times D$
CrossModal-LP[28]	$\sim 3\text{min}$	✓	$N \times D$
Standard LP[41]	$\sim 3\text{min}$	✓	$N \times D$
Tip-adapter-F[62]	$\sim 7\text{min}$	✓	$N \times K + N \times (K + 1) \times D$
Tip-Adapter[62]	2.1s	✗	$N \times K + N \times (K + 1) \times D$
APE [67]	24.6s	✗	$N \times K + N \times (K + 1) \times D$
GDA [55]	1.6 s	✗	$2 \times N \times D$
ProKeR (Ours)	4.7s	✗	$N \times K + N \times (K + 1) \times D$
ProKeR with RFF (Ours)	4.2s	✗	$N \times (D + R)$

Additional slides



EuroSAT

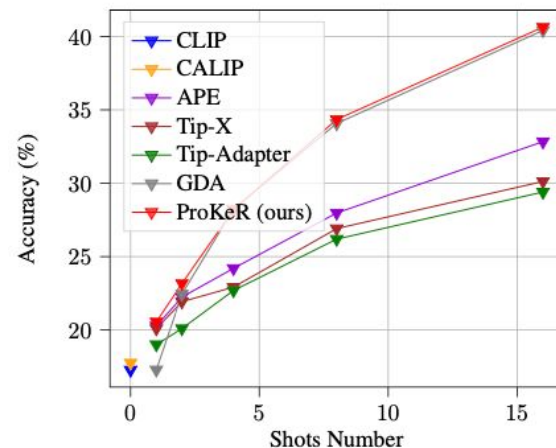


Example of class names:

['Annual Crop Land', 'Forest', 'Highway or Road', 'Industrial Buildings']



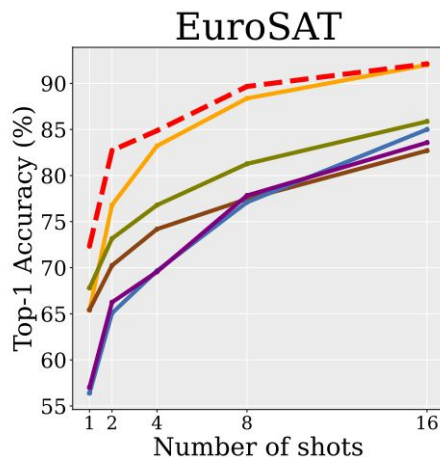
FGVCAircraft



Example of class names:

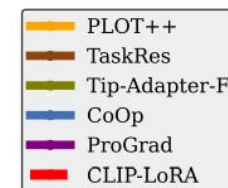
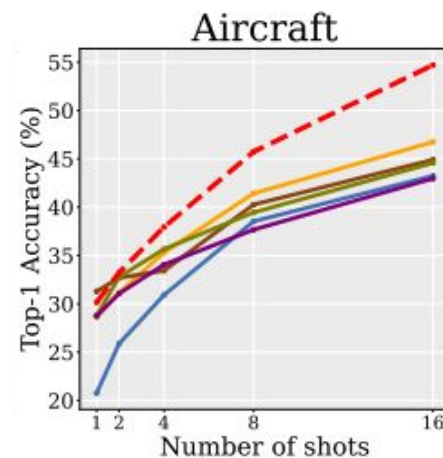
['707-320', '727-200', 'PA-28', 'Dornier 328']

Additional slides



Example of class names:

['Annual Crop Land', 'Forest', 'Highway or Road', 'Industrial Buildings']



Example of class names:

['707-320', '727-200', 'PA-28', 'Dornier 328']

$$\min_{\phi \in \mathcal{H}} \sum_{i=1}^{NK} \|\phi(\mathbf{S}_i) - \mathbf{L}_i\|_2^2 + \lambda \|\phi - f_{\text{clip}}\|_{\mathcal{H}}^2$$



$$\min_{\phi \in \mathcal{H}} \sum_{i=1}^{NK} k'(\mathbf{S}_i, \mathbf{z})(\phi(\mathbf{S}_i) - \mathbf{L}_i)^2 + \lambda \|\phi - f_{\text{clip}}\|_{\mathcal{H}}^2$$

Local Kernel Ridge Regression for Scalable, Interpolating, Continuous Regression. TMLR 2024

Additional slides

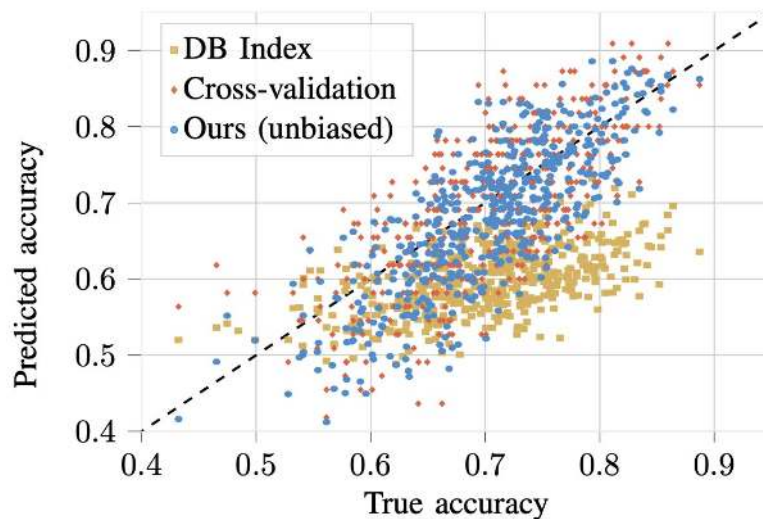


Figure 10: Scatter plot of 5-class few-shot problems with 10 samples per class from ImageNet

Additional slides

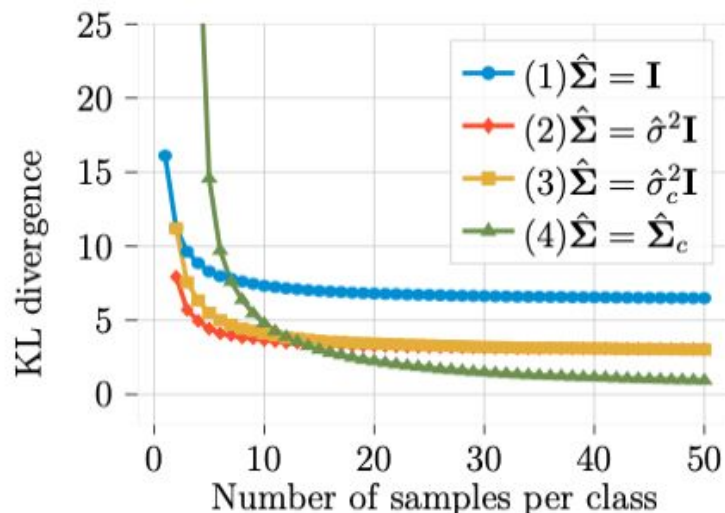


Figure 4.2 – We compute the average KL divergence between a Gaussian distribution fitted with a limited number of samples and its closest Gaussian approximation based on a larger sample set. This average is taken over 10^3 few-shot classification tasks, each involving 5 classes, using the test set of the Meta-dataset ImageNet.

Additional slides

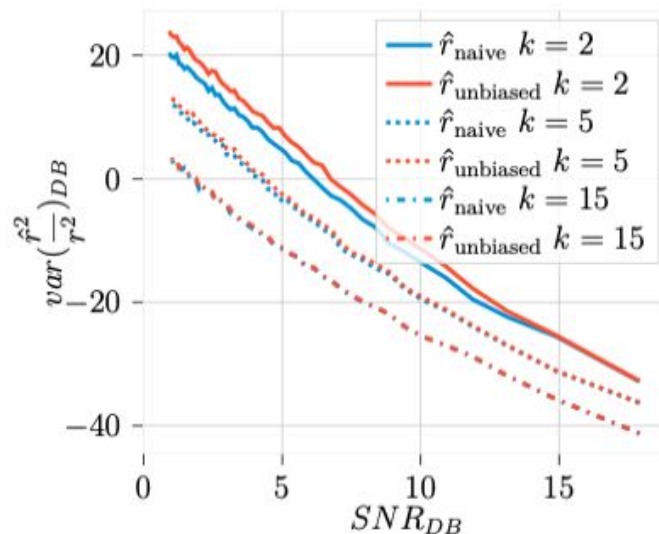


Figure 4.8 – Variance of both estimators against different SNR values in log-log scale. The estimates are normalized by the true distances. The unbiased estimator shows slightly higher variance compared to the naive estimator. The variance of the two estimators are asymptotically of $\mathcal{O}(\frac{1}{\text{SNR}})$.

Additional slides

Lemma 1. Given two classes represented univariate isotropic distributions $\mathcal{N}_a(\mu_a, \sigma^2)$ and $\mathcal{N}_b(\mu_b, \sigma^2)$ with shared and known standard deviations, the true probability of error P_e is bounded by:

$$P\left(\left|P_e - \hat{P}_e\right| \leq \frac{\phi'(0)}{\sqrt{2k}} \left| \phi^{-1}\left(1 - \frac{\alpha}{2}\right) \right| \right) \geq 1 - \alpha, \quad (4.7)$$

with a probability $1 - \alpha$, where $\hat{P}_e$ is the estimated probability error from a sequence of k i.i.d random variables drawn from the two distributions, with empirical mean estimates $\hat{\mu}_a$ and $\hat{\mu}_b$, ϕ is the cumulative distribution function of $\mathcal{N}(0, 1)$, ϕ^{-1} is its inverse and ϕ' is its derivative, with $\phi'(0) = \frac{1}{\sqrt{2\pi}}$.

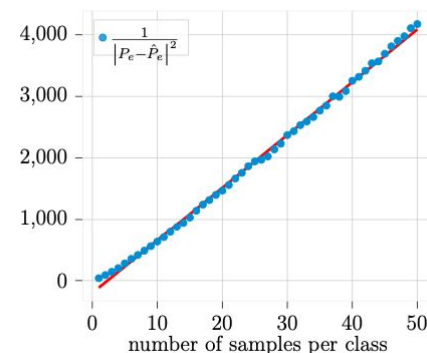


Figure 4.9 – 2-class, $R^2 = 0.9979$

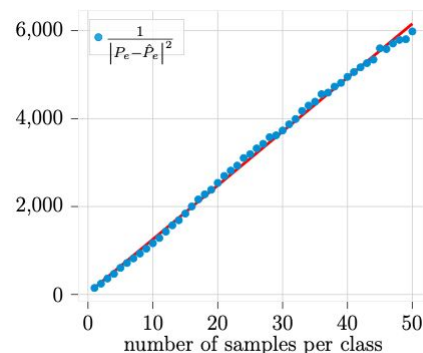
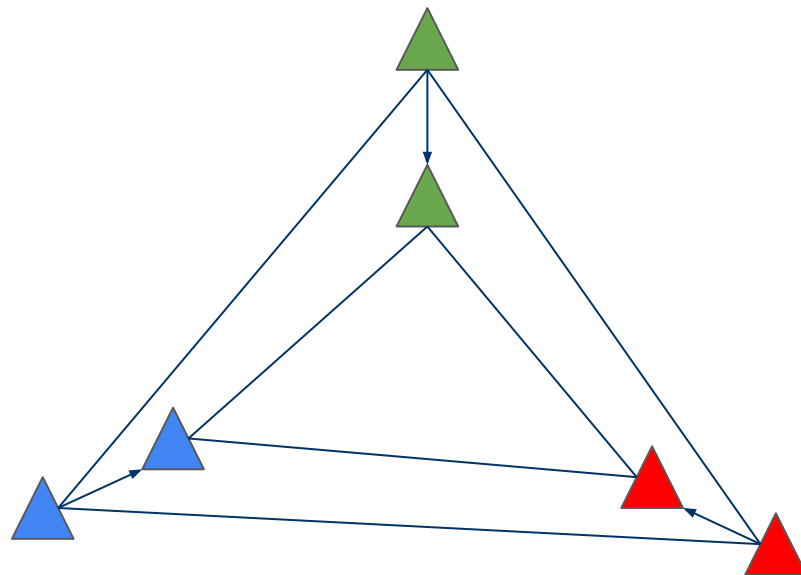
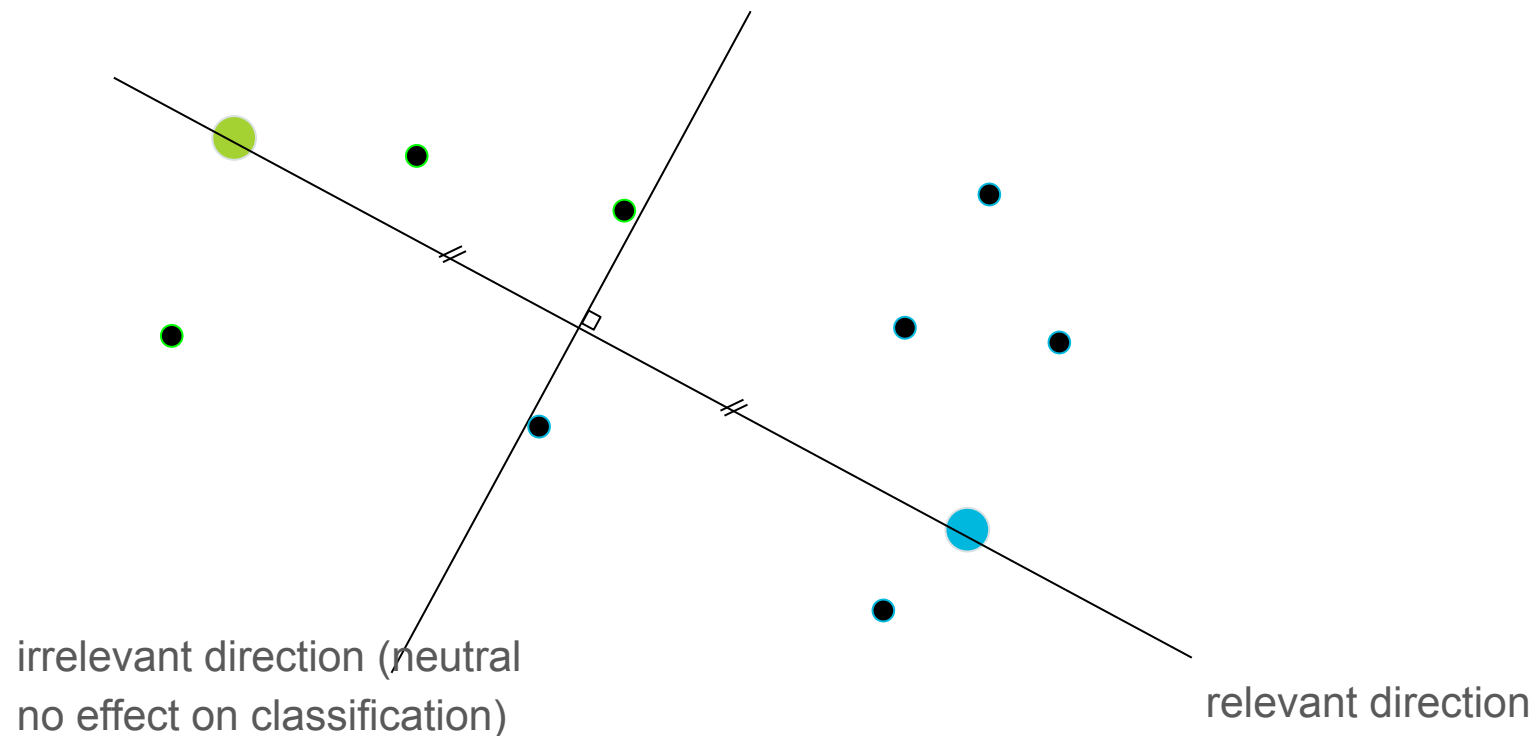


Figure 4.10 – 5-class, $R^2 = 0.9981$

Non-metric multidimensional scaling



Additional slides



Additional slides



(a) Level 0, class name is “Mugwort”



(b) Level 5, class name is “Vertebrates”



(c) Level 9, class name is “Plants”

Figure 5.10 – Examples of query samples from the proposed hierarchical sampling for iNaturalist. Positive query samples are highlighted in green while negative query samples are framed in red.

Additional slides

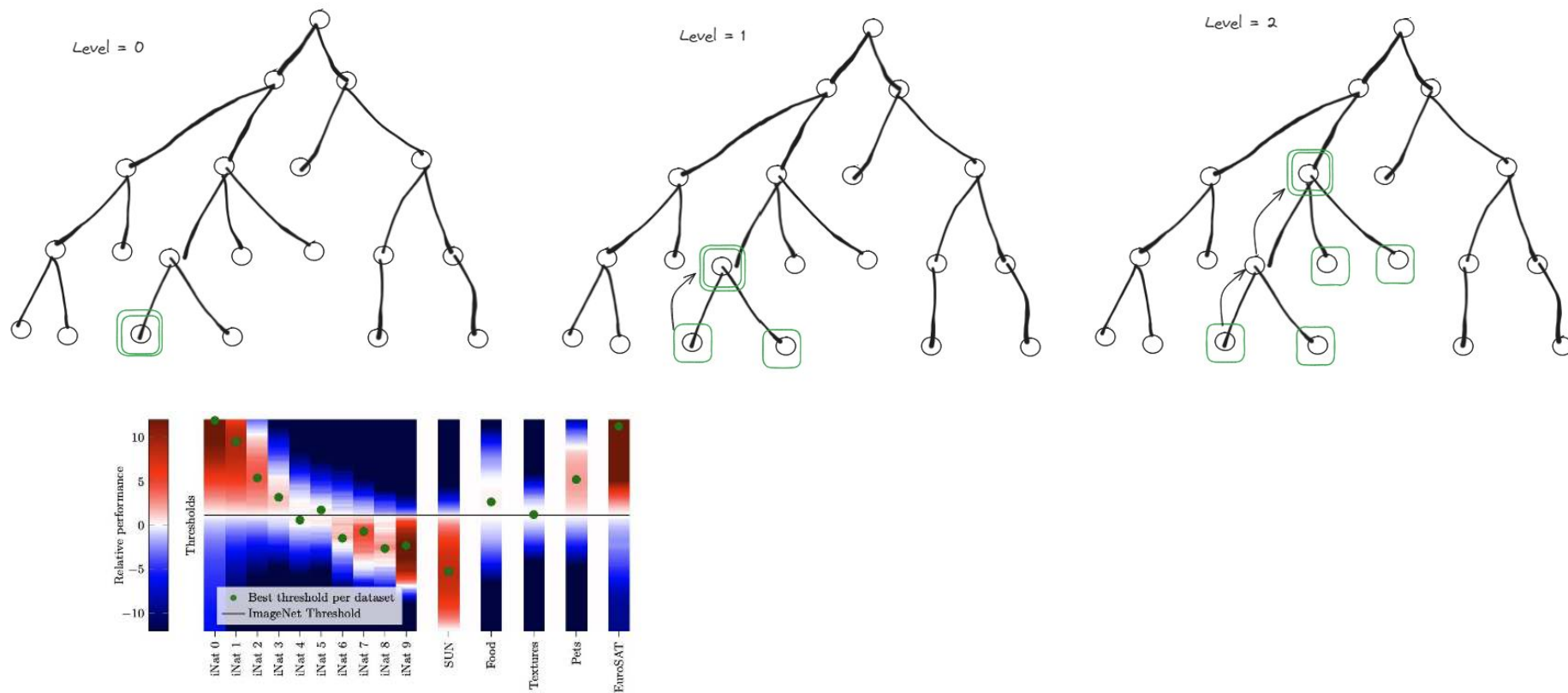


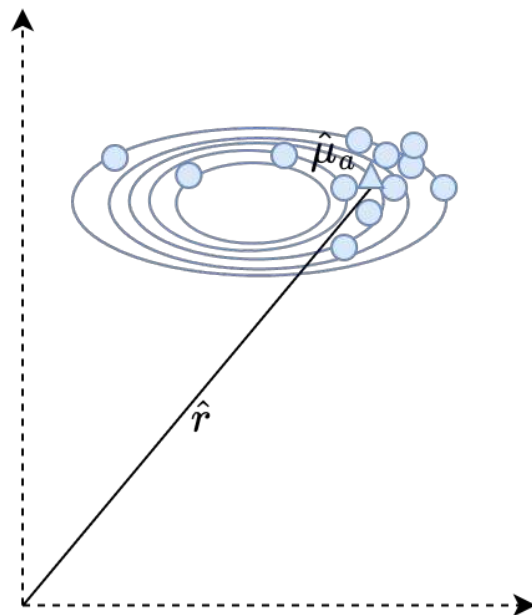
Fig. 3: Impact of varying a single threshold for all tasks. For each dataset, we vary the threshold (lowest to highest from bottom to top) and report the relative performance to the one obtained from ImageNet. For iNaturalist, we report the performance per level of granularity.

Additional slides

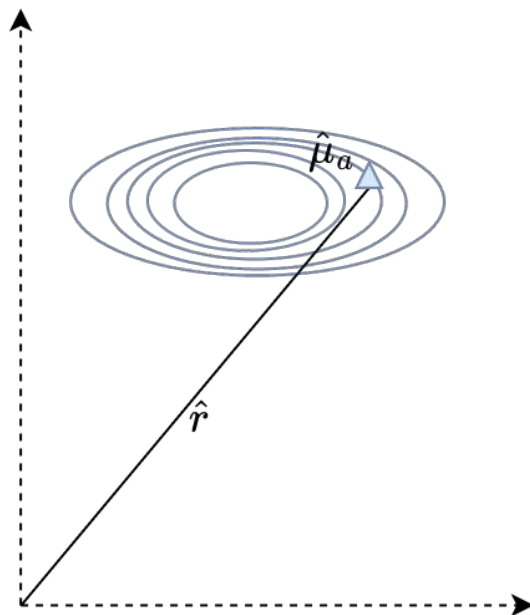
Table 1. Few-shot classification performance of pairwise combinations of a variety of training and adaptation algorithms. All evaluation tasks are 5-way 5-shot tasks sampled from Meta-Dataset (excluding ImageNet). We sample 2000 tasks per dataset in Meta-Dataset and report the average accuracy over all datasets along with the 95% confidence interval. The algorithms are listed according to their partial order according to Definition 3.2 from top to bottom and from left to right. * denotes training algorithm that uses transductive BN (Bronskill et al., 2020) that produces a much higher, unfair performance using Fintune and TSA as adaptation algorithms. †: TSA and eTT are both architecture-specific partial-finetune algorithms, thus TSA can be used for CNN only and eTT for original ViT only.

Training algorithm	Training dataset	Architecture	MatchingNet	MetaOpt	NCC	Adaptation algorithm				
						LR	URL	CC	TSA/eTT [†]	Finetune
PN	miniImageNet	Conv-4	48.54±0.4	49.84±0.4	51.38±0.4	51.65±0.4	51.82±0.4	51.56±0.4	58.08±0.4	60.88±0.4
MAML*	miniImageNet	Conv-4	53.71±0.4	53.69±0.4	55.01±0.4	55.03±0.4	55.66±0.4	55.63±0.4	62.80±0.4	64.87±0.4
CE	miniImageNet	Conv-4	54.68±0.4	56.79±0.4	58.54±0.4	58.26±0.4	59.63±0.4	59.20±0.5	64.14±0.4	65.12±0.4
MatchingNet	miniImageNet	ResNet-12	55.62±0.4	57.20±0.4	58.91±0.4	58.99±0.4	61.20±0.4	60.50±0.4	64.88±0.4	67.93±0.4
MAML*	miniImageNet	ResNet-12	58.42±0.4	58.52±0.4	59.65±0.4	60.04±0.4	60.38±0.4	60.50±0.4	71.15±0.4	73.13±0.4
PN	miniImageNet	ResNet-12	60.19±0.4	61.70±0.4	63.71±0.4	64.46±0.4	65.64±0.4	65.76±0.4	70.44±0.4	74.23±0.4
MetaOpt	miniImageNet	ResNet-12	62.06±0.4	63.94±0.4	65.81±0.4	66.03±0.4	67.47±0.4	67.24±0.4	72.07±0.4	74.96±0.4
DeepEMD	miniImageNet	ResNet-12	62.67±0.4	64.15±0.4	66.14±0.4	66.14±0.4	68.66±0.4	69.76±0.4	74.21±0.4	74.83±0.4
CE	miniImageNet	ResNet-12	63.27±0.4	64.91±0.4	66.96±0.4	67.14±0.4	69.78±0.4	69.52±0.4	74.30±0.4	74.89±0.4
Meta-Baseline	miniImageNet	ResNet-12	63.25±0.4	65.02±0.4	67.28±0.4	67.56±0.4	69.84±0.4	69.76±0.4	73.94±0.4	75.04±0.4
COS	miniImageNet	ResNet-12	63.99±0.4	66.09±0.4	68.31±0.4	69.26±0.4	70.71±0.4	71.03±0.4	75.10±0.4	75.68±0.4
PN	ImageNet	ResNet-50	63.68±0.4	65.79±0.4	68.40±0.4	68.87±0.4	69.69±0.4	70.81±0.4	74.15±0.4	78.42±0.4
S2M2	miniImageNet	WRN-28-10	64.41±0.4	66.59±0.4	68.67±0.4	69.16±0.4	70.88±0.4	71.38±0.4	74.94±0.4	76.89±0.4
FEAT	miniImageNet	ResNet-12	65.42±0.4	67.15±0.4	69.06±0.4	69.21±0.4	71.24±0.4	72.07±0.4	75.99±0.4	76.83±0.4
IER	miniImageNet	ResNet-12	65.37±0.4	67.31±0.4	69.30±0.4	70.01±0.4	72.48±0.4	72.85±0.4	76.70±0.4	77.54±0.4
Moco v2	ImageNet	ResNet-50	65.47±0.5	68.63±0.4	71.05±0.4	71.49±0.4	74.46±0.4	74.57±0.4	79.70±0.4	79.98±0.4
Exemplar v2	ImageNet	ResNet-50	67.70±0.4	70.07±0.4	72.55±0.4	72.93±0.4	75.26±0.4	76.83±0.4	80.22±0.4	81.75±0.4
DINO	ImageNet	ResNet-50	73.97±0.4	76.45±0.4	78.30±0.4	78.72±0.4	80.73±0.4	81.05±0.4	83.64±0.4	83.20±0.4
CE	ImageNet	ResNet-50	74.75±0.4	76.94±0.4	78.96±0.4	79.57±0.4	80.89±0.4	81.51±0.4	84.07±0.4	84.92±0.4
BiT-S	ImageNet	ResNet-50	75.44±0.4	77.86±0.4	79.84±0.4	79.97±0.4	81.79±0.4	81.91±0.4	84.84±0.3	86.40±0.3
CE	ImageNet	Swin-B	75.17±0.4	77.81±0.4	80.06±0.4	81.04±0.4	82.55±0.4	82.46±0.4	-	88.16±0.3
DeiT	ImageNet	ViT-B	75.82±0.4	78.34±0.4	80.62±0.4	81.68±0.4	82.80±0.3	83.13±0.4	84.22±0.3	87.62±0.3
CE	ImageNet	ViT-B	76.78±0.4	78.81±0.4	80.65±0.4	81.13±0.3	82.69±0.3	82.77±0.3	85.60±0.3	88.48±0.3
DINO	ImageNet	ViT-B	76.44±0.4	79.11±0.4	81.23±0.4	82.01±0.4	84.16±0.3	84.44±0.3	86.25±0.3	88.04±0.3
CLIP	WebImageText	ViT-B	78.06±0.4	81.20±0.4	83.04±0.3	83.22±0.3	84.11±0.3	84.20±0.3	87.66±0.3	90.26±0.3

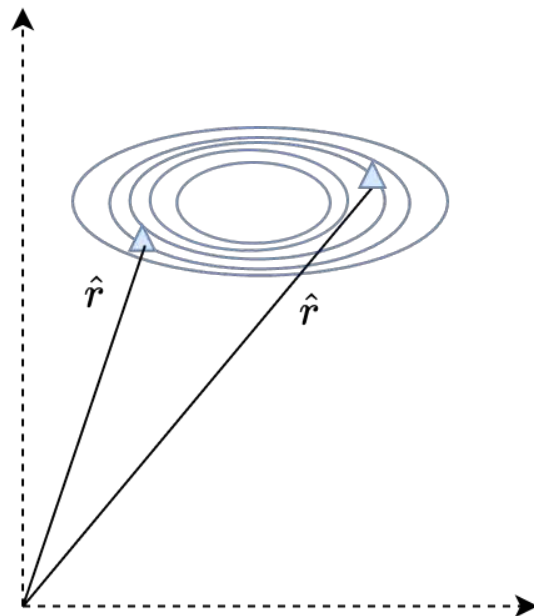
A Statistical Model for Predicting Generalization in Few-Shot Classification



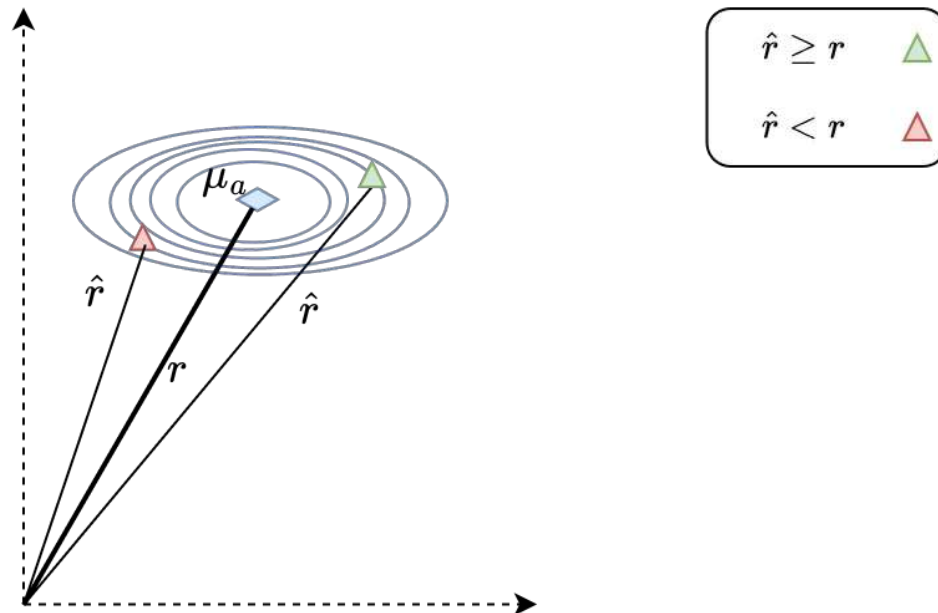
A Statistical Model for Predicting Generalization in Few-Shot Classification



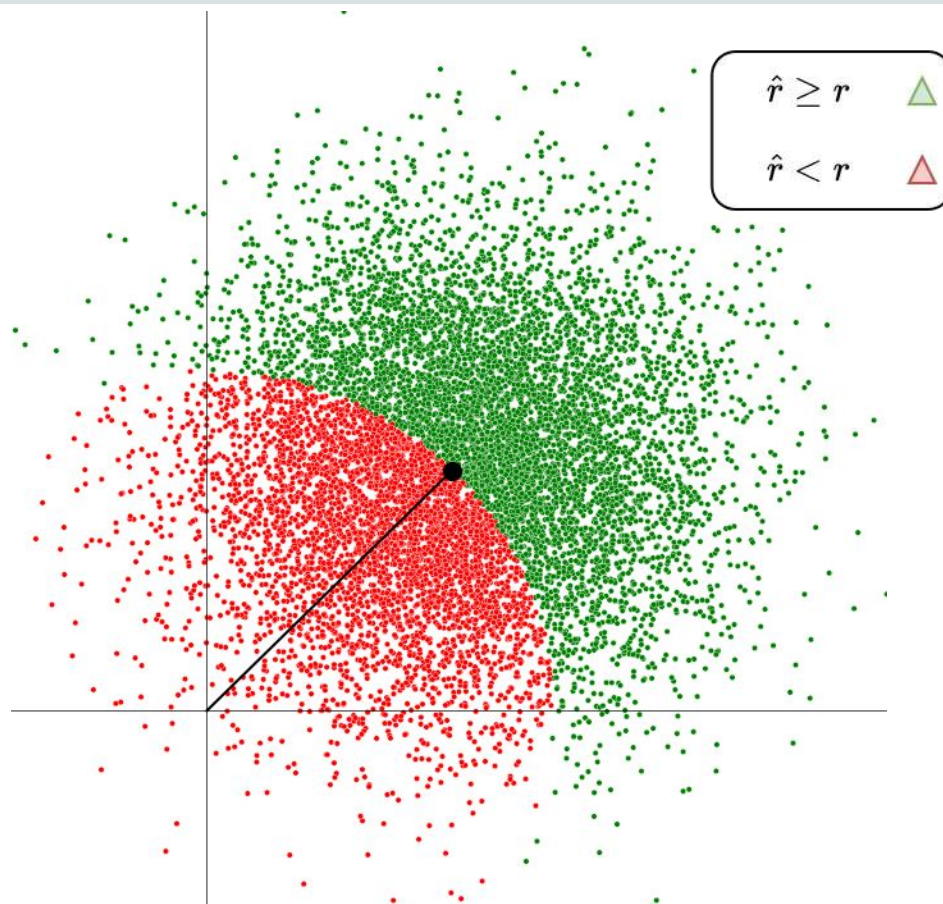
A Statistical Model for Predicting Generalization in Few-Shot Classification



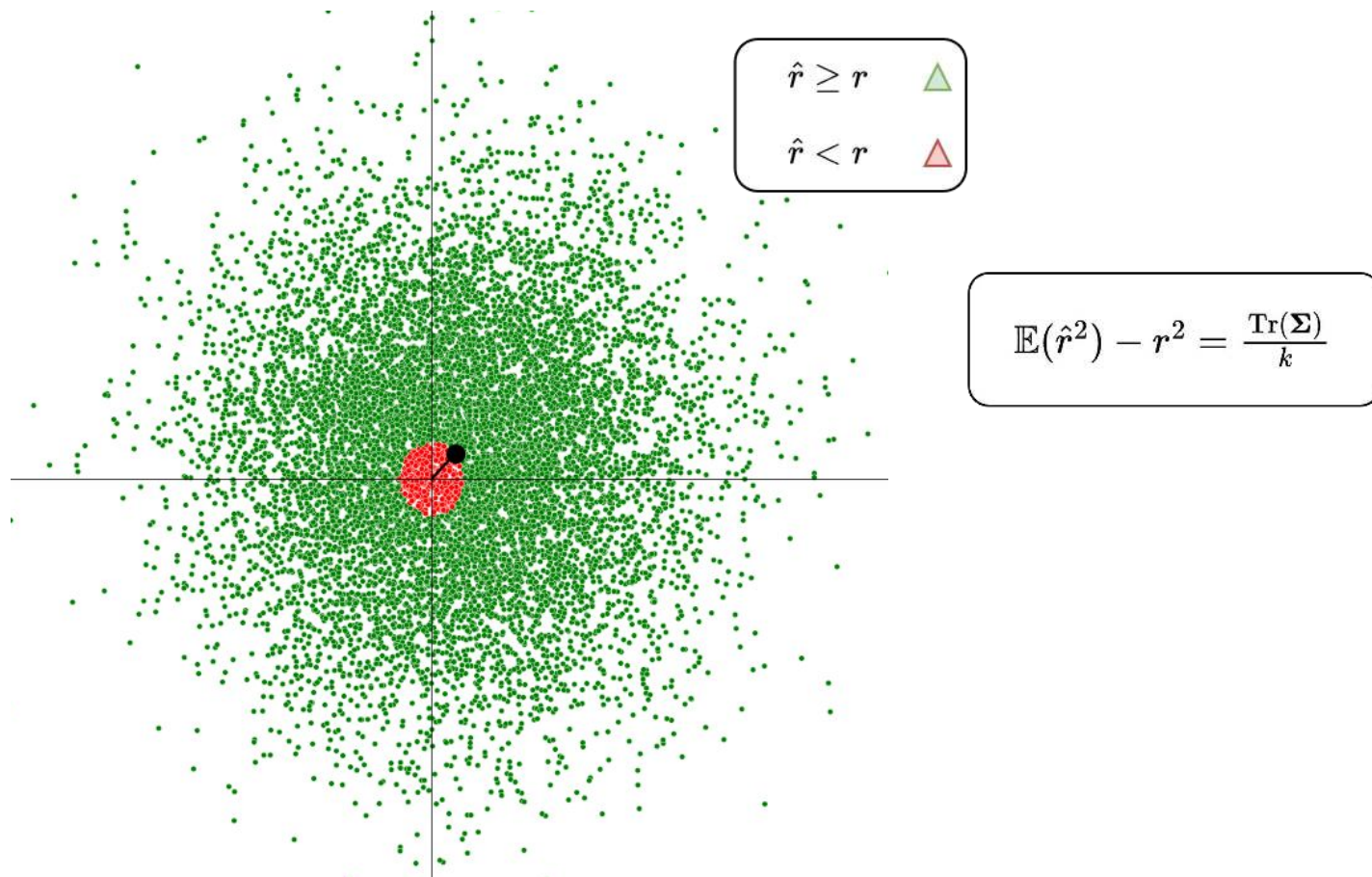
A Statistical Model for Predicting Generalization in Few-Shot Classification

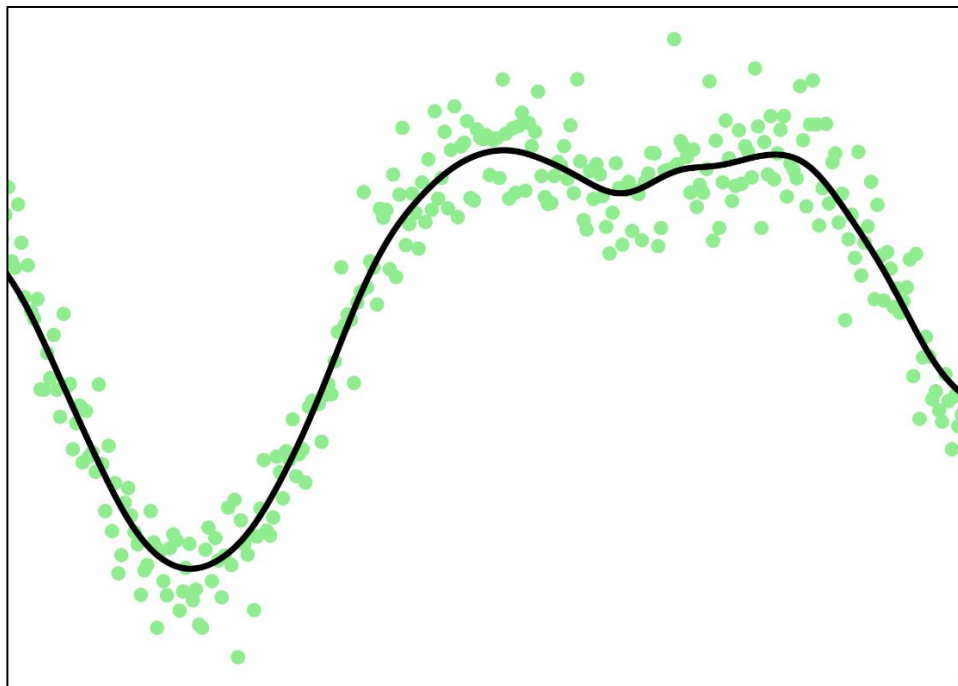


A Statistical Model for Predicting Generalization in Few-Shot Classification

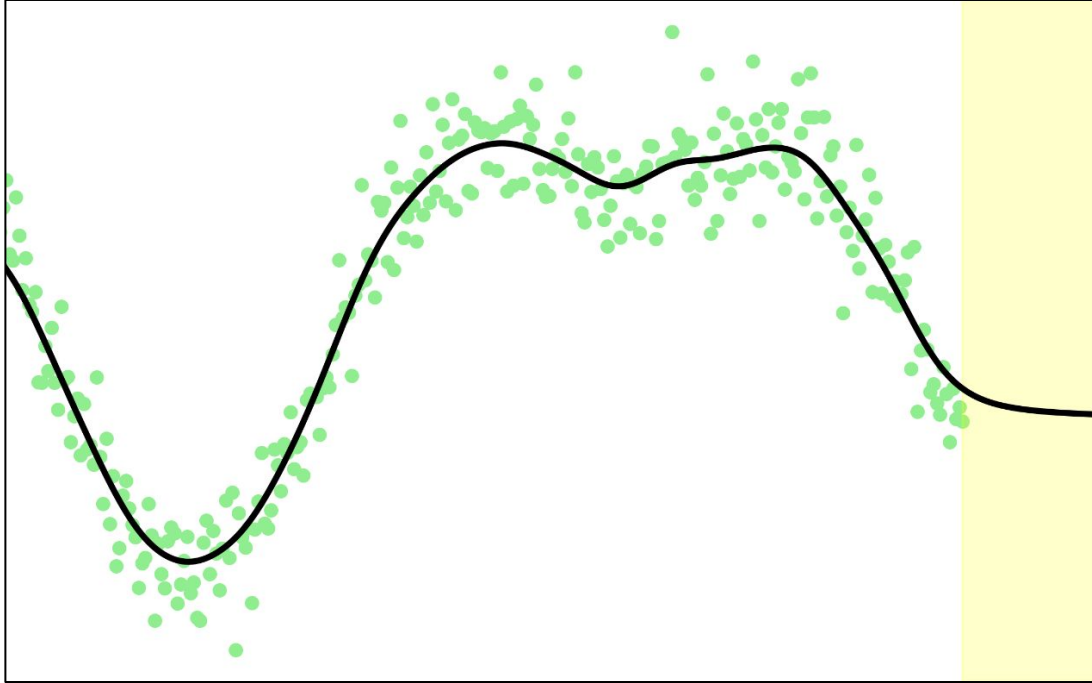


A Statistical Model for Predicting Generalization in Few-Shot Classification





A Kernel Perspective on Training-Free Few-Shot Adaptation of Large Vision-Language Models



A Kernel Perspective on Training-Free Few-Shot Adaptation of Large Vision-Language Models

